From Pizza.owl to Executable Intelligence

EXECUTABLE KNOWLEDGE ARCHITECTURE (EKA)

Mastering Ontology Engineering

with Protégé and Pizza.owl

VOLUME 3

Chapters 14-16

SEMANTIC LOGIC

OWL Restrictions, Reasoning and Governance

XIAOQI ZHAO

Chapters List -- Volume 3

This Volume's Chapters (Volume 3 of 7)

Full Book Roadmap

- Volume 1 -- Chapters 00-08: Ontology Foundations in Protégé
- Volume 2 -- Chapters 09-13: Object Properties, Characteristics, Domain and Range
- **Volume 3 (this book)** -- Chapters 14-16: Semantic Requirements and EKA Governance
- Volume 4 -- Chapters 17-24: Semantic Knowledge Development Lifecycle (SKDL) (In writing...)
- Volume 5: EKA Part 1 — Executable Knowledge Implementation (In Plan)
- Volume 6: EKA Part 2a — Advanced Reasoning & Interdisciplinary Depth (In Plan)
- Volume 7: EKA Part 2b — Synthesis & Methodology (In Plan)

Appendix

Last Updated at 2026-09-11

Executable Knowledge Architecture (EKA)

Mastering Ontology Engineering with Protégé and Pizza.owl

Table of Content

From Ontology Learning to Executable Knowledge Architecture

The Semantic Web has existed for more than two decades, yet for many architects, engineers, analysts, and AI practitioners, ontology engineering still feels abstract, academic, or difficult to approach systematically.

One of the reasons is simple:

> Most ontology learning materials focus either too heavily on theory or too narrowly on tool operations.

As a result, many learners can:

- remember OWL terminology,
- click through Protégé demonstrations,
- or understand isolated semantic concepts,

but still struggle to answer the most important question:

> Why does ontology engineering actually matter in modern enterprise and AI systems?

This book was created to answer that question.

Why the Pizza.owl Tutorial Matters

Among all ontology learning materials ever created, Michael DeBellis' Pizza OWL tutorial remains one of the most influential and practical introductions to OWL ontology engineering.

Using a familiar pizza domain, the tutorial teaches learners how to:

- think semantically,
- construct ontology hierarchies,
- model relationships,
- apply OWL logic (rules), and
- and understand reasoning behavior inside Protégé

What makes the Pizza tutorial special is that it gradually introduces semantic complexity in a highly structured manner.

Rather than overwhelming learners immediately with advanced ontology theory, it guides readers step by step through the evolution of a real semantic model.

That educational philosophy strongly influenced the structure of this book.

This eBook serves as the comprehensive companion guide to the Protégé OWL Pizza Tutorial Hands-on Series

Why This Book?

Most ontology materials focus either on abstract theory or narrow tool operations. This book bridges that gap by connecting semantic engineering to the **Executable Knowledge Architecture (EKA)** framework.

It is designed to help you engineer machine-understandable meaning for AI systems, enterprise knowledge platforms, and digital twins.

About This Book

This eBook was written as a companion knowledge guide to the YouTube playlist:

```
**Protégé OWL Pizza Tutorial Hands-on Series**
(https://www.youtube.com/playlist?list=PL6DEHvciXKeUx4P32B3hKMK1t6mC8RhsW)
```

Since publishing the Protégé OWL Pizza Tutorial series in 2023, I've realized how valuable practical ontology learning remains for architects and AI practitioners. What started as a hands-on walkthrough of Michael DeBellis' tutorial gradually became an important foundation for my EKA (Executable Knowledge Architecture) thinking — connecting ontology, knowledge graphs, and executable intelligence. The playlist may use pizza as the example domain, but its real purpose is helping learners develop semantic thinking and understand how machine-readable knowledge is engineered.

However, the goal is not merely to transcribe the videos.

Instead, this book expands the tutorial into a much deeper professional learning experience by combining:

- ontology engineering theory,
- hands-on Protégé practice,
- semantic modeling methodology,
- enterprise architectural thinking,
- and modern knowledge graph perspectives.

Every chapter corresponds closely to the learning progression of the original video series so readers can learn both visually and conceptually in parallel.

This synchronization is intentional.

Ontology engineering is best learned progressively.

Why Ontology Matters Today

We are entering a new era of intelligent systems.

Modern enterprises increasingly rely on:

- AI systems,
- semantic search,
- enterprise knowledge graphs,
- digital twins,
- intelligent automation,
- contextual reasoning,
- and machine-readable knowledge.

Traditional architectures are built purely on:

- servers and databases,
- documents,
- diagrams
- and APIs

are no longer sufficient for advanced semantic intelligence.

The missing layer here is:

```
Meaning
```

Ontology engineering provides that layer.

OWL ontologies allow organizations to formally define:

- concepts,
- relationships,
- constraints,
- semantics,
- and inference logic

in a machine-readable and machine-processable form.

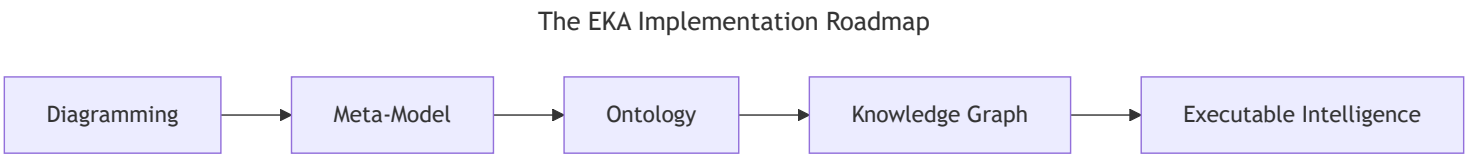
This transforms static information into executable knowledge.

Ontology Within the EKA Framework

This book also introduces ontology engineering through the lens of the EKA (Executable Knowledge Architecture) framework.

EKA represents an architectural approach for transforming enterprise knowledge into executable intelligence systems.

Within EKA, the implementation roadmap is:



Ontology occupies the critical semantic transformation layer.

This is where:

- visual architecture,
- conceptual models,
- and enterprise abstractions

become:

- machine-readable semantics,
- reasoning structures,
- and graph-ready knowledge.

Protégé therefore becomes more than just an ontology editor.

It becomes an engineering environment for executable semantics.

This perspective fundamentally changes how ontology engineering should be understood.

This Book Is NOT Only About Pizza

Although the tutorial domain uses pizza examples, the real purpose of the tutorial is much larger.

The Pizza ontology teaches foundational semantic engineering patterns that scale into real-world domains such as:

- enterprise architecture,
- healthcare,
- finance,
- manufacturing,
- government,
- cybersecurity,

- digital twins,
- and AI knowledge systems.

The same semantics principles used to model:

```
(Pizza)-[:hasTopping]->(CheeseTopping)
```

can later scale into:

```
(Application)-[:supportsCapability]->(businessFunction)
```

or:

```
(Device)-[:connectedTo]->(Sensor)
```

Ontology engineering is domain-independent!

The Pizza example simply provides an approachable learning environment.

Who This Book Is For

This book is intended for:

- Enterprise Architects
- Solution Architects
- Data Architects
- Knowledge Engineers
- AI Engineers
- Semantic Web Learners
- Knowledge Graph Practitioners
- Protégé Beginners
- Meta-Modeling Practitioners
- EKA Learners
- Digital Transformation Professionals

No prior ontology experience is required.

However, reader with backgrounds in:

- UML,
- ER Modeling,
- Enterprise Architecture,
- Databases,
- Graph Technologies,
- or Software Engineering

will often recognize important conceptual parallels throughout the book.

What Makes This Book Different

Most ontology books fall into one of two extremes:

Style	Problem
Highly academic	Difficult for practitioners
Pure tool tutorials	Lack conceptual depth

This book intentionally bridges both worlds.

The objective is to create `A Practical Ontology Engineering Handbook` that remains:

- technically rigorous,
- professionally structured,
- architecturally meaningful,
- and implementation-oriented.

The book therefore combines:

- OWL theory,
- Protégé operations,
- semantic engineering methodology,
- EKA architecture thinking,
- and knowledge graph perspectives

into a single progressive learning journey.

How to Use This Book - The Learning Journey

The recommended learning approach is:

1. **Watch**: The corresponding YouTube video chapter.
2. **Read**: The matching eBook chapter for deeper conceptual context.
3. **Practice**: Manual Protégé exercises and hands-on semantic modeling.
4. **Reflect**: Map these semantic principles to your own enterprise architecture.

Ontology engineering is not mastered through passive reading alone.

It requires semantic thinking practice.

The goal is not merely learning Protégé.

The goal is learning how to engineer machine-understandable meaning.

Structural Integrity

- **Foreword**: By Timothy W. Cook (Founder of SDC).
- **Core Curriculum**: Covers everything from basic class hierarchies and object properties to advanced SHACL, SPARQL, and reasoning behavior.
- **Error Tracker**: An included log to help you troubleshoot common modeling pitfalls.

Acknowledgements

This book has benefited greatly from the generous feedback, technical reviews, and professional discussions contributed by members of the ontology and semantic technologies community.

The full acknowledgements are maintained here: ACKNOWLEDGEMENTS.md

Our Intellectual Heritage & Licensing

This work is a direct descendant of the **Protégé 4 Tutorial (version 1.3) by Matthew Horridge**. We are deeply indebted to the original visionaries who developed the preceding versions and established the core teaching material: **Holger Knublauch, Alan Rector, Robert Stevens, Chris Wroe, Simon Jupp, Georgina Moulton, Nick Drummond, and Sebastian Brandt.**

In this modern adaptation, we have incorporated **Michael DeBellis's** revisions, which successfully bridged the transition to Protégé 5.5 and integrated advanced practices such as SWRL, SPARQL, and SHACL. Furthermore, we acknowledge the contributions of **Lorenz Buehmann, André Wolski, Dick Ooms, Colin Pilkington, Livia Pinera, Jans Aasman, Yan Xu, and the team at Franz Inc.**, whose work in applying these ontologies to real-world triple stores like AllegroGraph and Gruff has significantly shaped our current perspective on **Executable Knowledge Architecture (EKA)**.

Special Thanks

We extend our sincere gratitude to **Michael DeBellis** for creating one of the most influential practical OWL learning resources available to the ontology community. (Visit Michael's homepage here - https://www.michaeldebellis.com/). We also thank him for his meticulous technical audit for this series; his expert insights -- particularly regarding the crucial engineering distinction between modeling tool and persistent graph databases -- have been instrumental in ensuring this content meets the standards of modern production-grade environments.

We are also honored to feature a foreword and ongoing review by **Timothy Cook (Founder of SDC)**. His guidance and endorsement have been vital in validating our approach, helping to bridge the gap between traditional semantic modeling and the sophisticated demands of modern enterprise architecture / knowledge architecture.

Finally, we thank **Stanford University and the Protégé community** for their enduring commitment to advancing open ontology engineering tools and Semantic Web technologies.

By preserving this attribution chain, we honor the intellectual legacy of the individuals and organizations that have sustained this field. We stand on the shoulders of these giants, aiming to provide a bridge from traditional semantic modeling to the future of `Executable Knowledge Architecture`.

Related Resources

Official Resources:

- Protégé Official Website: https://protege.stanford.edu/
- Protégé Pizza Repository: https://github.com/yasenstar/protege_pizza/tree/main
- EKA Official Website: https://xiaoqi.com/

Learning Resources:

- YouTube Channel - Xiaoqi(Yasen) Zhao: http://www.youtube.com/@yasenzhao
- Udemy Course - Xiaoqi Zhao: https://www.udemy.com/user/xiaoqi-zhao

Final Thoughts

Ontology engineering is NO LONGER a niche academic discipline.

It is rapidly becoming part of the foundational infrastructure for:

- AI systems,
- enterprise knowledge platforms,
- semantic interoperability,
- and executable intelligence architectures.

Understanding ontology today is increasingly similar to understand databases or APIs twenty years ago.

It is becoming a core architectural capability.

The journey begins with a simple Pizza ontology.

But the destination is much larger:

```
Executable knowledge
```

Welcome to the World of Ontology Engineering!

> "You" are the learner of this book, so while you're reading, I'll say to "you" directly instead of "learner" / "reader" from now on.

Last updated at: 2026-09-11

Acknowledgements

This project would not have reached its current level of maturity without the generosity of numerous members of the ontology, semantic web, enterprise architecture, and knowledge graph communities.

Many people contributed through technical discussions, detailed chapter review, engineering suggestions, issue reports, and encouragement throughout the writing process.

The following individuals deserve special recognition.

Michael DeBellis

Author of the original *Protégé 5 New OWL Pizza Tutorial*.

This book is built upon Michael's excellent tutorial, which has served for many years as one of the most practical introductions to OWL ontology engineering.

Beyond creating the original tutorial, Michael has kindly provided valuable feedback on the direction of this book and has been supportive of extending the tutorial into a broader engineering perspective through the Semantic Knowledge Development Lifecycle (SKDL) and Executable Knowledge Architecture (EKA).

Without his original educational work, this project would not exist.

Timothy W. Cook

Founder & CEO, Axius SDC, Inc.

Tim has generously reviewed multiple chapters of this book during its development, particularly those concerning semantic governance, ontology engineering practice, and enterprise-scale knowledge systems.

His feedback has helped strengthen several important engineering themes throughout the SKDL framework, including:

- treating ontology as a meaning-driven system rather than merely a validation system;
- distinguishing semantic inference from policy enforcement;
- clarifying the role of semantic governance within long-lived knowledge systems;
- improving the engineering narrative surrounding reusable semantic knowledge.

His perspective comes from extensive real-world work in semantic interoperability, healthcare data standards, and enterprise semantic governance.

I am deeply grateful for his thoughtful reviews, constructive criticism, and continued encouragement throughout the writing of this book.

The Ontology Community

Many members of the broader ontology and semantic web community have contributed indirectly through discussions, questions, publications, conference presentations, open-source software, and years of accumulated research.

Protégé, OWL, RDF, Description Logic, and the Semantic Web ecosystem are the result of decades of collaborative work by thousands of researches and practitioners.

This book stands on the collective foundation.

Readers - YOU

Finally, sincere thanks to every reader who as reported errors, suggested improvements, opened GitHub issues, starred the repository, shared the project with others, or simply spent time learning ontology engineering through this book.

Every question, correction, and discussion helps improve future editions.

Knowledge grows through reuse, collaboration, and continuous refinement -- the very principles promoted throughout this book.

Last updated at 2026-09-11

Legal & Licensing

This eBook is a derivative work of the " `Pizza.owl` Tutorial", a foundational resource for the ontology engineering community.

We are committed to transparency and the preservation of the intellectual legacy established by the original authors.

Licensing Terms

This work is governed by a dual-licencing strategy to balance educational openness with software best practices:

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Attribution & Intellectual Heritage

This work stands on the shoulders of the community that created the original Protégé 4 Pizza Tutorial. We acknowledge the visionary contributions of **Matthew Horridge, Holger Knublauch, Alan Rector, Robert Stevens, Chris Wroe, Simon Jupp, Georgina Moulton, Nick Drummond, and Sebastian Brandt.**

We also extend our gratitude to **Michael DeBellis** for his invaluable modern revisions, technical audit, and his ongoing dedication to providing practical OWL learning resources for the ontology community.

For further details regarding the full attribution chain, the history of this project, or the specific licensing of individual files, please refer to the README.md and LICENSE.md files in the official GitHub repository (https://github.com/yasenstar/protege_pizza).

Last updated at 2026-09-11

Foreword from Timothy W. Cook

For more than twenty years, ontology engineering has carried a quiet reputation: powerful in theory, hard to reach in practice. Michael DeBellis' Pizza tutorial did more than almost any other resource to change that, by letting people learn semantics with their hands instead of only their heads. What Xiaoqi Zhao has done here is take that well-loved starting point and carry it all the way to where the work actually has to live, which is inside real systems, under real governance.

This book earns its subtitle. In a single progression it moves from the first class you build in Protégé to the question every enterprise eventually faces: how machine-readable meaning becomes something a system can act on safely. Along the way it never loses the teacher's discipline of one idea at a time.

I want to point readers in particular at Chapters 12 and 13, because their titles tell you something. They do not *only* say "object property characteristics" and "domain and range." They say *governing* semantic relationship behavior and *governing* semantic boundaries. That word is the whole game. A functional or transitive property is not a checkbox in a tool. It is a rule about how a relationship is allowed to behave. A domain and a range are not decoration. They are a statement about what may legitimately connect to what. Xiaoqi teaches these as governance, and that is exactly the right approach.

Chapter 13 also does something I rarely see in an introductory text: it tells the truth about a sharp edge. OWL domain and range do not reject a bad assertion the way a database constraint would. The reasoner infers, under an open-world assumption, rather than refuses, and as the chapter notes, this surprises newcomers who expected enforcement. That surprise is where academic ontology meets the enterprise information system. The moment data leaves the editor and travels between systems, a boundary that only infers is not yet a boundary you can trust. It has to be able to refuse what does not belong, deterministically, with its rules bound into the data rather than living in a separate document or a separate dashboard.

That is the work my colleagues and I have spent years on in the Semantic Data Charter. SDC treats meaning and the constraint definitions as shareable, bound assets. They are one inseparable data payload, so admissibility can be checked at the point of use, in the open, against published specifications. Readers will recognize this as related to the layer Xiaoqi names Γ in his EKA tuple: the governance layer that turns a clever model into a trustworthy one. EKA is the architecture that orchestrates the pieces; a substrate like SDC is one open, standards-based way to make that governance layer real in production. The two meet exactly where this book spends its most careful chapters.

There is a larger reason to welcome a book like this now. Ontology engineering is becoming a core architectural capability, the way databases and APIs did before it, and the field is poorly served when it stays split between texts that are rigorous but unreachable and tutorials that are reachable but shallow. Xiaoqi has taken the harder middle path: technically honest, professionally structured, and grounded in the standards that real organizations have to answer to. That is the bridge between theory and practice, and we need more people building it.

Read this book with Protégé open. Do the exercises. And when you reach those governing chapters, remember that the discipline you are learning on a pizza is the same discipline that will one day decide whether a clinical, financial, or regulatory system is allowed to act. As Xiaoqi writes, meaningful intelligence depends on meaningful boundaries. This book teaches you how to build them.

Timothy W. Cook
Founder, Axius SDC, Inc.
Creator, the Semantic Data Charter (SDC)
June 2026

Last Updated at 2026-09-11

Preface from Michael DeBellis

I sometimes find it easy to be pessimistic about the human race. The world gives us plenty of reasons for discouragement. But one thing that illustrates some of the best in people is the open source movement.

It is remarkable how much of the software infrastructure that powers the modern internet is open source. The Linux operating system, Apache web servers, Kubernetes, Kafka, Docker containers, and countless programming languages, libraries, databases, and development tools were created as open-source projects or converted from internal projects to software freely available to all developers. Organizations and individuals understand that software becomes more valuable when it is shared openly so that the developer community can study it, improve it, and build on it. It's a very positive thing that corporations such as Google and LinkedIn are motivated to contribute software in which they've invested heavily into the open source infrastructure. Even more so that so many individuals will contribute their time to maintain and expand open source tools for the simple joy of seeing things work and providing tools that colleagues will find useful.

That generosity is not limited to code. Documentation, examples, tutorials, walkthroughs, bug reports, and teaching materials are also essential parts of the open-source ecosystem. They are just as important, because good software is wasted if no one can understand or learn to use it.

Tutorials are often underappreciated. As I said when I first created my revised version of the Protégé Pizza tutorial, no one gets a PhD for writing a tutorial. Tutorials do not usually receive the same professional recognition as research papers, software frameworks, or commercial products. Yet, for many learners, a good tutorial is the difference between a technology remaining abstract vs. becoming usable.

That is why I am pleased to see Xiaoqi Zhao's *Mastering Ontology Engineering with Protégé and Pizza.owl: From Semantic Foundations to Executable Knowledge Architecture (EKA)*. It continues a tradition that has always been central to the Semantic Web community: building on prior work, acknowledging intellectual lineage, and helping new learners move from abstract ideas to practical results.

The Pizza tutorial itself has a long history. The original tutorial was developed by members of the Protégé and ontology engineering community. It became one of the best-known introductions to OWL ontology development. Its genius was the choice of a domain that was familiar enough to get out of the learner's way. Almost everyone understands pizzas, toppings, bases, vegetarian choices, and spicy

ingredients. Because the domain is simple, learners can focus on the real subject: classes, properties, axioms, inference, queries, and rules.

My own revision of the Pizza tutorial was motivated by a practical teaching need. I was preparing a hands-on workshop and realized that the older tutorial no longer matched the current Protégé user interface closely enough for beginners. Especially since the majority of the students in that workshop were Library Science experts, not developers. Experienced OOP developers can translate instruction such as “add superclass” into revised user interfaces such as “add subclassOf”. New learners, especially those coming from fields such as library science and information architecture, should not have to do that translation while also trying to understand the concepts of OWL and other Semantic Web languages. The tutorial needed to match the actual tool.

What began as a small update became a much larger project. I updated the examples for Protégé 5, revised screenshots and instructions, and added material on topics such as SPARQL, SHACL, SWRL, IRIs, namespaces, and WebProtégé. My goal was not just to teach people where to click, but to help them understand why the clicks mattered.

Xiaoqi’s eBook takes that work in a new direction. It uses the Pizza tutorial as a foundation, but it is not simply a transcription or repetition of the older material. It connects OWL ontology engineering to a broader architectural vision that he calls Executable Knowledge Architecture (EKA).The eBook encourages learners to see ontology development not as an isolated academic exercise, but as part of a larger architecture for formal knowledge, knowledge graphs, reasoning, validation, and intelligent systems.

I’ve spent half of my career as a consultant and the other half doing research. One of the most important lessons I learned as a consultant is that the hard part isn’t writing software, the hard part is integration. That’s true whether we’re integrating rules with COBOL programs, LLMs with enterprise data, or ontologies and knowledge graphs into distributed computing environments.

That broader perspective is timely.

For a while, it seemed that the Semantic Web might remain a powerful idea that never achieved its hoped-for impact. The original vision was ambitious: not just a web of documents, but a web of data and meaning, where machines could understand and integrate information across sources. The challenge was that adding semantics requires work. It requires modeling. It requires shared identifiers. It requires explicit commitments about meaning. It requires different groups in large organizations to talk to each other and define shared concepts and domain boundaries.

Many organizations decided that this was too much effort. It was easier to publish documents, tables, and data dumps and let humans, search engines, and especially machine learning systems make sense

of them.

Ironically, the rise of Large Language Models (LLMs) has created renewed interest in the very technologies that some people thought machine learning would make unnecessary. LLMs are extraordinary tools. Used well, they can provide enormous productivity benefits. But they also have well-known limitations. Their reasoning is opaque. They can generate fluent but incorrect answers. They may struggle with provenance, controlled terminology, and domain-specific meaning.

These problems are not merely accidental bugs. They follow naturally from the fact that LLMs are statistical models rather than explicit knowledge models. Improvements in scale, training, and architecture will continue to make them more capable, but many practical applications still need complementary structures: curated data, explicit semantics, provenance, and constraints.

That is why knowledge graphs and ontologies are becoming more important. Retrieval Augmented Generation (RAG) was an early response to the problem of grounding LLMs in curated information. But as applications become more demanding, simple document retrieval is often not enough. Many domains require richly interconnected knowledge and automated reasoning. RDF, OWL, SPARQL, SHACL, and reusable ontologies such as Dublin Core, PROV-O, and SKOS provide mature standards for representing and working with exactly that kind of explicit knowledge representation. Explicit knowledge representation based on logic and set theory is the perfect complement to the powerful inductive statistical models that power LLMs.

This renewed interest also means that the word “ontology” is now used in many different ways. Some commercial systems use the term for models that are closer to enterprise object models than to ontologies in the Semantic Web sense. I understand why some researchers and ontology engineers find that frustrating. I sometimes do too. But I also see it as a sign of progress. I have seen this pattern since the early days of expert system adoption and later with the adoption of Object-Oriented Programming and Agile methods: when ideas move from research to industry, terms such as rule-based, knowledge base, *Rapid Application Development (RAD)*, object model, and ontology are stretched, simplified, and repurposed. That can create confusion, but it also shows that industry has recognized that the underlying ideas matter. Eventually as practitioners understand the core research ideas, they make educated decisions and can separate real technology from marketing hype. Tutorials such as this eBook are an essential part of that process. Learners need to understand not only that ontologies are useful, but what ontologies and other semantic technology provide that is new and why these new capabilities matter.

One of the things I appreciate about this eBook is that it tries to connect beginner-level Protégé modeling with larger architectural questions. A learner begins with classes such as *Pizza*, *PizzaTopping*, *CheeseTopping*, and *VegetableTopping*. But those beginner examples lead naturally to deeper questions. What is the difference between a class and an individual? When should a class be primitive,

and when should it be defined? What does a reasoner actually infer? Why does OWL's open-world assumption behave differently from a database constraint? When should we use OWL reasoning, and when should we use SHACL validation? How do ontologies relate to RDF graphs, triple stores, SPARQL queries, and larger distributed system architectures?

Those are not minor details. They are the conceptual foundation of ontology engineering.

My strongest advice to readers is to keep Protégé open while reading. Do the exercises. Run the reasoner frequently. That last part is really critical. For training ontologies, the reasoner executes immediately. Running it after every change means you'll see an error as soon as it is created. When that happens, the reasoner can almost always focus you on the specific problem. If you wait until there are several errors the reasoner feedback is usually much less helpful and finding errors can take hours instead of minutes.

When something surprising happens, slow down and understand why. In OWL, surprises are often the most valuable teaching moments. A reasoner that classifies something unexpectedly, fails to classify something you expected, or reports an inconsistency is showing you the difference between what you intended to model and what you actually modeled.

That distinction is at the heart of ontology engineering.

A good ontology is not just a diagram. It is not just a taxonomy. It is not just a data model. It is an explicit, formal model of meaning in a domain. Building one requires careful thought, iteration, testing, and patience. The Pizza tutorial remains useful because it gives learners a simple domain in which to practice those habits before applying them to medicine, finance, manufacturing, enterprise architecture, LLM integration and other business, engineering, and scientific domains.

I am glad to see this new contribution to the Pizza tutorial lineage. It reflects the best spirit of open educational work.

The journey begins with pizza. But the real subject is meaning.

Michael DeBellis

2 July 2026

San Francisco, CA

https://www.michaeldebellis.com/blog

Last updated at 2026-09-11

Preface from Author

From Static Diagrams to Executable Knowledge: A Personal Journey

For nearly thirty years, I have lived and breathed enterprise information technology. My career began long before “knowledge graph” was a common term, working across telecom, automotive, finance, and IT outsourcing consulting. For most of those years, my primary hands-on tools for making sense of complexity was static diagramming, mostly with Visio, draw.io, or even office suite like Excel and Powerpoint. It was powerful, but it was also… silent. The diagrams described architecture, but they could not execute it. They could not reason, validate, or adapt.

That was the before.

The shift (2021 – 2022) began when I moved from old toolboxes to Archi, the open-source ArchiMate modeling tool. Suddenly, I was not just drawing; I was building a **repository**. I discovered the power of a shared, collaborative model – a single source of truth about an enterprise. But even Archi, as good as it is, had limitations. The model was still primarily for human consumption. It was a rich, structured "graph" database, but it was not a reasoning engine.

It was around this same time that I discovered the **Essential EAS Open Source (Ontology)** – a glimpse of what happens when an EA repository is built on formal semantics. That was the first crack in the old paradigm, learnt and practiced with its modeling tool, Protégé - the first time I met this word.

The catalyst (2023) was the `Pizza.owl` tutorial. Through creating my first video series on the topic, I got in touch with Michael DeBellis. More importantly, I got my hands on Protégé. For the first time, I was not just modeling structure; I was formalizing **meaning**. I was defining classes, properties, and rules that a machine could use to infer new knowledge, check for contradictions, and navigate relationships intelligently.

The pizza example was simple, but the potential was explosive. I realized that the future of enterprise architecture was not in prettier diagrams, but in **executable knowledge**.

With that learning, back to the Essential EAS tool, I realized how powerful of treating enterprise architecture from semantic perspective, and how flexible the tool provided, e.g. if you need one meaning (triple) that EAS built-in meta-model doesn't exist, just create one new from it's slots and that's now fitting the knowledge of your enterprise reality, more and more.

The synthesis & the birth of EKA & SKDL (2023 – 2026) became a period of intense exploration. My YouTube channel (and other video sharing platform, like BiliBili, DouYin, etc..) and online courses grew

to host not just the `Pizza.owl` series, but courses on ArchiMate, meta-modeling, diagramming as code (PlantUML, D2, OpenSCAD), Neo4j graph databases, and ontology engineering. Each course was a piece of a larger puzzle. programming taught me logic, PlantUML taught me that diagrams could be generated from text (code), Neo4j showed me how to traverse relationships at scale, and OpenSCAD reinforced the power of parametric, declarative design.

Through years of distillation, everything crystallized into two simple, powerful core concepts for readers to take away:

1. **EKA (Executable Knowledge Architecture)**: The overarching paradigm that treats enterprise architecture and knowledge graphs not as static documents, but as executable, machine-readable specifications.
2. **SKDL (Semantic Knowledge Development Lifecycle)**: The structured 7-stage approach I formulated to guide practitioners seamlessly from raw domain concepts all the way to advanced semantic reasoning and governance.

Why this book? This book is not a transcript of my videos. It is the theory behind the practice. It is the result of years of hands-on work, structured precisely around the SKDL stages. I wrote it because I saw a gap:

> *countless tutorials show you how to click buttons in Protégé, or write queries in Neo4j, but almost none provide a complete lifecycle methodology for architectural thinking.*

This book aims to bridge that gap, using the familiar `Pizza.owl` as a safe and repeatable domain to walk you step-by-step through the SKDL framework.

You will start with pizza. But if you follow the journey across the volumes, you will be able to model anything: an enterprise architecture, a digital twin, a regulatory framework, or an AI knowledge system.

The goal is not just to teach you a tool. The goal is to give you a new way to think about knowledge through **EKA** and **SKDL**: as something that can be **formalized, reasoned over, governed, and executed.**

Welcome to the journey.

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September, 2026 (Updated)

Last updated at 2026-09-11

Chapter 14 -- Semantic Requirements: Existential Restriction and the Logic of "Some"

> **Editor's note:** This chapter is Part 1 of a three-chapter unit on OWL property restrictions (originally drafted as a single Chapter 14). Chapter 14 covers *why* restrictions matter and the existential restriction (`some`). Chapter 15 covers the universal restriction (`only`), their combination, and a Neo4j implementation walkthrough. Chapter 16 synthesizes both into the EKA framework, reviews common mistakes, and closes the unit.

Table of Contents:

14.1 Chapter Introduction -- From Semantic Boundaries to Semantic Requirements

By the end of Chapter (13), you had already begun understanding an important truth about ontology engineering:

> semantic relationships alone are not sufficient to fully describe meaning!

Through **Property Domain and Range**, ontology engineers learned how to establish:

> semantic boundaries.

For example, `hasTopping` may define:

Domain	Range
`Pizza`	`PizzaTopping`

This tells ontology something important:

> pizzas may have toppings.

Likewise:

> topping belong to pizzas.

However, a subtle but important limitation still remains.

Consider the following question:

> Does every pizza necessarily have toppings?

From the perspective of:

> Domain and Range alone,

the answer is:

> **not necessarily.**

Why?

Because Domain and Range describe:

> where relationships are logically valid,

but they do not express:

> what relationships are semantically required.

This distinction represents one of the most important maturity transitions in ontology engineering.

In earlier chapters, ontology primarily focused on:

> semantic structure.

You created:

- classes
- subclass hierarchies
- object properties
- inverse properties
- property characteristics
- semantic boundaries

Ontology therefore answered questions such as:

- What things exist?
- How are concepts related?
- What semantic relationships are valid?

Chapter 14 now introduces a deeper semantic question:

> **What relationships must exist?**

This shift is profound.

Because ontology is no longer merely describing:

> allowable semantic connections.

Ontology now begins defining:

> **semantic obligations.**

For example:

Saying:

> Pizza `hasBase` PizzaBase

describes:

> a valid relationship.

But saying:

> every Pizza must have at least one PizzaBase

introduces something fundamentally different:

> a semantic requirement.

Ontology therefore begins evolving from:

> connected semantics

toward:

> **logical semantics.**

This chapter begins with one of OWL's most powerful modeling constructs:

> **Existential Restriction**

often expressed through:

> `someValuesFrom`

At first glance, existential restrictions may appear technically simple.

Yet conceptually, they represent a major milestone!

Because this is the point where ontology starts moving toward:

> **machine-understandable logic.**

Rather than merely storing knowledge, ontology begins expressing:

> how knowledge should behave.

Within the broader direction of:

> **Executable Knowledge Architecture (EKA)**

this chapter represents another maturity leap:

from:

> governed semantic relationships

toward:

> **governed semantic requirements.**

> **Within this chapter, using `someValuesFrom` is interchangeable with `some` ; `allValuesFrom` is interchangeable with `only` .**

14.2 Why RDF and RDFS Are Not Enough

To fully appreciate why `existential restrictions` matter, it is helpful to revisit a theme introduced earlier in:

> **Chapter 08 -- RDF as a Language**

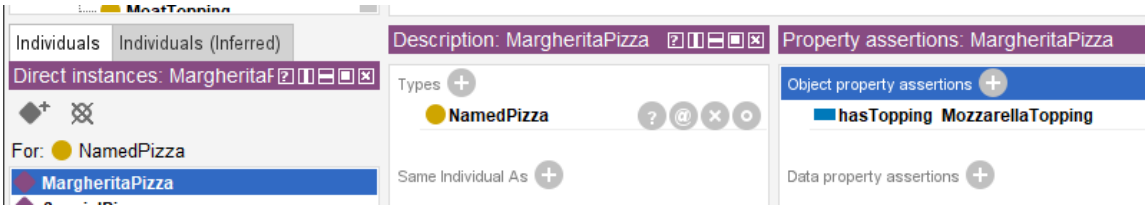
As discussed previously, RDF provides a remarkably elegant mechanism for representing knowledge.

Through simple triple "subject → predicate → object", RDF allows semantic relationships to be represented consistently.

For example:

> MargheritaPizza → `hasTopping` → MozzarellaTopping

In Protégé, we model `Pizza.owl` as below screen:



This structure gives ontology an important foundation:

> connected meaning.

Later, RDFS (RDF Schema) extends this capability by introducing:

- class hierarchies
- subclass relationships
- Domain definitions
- Range definitions

These additions improve semantic understanding considerably.

For example:

RDFS can express:

> Pizza is a subclass of Food

or:

> `hasTopping` connects "Pizza → PizzaTopping".

At this stage, ontology becomes:

> semantically structured.

However, an important limitation still exists!

RDFS can describe:

> semantic possibility.

But it cannot adequately express:

> semantic **necessity**!

This distinction becomes critically important.

Consider the following statement:

> Pizza `hasBase` PizzaBase

What does this actually mean?

It tells ontology:

> pizza **may** logically has pizza bases.

But does ontology know that:

> every pizza **must** have a base?

NO!

RDFS alone cannot fully express this requirement.

Likewise:

> MargheritaPizza `hasTopping` MozzarellaTopping

does not automatically mean:

> every MargheritaPizza requires mozzarella.

Ontology still lacks a way to formalize:

> semantic **obligation**.

This limitation becomes increasingly problematic in real enterprise modeling.

Imagine an enterprise architecture ontology.

Suppose an organization defines:

> Application `hostedOn` Infrastructure

This relationship may be semantically valid.

However, enterprise architects often need stronger semantic meaning, like:

> every production application must be `hosted on` infrastructure.

Notice the difference.

The first statement describes: **possibility**;

The second introduces: **requirement**.

This is precisely where:

> **OWL (Web Ontology Language)**

extends semantic capability beyond RDF and RDFS.

OWL introduces:

> logical expressiveness.

Ontology now becomes capable of describing:

- Rules,
- Requirements,
- Obligations, and
- **machine-interpretable semantic logic**.

This transition is important enough to view as a fundamental evolution:

- **RDF** - connected data - *"Something is connected to something."*
- **RDFS** - structured semantics - *"These kinds of things may be connected in these ways."*
- **OWL** - logical semantics - *"These kinds of things **must** be connected in these ways."*

14.3 Property Restrictions -- Introducing Semantic Logic in OWL

By the end of Chapter (13), ontology had already become significantly more expressive.

You could now model:

- classes and subclass hierarchies
- object properties and inverse properties
- property characteristics
- semantic boundaries through Domain and Range

However, ontology skill lacked an important capability:

> **the ability to formally describe semantic conditions.**

Consider the following statement:

```
Pizza  hasTopping  PizzaTopping
```

This tells ontology something useful already:

> pizza may have toppings.

Yet, ontology still cannot answer a deeper semantic question:

> what makes a pizza become a specific kind of pizza?

For example:

What semantically distinguishes `MargheritaPizza` from `AmericanPizza` or `VegetarianPizza` ?

Either pizza above may have some topping, but merely defining classes and relationships like this way is insufficient.

Ontology now requires a mechanism capable of expression:

> **semantic constraints**

and:

> **logical requirements.**

This need introduces one of the most important constructs within:

> **OWL (Web Ontology Language)**

knows as:

> **Property Restrictions.**

Property restrictions allow ontology engineers to formally describe:

- how properties should behave?
- what relationships are expected? and
- what semantic conditions must hold?

In other words, property restrictions move ontology from:

> semantic structure

toward:

> **semantic logic.**

This distinction is fundamental.

Because until now, ontology primarily described:

> what thinks exist

and

> how things connect.

With property restrictions, ontology begins expressing:

> **what things mean.**

This is one of the first moments where ontology becomes **logically interpretable**.

And consequently **reasoner-friendly.**

Within OWL, property restrictions are represented as:

> logical class descriptions.

Meaning: A class can now be defined not merely through **a name**, but through **semantic conditions.**

For example:

Instead of manually declaring:

> `CheesyPizza`

ontology can describe it logically as:

> a `Pizza` that has **at least one** `CheeseTopping` .

This subtle shift changes ontology dramatically.

Meaning is no longer **manually assigned.**

Meaning becomes **logically inferable.**

To better understand property restrictions, it is useful to view them as a family of semantic mechanisms.

```
/
├─ Quantifier Restriction
│  ├─ Existential Restrictions
│  └─ Universal Restrictions
├─ Cardinality Restrictions
└─ hasValue Restrictions
```

Broadly speaking as above tree-view, OWL property restrictions can be grouped into three major categories:

1. Quantifier Restrictions

Quantifier restrictions describes:

> **whether certain relationships exist**

or:

> whether relationships are restricted to certain types.

These restrictions focus on semantic existence and semantic scope.

Quantifier restrictions include:

Existential Restriction

(`someValuesFrom`)

Meaning: there exists at least one relationship satisfying a condition.

Example:

> `Pizza` `hasTopping` some `CheeseTopping`

Meaning:

> every pizza must have at least one cheese topping.

Universal Restriction

(`allValuesFrom`)

Meaning: all (every) successor individual must belongs to a specified class.

Example:

> `Pizza` `hasTopping` only `PizzaTopping`

Meaning:

> every topping of a pizza must be a `PizzaTopping` .

Notice the important distinction.

- Existential restriction expresses: **minimum semantic existence.**
- Universal restriction expresses: **semantic limitation.**

These two concepts are frequently confused by beginners.

Yet they represent fundamentally different forms of semantic reasoning.

Chapter (14) begins with:

> **Existential Restriction**

because ontology must first understand:

> required existence

before learning:

> restricted universality.

Do you see any similarity for this ontology (machine) learning approach with how human being learn? True, they are in the common approach.

2. Cardinality Restrictions

While quantifier restrictions focus on existence, cardinality restrictions focus on **quantity.**

Ontology may sometimes need to define:

> how many relationships are permitted or required.

OWL therefore supports several forms of:

> cardinality constraints.

Including:

(1) Minimum Cardinality

(`minCardinality`)

Meaning: at least N relationships must exist.

Example:

> `Pizza` `hasTopping` minimum 2

Meaning:

> a pizza must have at least two toppings

(2) Maximum Cardinality

(`maxCardinality`)

Meaning: no more than N relationships may exist.

Example:

> `Person` `hasBiologicalMother` maximum 1

Meaning:

> a person cannot have more than one biological mother.

(3) Exact Cardinality

(`exactCardinality`)

Meaning: exactly N relationships are required.

Example:

> `Standard_Bicycle` `hasWheel` exactly 2

Meaning:

> standard bicycles must have precisely two wheels.

Cardinality restrictions become particularly important in:

- enterprise governance,
- data quality validation, and
- business rule enforcement.

Within enterprise architecture, examples may include:

> an `business_application` must have exactly one `system_owner`

or:

> a `business_process` must involve at least one `responsible_role` .

Ontology therefore becomes increasingly capable of expressing:

> operational logic,

rather than merely:

> descriptive semantics.

3. Value Restrictions

The third major category involves **specific required values** represented through `hasValue` .

Unlike existential restriction, which asks for:

> as least one qualifying relationship,

value restriction specifies:

> a particular exact value.

For example:

Suppose an enterprise policy states:

> all `production_applications` must belong to environment " `Production` ".

Ontology may express:

> `hasEnvironment value Production`

This means:

> the relationship must point to a specific predefined individual.

In `Pizza.owl` terms, imagine requiring:

> a `NamedPizza` must always have a particular base.

Rather than merely saying:

> some pizza base exists,

ontology would specify:

> one exact expected value.

Value restrictions therefore introduce:

> semantic precision

as well as:

> semantic determinism.

Together, these three categories establish the foundation of:

> OWL semantic logic.

They transform ontology from:

> semantic modeling

into:

> formal semantic definition.

Summarized view in below comparison table from also the evolution discussed in 14.2 from RDF/RDFS to OWL:

Layer	Core Capability	Limitation
RDF	Triples (connected meaning)	No structure
RDFS	Class hierarchies, Domain & Range (structured semantics)	Can only describe **possibility**, cannot express **necessity**
OWL	Logical expressiveness (rules, requirements, obligations)	--

Existential restrictions represent one of the earliest moments where you may begin seeing:

> ontology as **logic**.

Rather than merely:

> ontology as structure.

And this transition fundamentally changes how ontology can support:

- Reasoning (R),
- Governance (Γ), and eventually
- executable intelligence.

Among these restriction categories, **quantifier restrictions** are typically introduced first because they provide the conceptual foundation for semantic participation and logical class definition. We therefore begin with existential and universal restrictions.

14.4 Quantifier Restriction -- Understanding Existential and Universal Logic

Among all OWL restrictions, the most foundational are:

> **Quantifier Restrictions.**

These restrictions focus on a deceptively simple question:

> **what kind of relationships should exist?**

Quantifier restrictions allow ontology to reason about:

> semantic participation.

In other words:

ontology begins understanding not merely:

> whether concepts are connected,

but:

> how those connections should behave logically.

OWL provides two primary forms of quantifier restriction:

- **Existential Restriction**: represented by `someValuesFrom`
- **Universal Restriction**: represented by `allValuesFrom`

Although these may appear similar at first glance, their semantic meaning differs significantly.

14.4.1 Existential Restriction -- "At Least One Exist"

Existential restriction expresses:

> **there exists at least one qualifying relationship.**

In **Description Logic (DL)**, this is often represented conceptually as:

> $\exists R.C$

Meaning: there exists at least one (some) relationship R to something belonging to an individual belonging to class C.

=== Interesting Read: More mathematical notation ===

Expand above notation in DL, the concept of an existential restriction is formally defined using the following mathematical notation:

$$\exists R.C = \{x \in \Delta^{\mathcal{I}} \mid \exists y \in \Delta^{\mathcal{I}} : (x, y) \in R^{\mathcal{I}} \land y \in C^{\mathcal{I}}\}$$

Breakdown of this formula --

- $\Delta^{\mathcal{I}}$: The domain of interpretation, representing the set of all individuals in the ontology world.
- x, y: Individual elements within that interpretation domain.
- $R^{\mathcal{I}}$: The interpretation of the role (relationship) R, represented as a binary relation between individuals.
- $C^{\mathcal{I}}$: The interpretation of the concept (class) C, represented as the set of all individuals belonging to class C.
- $(x, y) \in R^{\mathcal{I}}$: Indicates that a relationship R exists between individual x and individual y.
- $y \in C^{\mathcal{I}}$: Indicates that individual y belongs to class C.
- $\land$: Logical conjunction (AND), meaning both conditions must hold simultaneously.

Conceptual visualization --

To understand this mapping, let's imaging a `parentTo` relationship connected to a `Person` class: $\exists parentTo.Person$

Meaning:

> there exists at least one child connected through `parentTo` to a `Person` .

In short, the expression $\exists R.C$ describes the set of all individuals x that have **at least one successor** y through relationship R, such that y belongs to class C.

=== END ===

Within our `Pizza.owl` ontology, consider:

> `hasTopping some MozzarellaTopping`

Semantically, this means:

> there exists at least one topping relationship to mozzarella topping.

This statement introduces something very important:

> semantic necessity.

Ontology now expects:

> at least one qualifying relationship.

This differs significantly from earlier chapters.

- Previously: relationship were optional.
- Now: relationships become logically meaningful (necessity).

A useful way to understand existential restriction is through the phrase:

> **AT LEAST ONE.**

Not:

> exactly one.

Not:

> all toppings.

Simply: there exists at least one!

This distinction matters enormously.

Because ontology reasoning depends heavily on:

> precise semantics.

Let's see an example in `Pizza.owl` :

Suppose a class defines:

> `MargheritaPizza`

as:

> `hasTopping some MozzarellaTopping`

Ontology now understand:

> mozzarella topping is semantically necessary by Margherita pizza.

Without satisfying this requirement:

> the pizza cannot fully conform to the class meaning, semantically.

This introduces another important shift.

Ontology modeling now begins answering:

> **What makes something what it is?**

A `MargheritaPizza` is not merely:

> a pizza.

It becomes:

> a pizza satisfying semantic conditions.

This distinction transform ontology engineering from:

> classification

toward:

> **formal semantic definition**.

Ontology therefore becomes increasingly capable of expressing:

> adaptive domain knowledge.

In enterprise context, this becomes extremely valuable.

See another example:

A `Critical_Application` may require:

> at least one `Disaster_Recovery_Mechanism` .

Or:

`Sensitive_Data_Process` may require:

> at least one `Compliance_Control` .

Ontology can now formally describe:

> mandatory semantic conditions.

Rather than merely:

> possible relationships.

This marks another important maturity leap toward:

> executable knowledge systems.

14.4.2 Universal Restriction -- "Only These Are Allowed"

Universal restriction expresses:

> **all successor individuals must satisfy a condition.**

Conceptually:

> $\forall R.C$

Meaning: every relationship R must point to class C.

=== Interesting Read: More mathematical notation ===

Expand above notation in DL, the concept of a universal restriction is formally defined using the following mathematical notation:

$$\forall R.C = \{x \in \Delta^{\mathcal{I}} \mid \forall y \in \Delta^{\mathcal{I}} : (x, y) \in R^{\mathcal{I}} \rightarrow y \in C^{\mathcal{I}}\}$$

Breakdown of this formula --

- $\Delta^{\mathcal{I}}$: The domain of interpretation, representing the set of all individuals in the ontology world.
- x, y: Individual elements within that interpretation domain.
- $R^{\mathcal{I}}$: The interpretation of the role (relationship) R, represented as a binary relation between individuals.
- $C^{\mathcal{I}}$: The interpretation of the concept (class) C, represented as the set of all individuals belonging to class C.
- $(x, y) \in R^{\mathcal{I}}$: Indicates that a relationship R exists between individual x and individual y.
- $y \in C^{\mathcal{I}}$: Indicates that individual y belongs to class C.
- $\rightarrow$: Logical implication (IF…THEN), meaning if relationship R exists, the target individual must satisfy class condition C.

Conceptual visualization --

To understand this semantic interpretation, same as previous example, imagine a `parentTo` relationship connected to `Person` , now as $\forall parentTo.Person$

Meaning:

> all individuals connected through `parentTo` must belong to the class `Person` .

Importantly, this does **not** imply that a person necessarily satisfies `parentTo` relationship.

Instead, ontology expresses:

> if a `parentTo` relationship exists, it must only point to valid `Person` individuals.

In short, the expression $\forall R.C$ describes the set of all individuals x such that:

> every successor individual y connected through relationship R must belong to class C.

This is why universal restriction expresses:

> **semantic limitation**

rather than:

> semantic existence.

In short, the expression $\forall R.C$ describes the set of all individuals x whose **every successor** y through relationship R must belong to class C.

=== END ===

For example:

> `Pizza hasTopping` only `PizzaTopping`

Meaning:

> all toppings of a pizza must belong to PizzaTopping.

This does **not** guarantee toppings exist.

Instead, it governs:

> semantic limitation.

This distinction is critically important.

Because beginners often mistakenly assume `only` means **required**.

But semantically, it means **constrained**.

A pizza may still have:

> zero toppings.

Ontology here simply says:

> if topping exist,

they must belong to:

> `PizzaTopping` .

Understanding this distinction is one of the earliest maturity moments in ontology engineering.

Because ontology logic depends heavily on:

> semantic precision.

14.4.3 Why Chapter 14 Begins with Existential Restriction

Michael intentionally introduces:

> existential restriction

before universal restriction.

This teaching sequence is important.

Because ontology first needs to understand:

> **semantic existence**

before introducing:

> **semantic limitation**

In practical modeling:

it is usually easier to first ask:

> what relationships are required?

before asking:

> what relationships are allowed?

This chapter therefore focuses on:

> **Existential Restriction**

using:

> `someValuesFrom`

before later chapters gradually expand toward:

> more advanced logical constraints.

14.4.4 OWL Restriction Syntax Pattern -- Understanding the Grammar of Semantic Constraints

After understanding the conceptual difference between:

> existential restriction (`someValuesFrom`)

and:

> universal restriction (`allValuesFrom`),

you should pause briefly to recognize an important fact, from language perspective, that:

> OWL restriction are not random syntax.

Instead, they follow a highly structured semantic grammar.

This is an important transition point in ontology learning.

Because many beginners mistakenly perceive expressions such as:

```
hasTopping some MozzarellaTopping
```

as:

> Protégé-specific notation

or simply:

> a software configuration format.

However, this interpretation is incomplete.

In reality, OWL restrictions represent:

> **formal semantic expressions**

which are used to describe:

> logical class conditions.

In other words:

OWL restriction statements behave more like:

> **semantic sentences**

than:

> user interface settings.

Understanding this grammar is important because future ontology modeling will increasingly depend upon:

> composing semantic logic

rather than merely:

> creating classes and relationships.

At a further high level, most OWL restriction expressions follow a common structural pattern:

$$ObjectProperty + RestrictionType + TargetClass$$

Conceptually:

$$Relationship + SemanticLogic + SemanticTarget$$

This structure can be understood as a reusable language pattern for constructing:

> **semantic constraints.**

Let us examine a common example from `Pizza.owl`:

```
hasTopping some MozzarellaTopping
```

This restriction can be decomposed into three semantic components:

Component	Meaning	Semantic Role
`hasTopping`	Object Property	Defines the relationship
`some`	Restriction Type	Defines existential logic
`MozzarellaTopping`	Target Class	Defines the semantic requirement

When interpreted together, ontology understands the expression as:

> a pizza must have **at least one** topping belonging to the class `MozzarellaTopping`.

Very importantly: the keyword `some` does **NOT** mean "several" or "an unspecified number"!

Within OWL semantics: `some` specifically represents **existential quantification**, meaning:

> **at least one qualifying relationship exists.**

Now consider a second example:

```
hasTopping only PizzaTopping
```

This expression follows the exact same grammar pattern:

Component	Meaning	Semantic Role
`hasTopping`	Object Property	Defines the relationship
`only`	Restriction Type	Defines universal logic
`PizzaTopping`	Target Class	Defines the semantic requirement

However, the semantic interpretation changes significantly.

Ontology now understand:

> all toppings associated with a pizza must belong to the class `PizzaTopping` .

Notice what changed.

The **relationship** remains identical.

The **target class** also remains similar.

Only the **restriction logic** changes.

Yet this small syntactic variation completely transforms semantic meaning.

This illustrates an important principle in ontology engineering:

> **small changes in OWL syntax may produce major differences in logical interpretation.**

As ontology modeling becomes more advanced, you will encounter increasingly sophisticated restriction patterns.

For example:

- Existential Quantifier Restrictions: `hasTopping some CheeseTopping`
- Universal Quantifier Restrictions: `hasTopping only PizzaTopping`
- Cardinality Restrictions: `hasTopping min 2`
- Exact Cardinality Restrictions: `hasWheel exactly 4`
- Value Restrictions: `hasEnvironment value Production`

Although these expressions appear different, they all follow a similar semantic grammar:

$$Property + Restriction + Target$$

Understanding this pattern is an important milestone!

Because ontology engineers gradually stop reading OWL as:

> syntax

and begin reading OWL as:

> **semantic language.**

This mindset shift is crucial.

Especially for enterprise ontology and knowledge graph engineering.

Since ontology increasingly becomes not merely:

> a data model,

but:

> **a language for expressing machine-interpretable meaning.**

With this grammar foundation established, you are now ready to begin applying existential restrictions practically inside `**Pizza.owl**` and Protégé.

14.5 Existential Restriction in `Pizza.owl` -- Understanding Tutorial 4.10.1 and 4.10.2

14.5.1 A Detail Look at Existential Restriction

After introducing the conceptual foundation of property restrictions, Michael now transitions toward one of this most important practical modeling techniques in OWL:

> **Existential Restriction**

through the construct `someValuesFrom` .

At this stage in the `Pizza.owl` tutorial, ontology modeling begins changing character.

Earlier chapters largely focused on **describing relationships.**

For example:

> Pizza `hasTopping` PizzaTopping

or:

> Pizza `hasBase` PizzaBase

These statements helped ontology understand:

> semantic structure.

However, ontology still lacked the ability to describe:

> **semantic requirements.**

This distinction is subtle but important.

Because knowing:

> a pizza may have toppings

is fundamentally different from saying:

> a pizza must contain at least one topping.

Ontology therefore begins transitioning from:

> semantic possibility

toward:

> **semantic obligation.**

14.5.2 From Semantic Requirements to Executable Queries: The EKA Knowledge Graph Bridge

This transition from "possibility" to "obligation" is not merely an academic logical exercise. Within the EKA framework, it has a very concrete and powerful implication for the K **- Knowledge Graph** layer.

Recall the EKA formalization: $EKA = (K, R, \Theta, \Phi, \Gamma)$.

- **Without Existential Restrictions (K is a "Descriptive Graph"):** The knowledge graph can only answer "what is" queries. For example, you can ask: *"Which pizzas have a `MozzarellaTopping`?"* This is a simple traversal of asserted relationships.
- **With Existential Restrictions (K becomes an "Intelligent Graph"):** Because the ontology now contains *logical rules* about what *must* exist, the knowledge graph can answer "what should be" and "what is missing" queries.

Example in a Graph Query (e.g., Cypher or SPARQL):

You have defined that a `CheesePizza` **must** have at least one `hasTopping` relationship to a `CheeseTopping`. Your knowledge graph contains the following individuals:

1. `PizzaA` (type: `CheesePizza`) - has a relationship `hasTopping` to `CheddarTopping`.
2. `PizzaB` (type: `CheesePizza`) - has **no** `hasTopping` relationships.

=== Additional Reading, if you have Neo4j on-hand ===

Assume your Neo4j graph contains nodes labeled `Pizza` and `CheeseTopping` (Note: `CheeseTopping` could be member of parent node `PizzaTopping`), connected by a relationship `:HAS_TOPPING`.

Based on above individuals information.

Run this Query 1:

```
MATCH (p:Pizza {type: 'CheesePizza'})
RETURN p.name AS PizzaName
```

You may get result as:

PizzaName
PizzaA
PizzaB

Both pizzas are returned, even though `PizzaB` has no toppings.

Then run this Question 2:

```
MATCH (p:Pizza {type: 'CheesePizza'})
WHERE (p)-[:HAS_TOPPING]-(>:CheeseTopping)
RETURN p.name AS ValidCheesePizzaName
```

You should see the result now as:

ValidCheesePizzaName
PizzaA

Only `PizzaA` is returned. `PizzaB` is excluded because it does not satisfy the "must have at least one cheese topping" rule.

From knowledge graph perspective:

- Query 1 represents a **descriptive knowledge graph**: it simply tells you what is asserted, but it cannot distinguish between a valid `CheesePizza` (with toppings) and an invalid or incomplete one.
- Query 2 represents the **existential restriction logic** (`hasTopping some CheeseTopping`) enhancing the knowledge graph: it queries only those individuals that *satisfy the semantic requirement*, this turns your ontology's logical rules into powerful, executable data quality filters.

In an EKA-enabled systems, Query 2 would be the basis for generating trustworthy insights, while Query 1 might be used for auditing or identifying data quality issues (i.e., finding `PizzaB` to flag it for correction).

=== END ===

Without the existential restriction (i.e., using only RDFS/SHACL), both `PizzaA` and `PizzaB` are valid members of `CheesePizza`. A query would simply return both.

With the existential restriction (using OWL and a reasoner), the semantic layer within EKA (R) can first validate the graph. It would identify `PizzaB` as **incomplete** or **inconsistent** based on the logical rule you defined in the ontology.

This allows the **Executable Intelligence (Φ)** layer to trigger actions. A governance workflow could automatically flag `PizzaB` for a data quality issue, or a smart query could be written to exclude it from results.

Therefore, Chapter (14) is the pivotal moment where your ontology stops being a mere "schema" for your knowledge graph and becomes its "logical guardian." The existential restriction (`someValuesFrom`) is your first tool to enforce business rules ("a product must have a category", "a critical application must have an owner") directly within your executable knowledge architecture.

14.5.3 Visualized Examine on Tutorial 4.10.1 and 4.10.2

Michael introduces existential restriction through a very deliberate modeling progression.

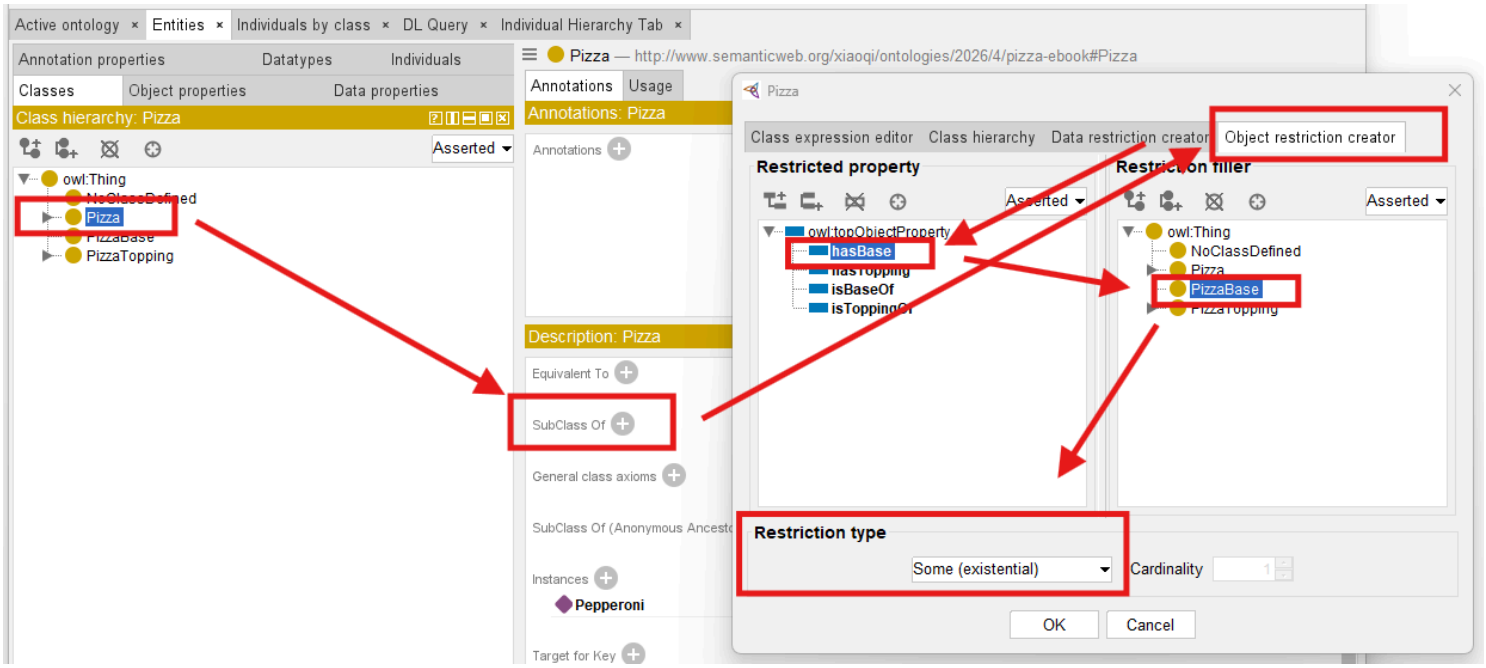
Rather than immediately creating complicated class logic, you first explore a simple but powerful idea:

> classes may be described through required relationships.

This introduces a major ontology engineering concept:

> **logical class definition.**

See below steps which described in Exercise 13 and result `Pizza hasBase some PizzaBase` :



Previously, you manually create classes primarily through naming and hierarchy.

For example:

> `MargheritaPizza`

may simply exist as:

> a subclass of `NamedPizza` .

However, ontology still does not truly understand:

> what semantically makes a `MargheritaPizza` a `MargheritaPizza` .

Existential restriction changes this.

Ontology can now begin expression:

> semantic identity.

For example:

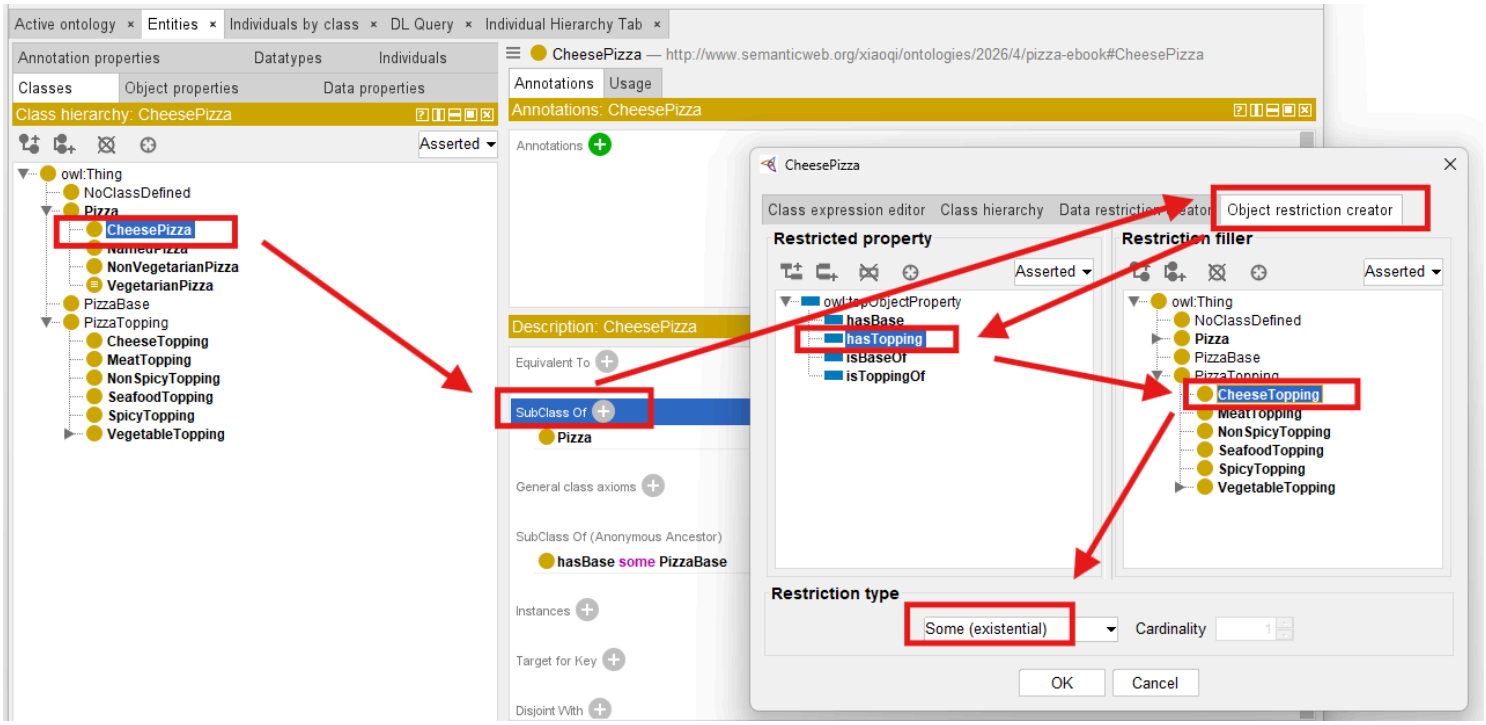
Instead of simply naming:

> `CheesePizza` ,

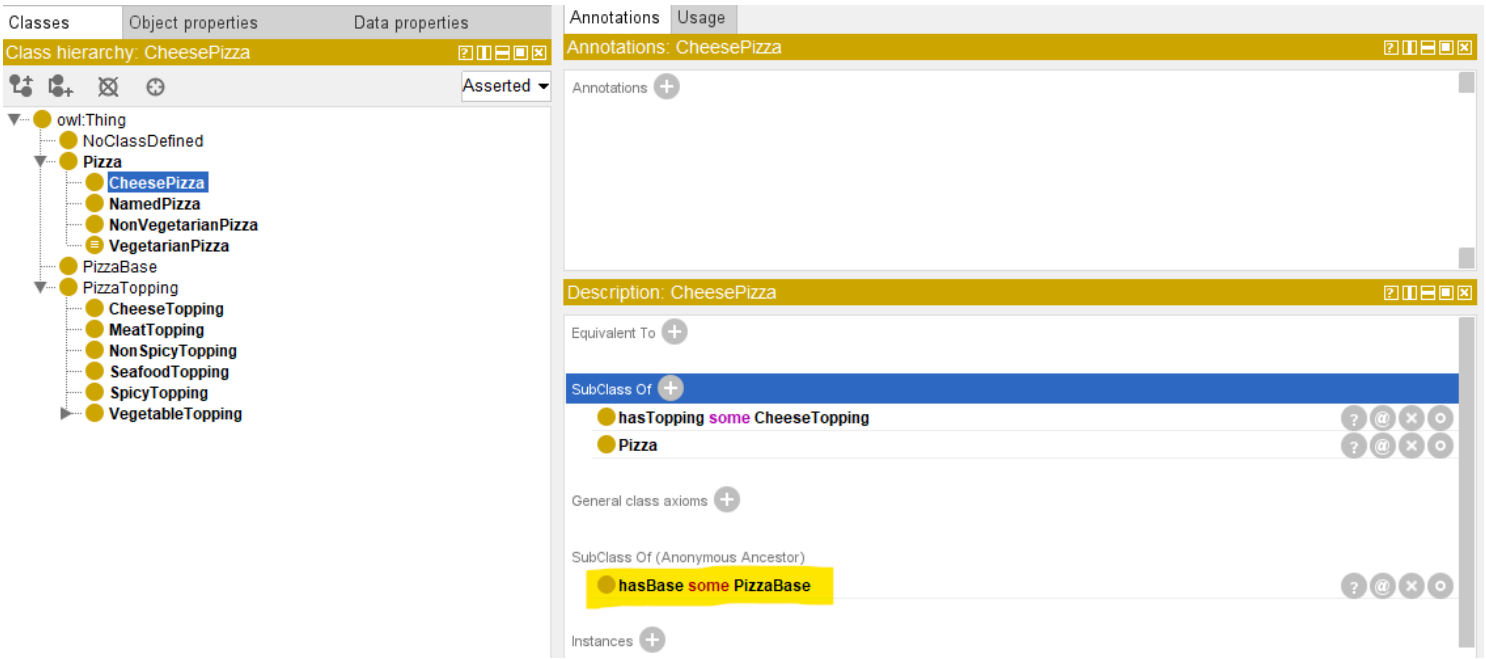
ontology may define:

> a pizza having at least one cheese topping.

See how to implement this in Protégé and resulting `CheesePizza hasTopping some CheeseTopping` :



Since `CheesePizza` is SubClass Of `Pizza` , in the result Description screen, you may see the previous Pizza / PizzaBase existential restriction is inherited, as below:



This ensure your ontology grows organically base on the known and true knowledge already have.

With such kind of configuration, your ontology is transformed fundamentally.

Meaning becomes:

> computable.

Reasoners may now determine:

> class membership automatically.

Ontology therefore becomes increasingly **inferential** rather than merely **descriptive!**

This represents one of the earliest moment where ontology begins feeling:

> intelligent.

Because class meaning no longer depends entirely upon:

> human categorization then manually configuration.

Instead:

> **logical semantics drive understanding based on machine-understandable rules.**

Within our `Pizza.owl` , existential restriction is intentionally introduced before more advanced logical restrictions because it establishes the foundational reasoning pattern:

> **required existence.**

Ontology first learns:

> something must exist.

Later chapters will teach ontology:

- what is allowed,
- what is limited, and
- how many relationships are acceptable.

This gradual semantic progression is one reason the `Pizza.owl` tutorial remains such an effective learning path.

Because ontology maturity develops incrementally.

14.5.4 View from Open World Assumption (OWA)

Interestingly, OWL reasoners do not immediately assume that a restriction has been violated simply because a required relationship has not yet been observed. In many cases, the absence of information does not necessarily imply semantic inconsistency. This behavior is closely related to one of the most important principles in ontology engineering:

> **Open World Assumption (OWA)**

which assumes that knowledge may still be incomplete.

We will explore this important concept in greater detail later in this chapter, as it fundamentally influences how OWL reasoners interpret existential restrictions and semantic requirements.

A quick example:

Suppose the ontology only knows `PizzaA` exists, with no `hasTopping` statements. Under OWA, the reasoner does not conclude `PizzaA` has no toppings. It concludes “unknown”. This is why `PizzaA` does not violate `hasTopping some CheeseTopping` — *absence of evidence is not evidence of absence.*

We will revisit OWA in detail when discussing vacuous truth (Chapter 15, section 15.1.3) and reasoner behavior (14.7.4).

14.6 Exercise 13 Walkthrough -- Creating Semantic Requirements in Protégé

With the conceptual foundation now established, you are fully ready to perform:

> Exercise 13

inside Protégé.

This exercise represents an important turning point.

For the first time, you begin configuring:

> semantic requirements

rather than merely:

> semantic structure (hierarchies and relationships).

Inside the `Pizza.owl` tutorial, you create existential restrictions using:

> `someValuesFrom`

through the Protégé interface (you already see screen samples in previous section).

At first glance, the interface interaction appears simple.

However, conceptually, something profound is happening.

Ontology is beginning to describe:

> **what a class must satisfy.**

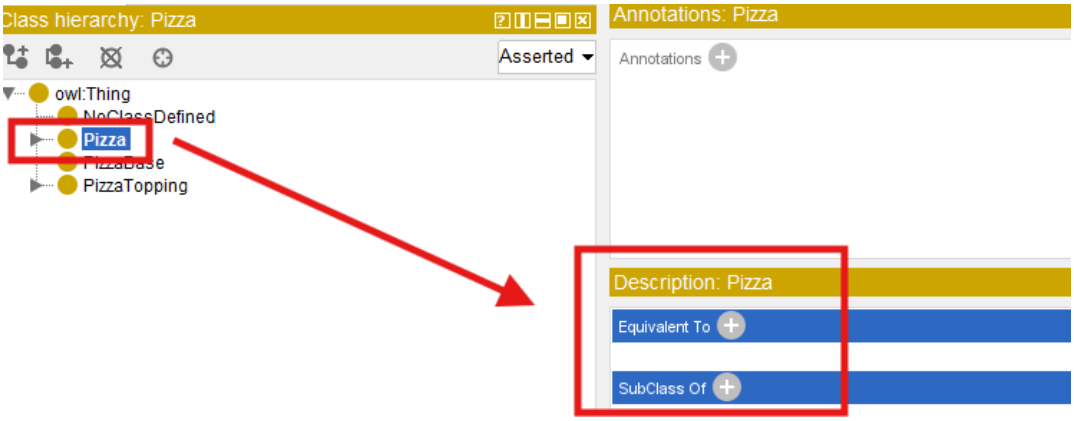
rather than merely:

> what relationships **may** exist.

Inside Protégé, you first select the target class.

Typically, Michael demonstrates this through `pizza` subclasses that require particular toppings.

You then navigates to `Equivalent Classes` or `**SubClass Restrictions**` depending on modeling preference.

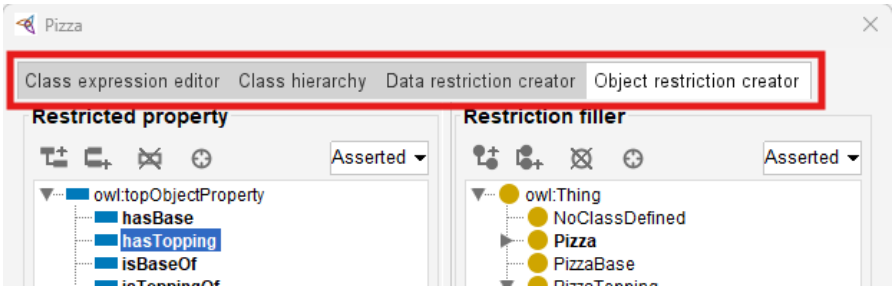


Next, Protégé provides access to:

> Object Restriction Editor.

"Editor" may not the accurate word, see below screen, you may perform more configurations here, including:

- Class expression editor
- Class hierarchy
- Data restriction creator
- Object restriction creator



At this stage, switch to actual tab name "Object restriction creator", you may perform following steps:

1. Select **Restriction Type**: `Some (existential)` means `someValuesFrom` , it's the default selection when you come to this screen, then
2. Configure **Property** in "Restricted property": `hasTopping` , and finally
3. Configure **Target Class** in "Restriction filler"

At first encounter, this may look like "Syntax".

However, ontology engineers should resist thinking about this merely as notation.

Because what has actually been created is:

> **semantic logic.**

Ontology now interprets this statement as:

> there must exist at least one topping relationship pointing toward mozzarella topping.

This changes how ontology understands "pizza identity".

A pizza now becomes classified not merely because:

> a human named it,

but because:

> semantic conditions are satisfied.

This subtle shift becomes extremely relies upon:

> inferred meaning.

rather than:

> manually asserted meaning.

=== Mathematical View ===

Let's try to map what you've configured to the previous mathematical formula:

$$\exists R.C = \{x \in \Delta^l \mid \exists y \in \Delta^l : (x, y) \in R^l \wedge y \in C^l\}$$

The statement:

> `CheesePizza` `hasTopping` some `CheeseTopping`

may have those elements referring to the variables as:

- Δ^l: the `Pizza` class domain
- R: `hasTopping` relationship
- C: `CheesePizza` class
- x, y: instance of `Pizza` (not `CheesePizza`)

A subtle but important note on x and class definitions:

> In the formula $\exists R.C = \{x \in \Delta^I \mid \exists y \in \Delta^I : (x, y) \in R^I \wedge y \in C^I\}$, the variable x ranges over instances of the class being defined. When you define `CheesePizza` as `Pizza and (hasTopping some CheeseTopping)`, the restriction is evaluated in the context of `Pizza`. This means x is an instance of `Pizza` (not necessarily an instance of `CheesePizza`). The restriction then asks: does this `Pizza` instance have at least one `hasTopping` relationship to a `CheeseTopping`? If yes, the reasoner can infer that the instance is also a `CheesePizza`. This is why x is an instance of `Pizza` in the formula – the restriction acts as a membership test for the subclass, not as a property of the subclass itself.

=== END ===

Exercise 13 therefore marks another maturity transition.

Ontology is beginning to evolve from:

> a semantic graph

toward:

> **semantic intelligence.**

14.7 Understanding Reasoning Outcomes -- Asserted vs. Inferred Semantics

After completing Exercise 13 and defining existential restrictions inside Protégé, learners may often encounter an interesting experience:

> the ontology reasoner appears to "know" things that were never manually entered.

At first glance, this may seem surprising.

Especially for readers who are more familiar with **transitional databases** or **graph databases**.

Because in those environments, data generally behaves according to a straightforward principle:

> what is **explicitly** stored is what exists.

Ontology reasoning behaves very differently.

Within OWL, semantic meaning is not limited only to:

> manually asserted facts.

Instead, reasoners are capable of producing:

> **inferred knowledge**

through logical interpretation.

This distinction represents one of the most important conceptual shifts in ontology engineering.

Because ontology gradually moves from:

> storing semantic information

toward:

> **deriving semantic meaning.**

To understand this transformation, you must first distinguish between:

Asserted Semantics and **Inferred Semantics**.

14.7.1 Asserted Semantics -- Explicitly Modeled Knowledge

Asserted semantics represent:

> information explicitly created by ontology engineers.

In other words:

there are semantic statements that humans intentionally model inside Protégé (or other ontology editors).

For example:

Suppose an ontology engineer manually creates (models):

```
PizzaA hasTopping MozzarellaTopping
```

This relationship becomes:

> asserted knowledge.

Because it was directly entered into the ontology.

Similarly:

if the ontology engineer manually states:

```
PizzaA rdf:type NamedPizza
```

this classification is likewise **asserted.**

Ontology does not need to infer it.

The semantic meaning already exists because:

> humans explicitly provided it.

From a modeling perspective, asserted semantics resemble:

> manually curated knowledge.

This is conceptually similar to:

> records stored in a database

or:

> relationships stored in a graph database.

However, ontology does not stop there.

14.7.2 Inferred Semantics -- Knowledge Derived Through Logic

Unlike asserted knowledge, inferred semantics are **not manually entered.**

Instead, they emerge through **semantic reasoning.**

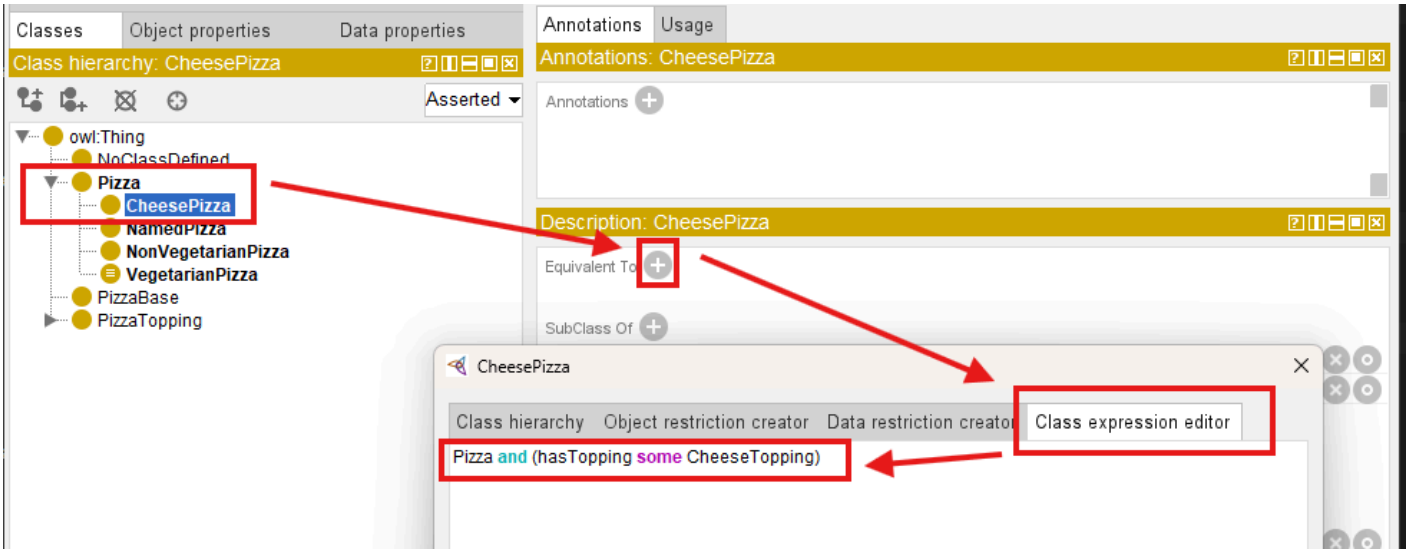
> **Note**
>
> Scared? Does this mean machine starts dominating semantic creation? Terminator? Skynet?... No worry, this is still the governed logic, rooted from human's initial design!

And, this is where existential restriction becomes especially important.

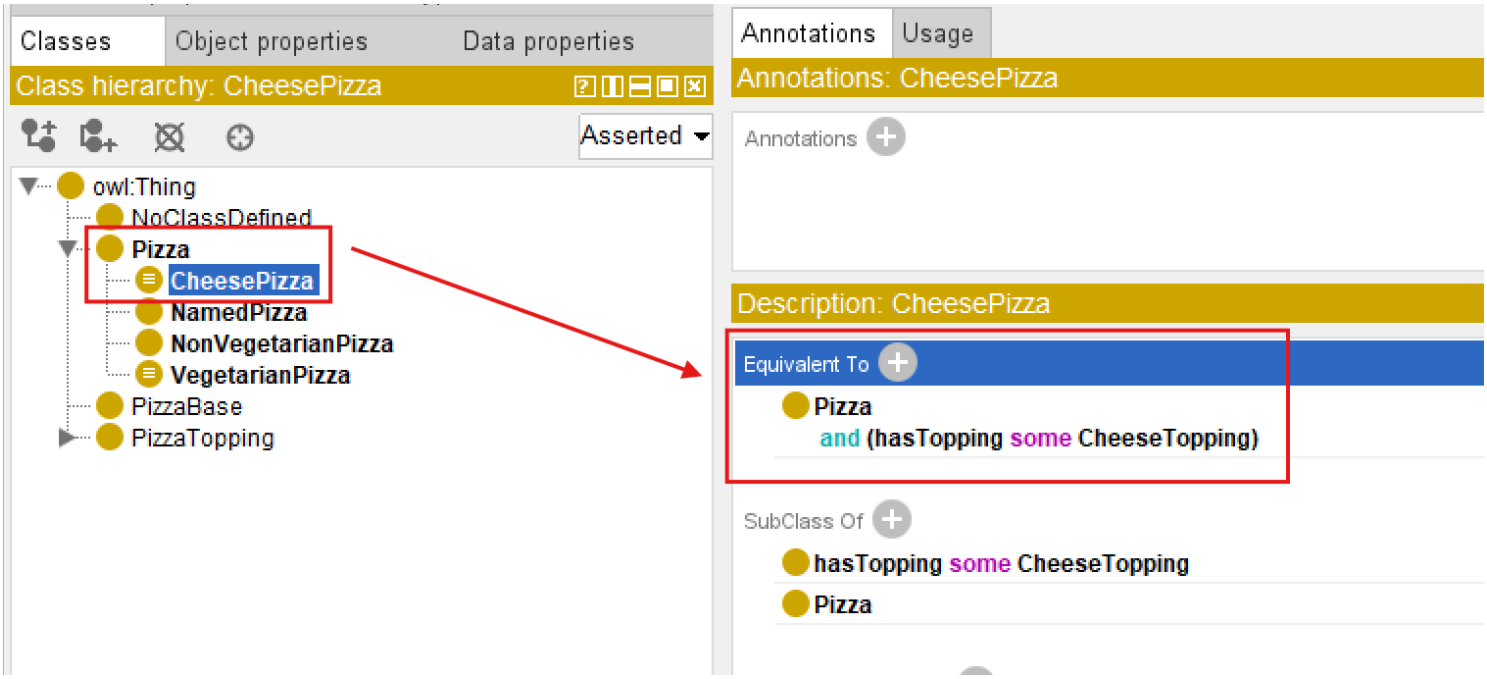
Consider the following ontology definition:

```
CheesePizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)
```

In Protégé, you may use below steps to add this definition:



After confirm, you may see this definition under `Equivalent To` :



This statement establishes a semantic condition.

Meaning:

> any pizza containing at least one cheese topping qualifies as a `CheesePizza` .

Note here we have the combination of more than one conditions (restrictions).

Now suppose ontology engineers explicitly model:

```
PizzaA hasTopping MozzarellaTopping
```

And ontology already understand:

```
MozzarellaTopping SubClassOf CheeseTopping .
```

Interestingly, magic happens:

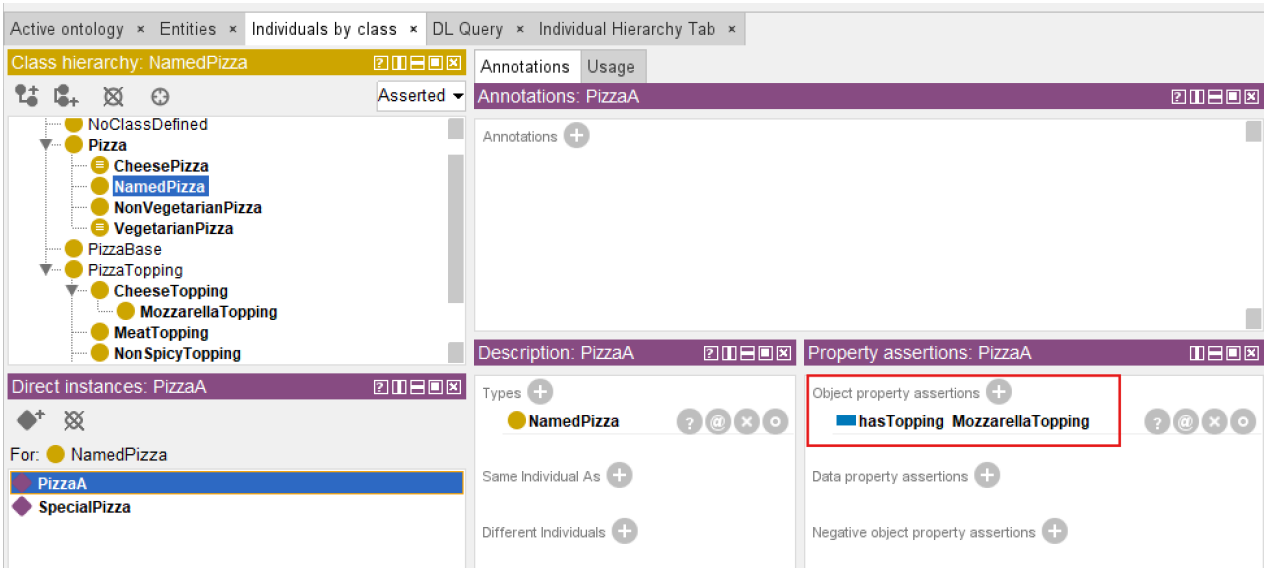
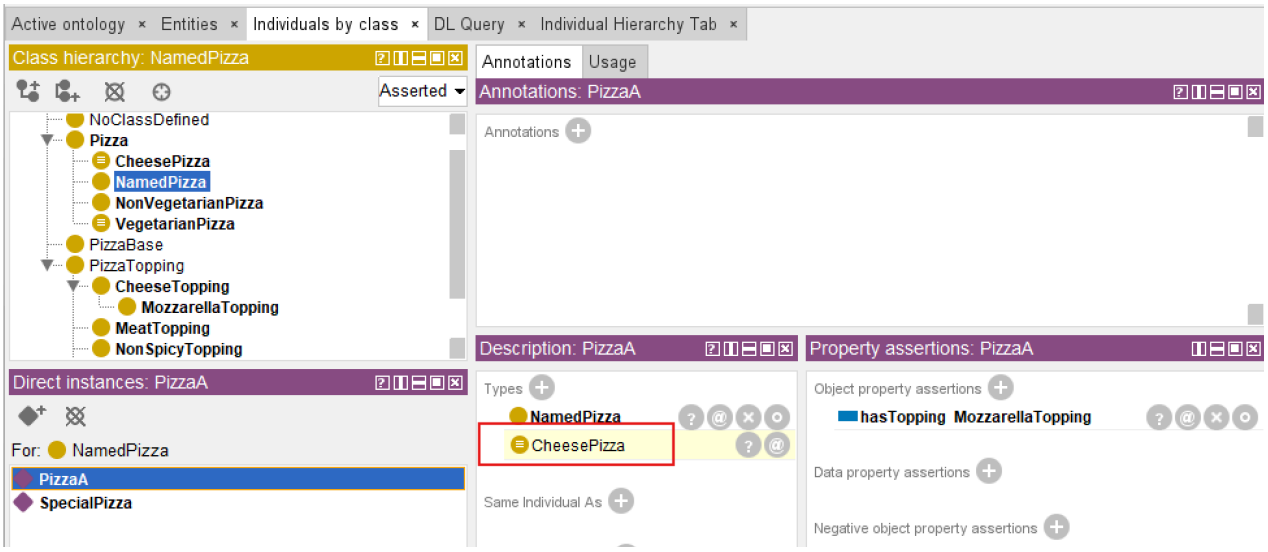
ontology engineers never explicitly declare:

```
PizzaA rdf:type CheesePizza
```

Yet after synchronizing the reasoner, Protégé may automatically infer:

```
PizzaA rdf:type CheesePizza
```

See from below comparison screen:

Before Reasoning	
After Reasoning	

This is a significant milestone!

Because ontology has now moved beyond:

> semantic storage

toward:

> **semantic interpretation.**

Meaning becomes **computable**.

The ontology reasoner analyzes:

- subclass hierarchy
- property restrictions
- existential conditions
- logical consistency, and
- **devices new knowledge**

This capability is one of the major distinctions separating ontology from:

> conventional graph data models.

Because ontology is capable of generating:

> **new semantic understanding**

rather than merely retrieving stored relationships.

=== Interesting Read: Necessary vs. Sufficient Conditions ===

The inference you just witnessed (`PizzaA` automatically classified as `CheesePizza`) depends on a fundamental logical distinction: the difference between **necessary** and **sufficient** conditions.

In Description Logic (the formal language behind OWL):

- $\texttt{SubClassOf}(\sqsubseteq)$ defines a *necessary* condition.
- $\texttt{EquivalentTo}(\equiv)$ defines *necessary and sufficient* conditions.

Mathematical form:

Concept	DL Notation	OWL Construct	Logical Direction
Necessary condition	$C \sqsubseteq D$	`SubClassOf`	$C(x) \rightarrow D(x)$
Necessary and sufficient condition	$C \equiv D$	`EquivalentTo`	$C(x) \leftrightarrow D(x)$

Example from this section:

You defined:

```
CheesePizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)
```

This means:

- **Necessary condition**: If something is a `CheesePizza` , it **must** have a cheese topping.
- **Sufficient condition**: If something has a cheese topping, it **can be inferred** to be a `CheesePizza` .

This is why the reasoner performed *automatic classification*. If you had used `SubClassOf` instead of `EquivalentTo` , the reasoner would **not** perform the reverse inference (from having a cheese topping to classifying as `CheesePizza`).

Why this matters for your modeling choices:

If you want...	Use...
Only inheritance (one-way)	`SubClassOf`
Logical definition with automatic classification (two-way)	`EquivalentTo`

This distinction is fundamental to ontology engineering. Choosing the wrong construct may lead to unexpected reasoning results (or missed inferences).

=== END ===

14.7.3 Why Existential Restriction Enables Inference

At this stage, you may naturally ask an important question:

> Why does existential restriction enable automatic classification?

The answer lies in the semantic meaning of the keyword:

> `some`

which represents:

> **existential quantification.**

Without OWL semantics, the expression `hasTopping some CheeseTopping` does not merely describe:

> a relationship.

Instead, it defines:

> **a logical test condition.**

More specifically, ontology interprets this expression as:

> there must exist **at least one** valid relationship through `hasTopping` pointing to an individual belonging to class `CheeseTopping` .

This subtle distinction is important.

Because ontology is not longer simply storing:

> semantic data,

but rather evaluating:

> **semantic conditions.**

In practice, ontology reasoners perform a process conceptually similar to searching for evidence.

The reasoner examines individuals inside the ontology and asks a logical question:

> "Can I find at least one relationship satisfying this existential condition?"

Suppose ontology contains:

```
PizzaA hasTopping MozzarellaTopping
```

and ontology already understands:

```
MozzarellaTopping SubClassOf CheeseTopping
```

The reasoner may then evaluate: `hasTopping some CheeseTopping` and conclude:

> YES - at least one valid successor individual exists.

This reasoning process may feel surprisingly intelligent, since it makes machine alike human.

However, semantically, the reasoner is performing something conceptually similar to:

> an automated semantic query.

Readers with graph databases experience may recognize an interesting parallel.

Inside Neo4j, a developer might write a Cypher query similar to:

```
MATCH (p:Pizza)-[:HAS_TOPPING]->(t:CheeseTopping)
RETURN p
```

or conceptually:

```
WHERE (p)-[:HAS_TOPPING]->(:CheeseTopping)
```

to locate pizzas having cheese toppings.

Ontology reasoners perform a related form of semantic evaluation.

However, instead of merely returning query results, the reasoner may additionally:

> **derive new semantic classifications.**

Meaning:

if a pizza satisfies the logical condition: `hasTopping some CheeseTopping`, the ontology reasoner may automatically conclude:

> this individual belongs to `CheesePizza` (class).

This introduces an important distinction between **graph traversal** and **semantic reasoning.**

- Graph database primarily discover **existing connections**.
- Ontology reasoners additionally derive **new meaning**!

Another important factor enabling inference is `EquivalentTo`.

Consider the following ontology definition:

```
CheesyPizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)
```

The keyword `EquivalentTo` represents:

> **necessary AND sufficient conditions**

This means the semantic rule works in **both directions.**

Ontology understands:

1. If something is a `CheesePizza`, it must satisfy the topping condition, and also
2. If something satisfies the topping condition, it may be classified as `CheesePizza`.

This bidirectional logic is extremely important.

Because it enables:

> **automatic classification.**

By comparison:

`CheesyPizza subClassOf (hasTopping some CheeseTopping)` would only define:

> a one-way semantic rule.

Meaning:

> every `CheesyPizza` must have cheese toppings,

but ontology would **not automatically infer** that:

> every pizza with cheese toppings becomes `CheesyPizza` .

This subtle modeling distinctions has major consequences for reasoning behavior.

And it highlights an important principles of ontology engineering:

> **small changes in logical constructors may produce significant different inference outcomes.**

Existential restriction therefore becomes one of the foundational mechanisms enabling ontology to evolve from:

> semantic structure

toward:

> **semantic intelligence.**

14.7.4 Reasoners Think Logically -- Not Intuitively

One of the most important mindset shifts in ontology engineering is learning that:

> ontology reasoners think logically,

not:

> **intuitively.**

From many learners, this initially feel unfamiliar and sometimes difficult to understand.

Because our humans naturally interpret missing information by making assumptions.

Ontology reasoners, however, do not (or we may also say "never") make assumption.

They follow:

> **formal logical evidence.**

This distinction becomes especially visible when working with **existential restrictions.**

Consider the following situation.

Suppose an ontology contains an individual `PizzaB` but no topping information has yet been defined.

A human reader may immediately think:

> "If no topping exists, then `PizzaB` is obviously not a `CheesyPizza` , although it has `Pizza` inside the name."

This conclusion feels natural, to humans.

After all, human reasoning frequently relies on:

> incomplete information and educated assumptions.

However, an OWL reasoner behaves differently.

The ontology reasoner does **not infer**:

> `PizzaB rdf:type CheesyPizza`

Yet importantly: the reasoner also does **not infer**:

> `PizzaB` is *not* a `CheesyPizza` .

Instead, ontology reaches a more conservative conclusion:

> **there is currently insufficient information to determine classification.**

In other words, the ontology simply says:

> **"I do not know yet."**

This behavior often surprises beginners.

Especially those with experience in

> relational databases,

> business rules engines,

or:

> graph databases.

Because many traditional systems implicitly follow a different assumption:

```
Missing information = false
```

Ontology reasoning, however, follows one of the most important principles in semantic modeling:

> **Open World Assumption (OWA)**

Under OWA: missing information does not imply **falsity.**

Instead, ontology assumes **knowledge may still be incomplete**.

Meaning:

> future information may still arrive, just not available for now yet.

For example:

The ontology may later receive `PizzaB hasTopping MozzarellaTopping` and ontology already understands `MozzarellaTopping SubClassOf CheeseTopping` .

At that moment, the reasoner may suddenly infer:

```
PizzaB rdf:type CheesyPizza
```

What has been changed?

Not the logical rule.

Only:

> **the available evidence (information).**

Ontology reasoning therefore behaves less like:

> guessing,

and more like:

> **evidence-based semantic evaluation.**

This distinction becomes clearer when comparing **Human Intuition** vs **OWL Reasoning**:

Human Intuition	OWL Reasoning Logic
No topping observed → probably not cheese pizza	No evidence observed (got) → classification postponed
Missing information suggest absence	Missing information may simply be incomplete
Infer based on assumptions	Infer based on formal evidence

Notice the key difference?

OWL reasoners do not ask "What is most likely true?"

Instead, they ask "What can be logically justified?"

This distinction becomes especially important when working with:

> existential restrictions.

Earlier in this chapter, we defined:

```
CheesyPizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)
```

The keyword `EquivalentTo` defines:

> **necessary and sufficient conditions.**

Meaning:

> ontology interprets the definition in **both directions.**

This enables the reasoner to conclude:

> if a pizza satisfies the existential condition, then it may be classified as `CheesyPizza` .

However, imagine the ontology instead used:

```
CheesyPizza SubClassOf (hasTopping some CheeseTopping)
```

This semantic meaning changes significantly.

Now ontology only understands:

> every `CheesyPizza` must satisfy the topping restriction.

But ontology can no longer conclude:

> every pizza satisfying the restriction automatically becomes `CheesyPizza` .

The inference direction becomes **one-way only.**

This subtle distinction between `EquivalentTo` and `SubClassOf` has major consequences for ontology behavior.

It also highlights an important engineering lesson:

> ontology reasoners execute logic exactly as modeled

while NOT:

> as human intended.

For ontology engineers, this carries important practical implications.

Semantic models must be designed **precisely** with careful attentions to:

- logical meaning,
- restriction semantics, and
- inference consequences.

At the same time, Open World Assumption provides an important advantage.

Because enterprise knowledge is often:

- incomplete,
- distributed, or
- continuously evolving.

OWL therefore remains highly suitable for enterprise knowledge graphs, semantic integration, and executable intelligence systems when meaning must gradually emerge from **accumulating knowledge** rather than requiring perfect information from the beginning.

As you continue through this chapter, Open World Assumption will become increasingly important for understanding:

> why ontology reasoners sometimes appear surprisingly conservative,

yet remain remarkably powerful in semantic interpretation.

14.8 Chapter 14 -> 15 Bridge

You have now built the first half of the semantic requirements toolkit: you know *why* RDF/RDFS alone cannot express requirements, you know the grammar of OWL restrictions, and you can create and reason over **existential restrictions** (`some`) -- the constraint that says "at least one relationship of this kind must exist."

But `some` only tells the reasoner what is *permitted* to exist. It says nothing about what is *forbidden*. A pizza declared to have `hasTopping some VegetableTopping` is only required to have at least one vegetable topping -- nothing stops the reasoner from also accepting a pepperoni topping on the same pizza. That gap is exactly where Chapter 15 begins, with the **universal restriction** (`only`) and the classic combination pattern that finally gives us a working definition of `VegetarianPizza EquivalentTo Pizza and (hasTopping only (CheeseTopping or VegetableTopping))` .

14.9 Key Concepts (Chapter 14)

Concept	Description
Property Restriction	An OWL class expression that constrains how a class relates to other classes or individuals through a property.
Existential Restriction (`some`)	Requires that *at least one* relationship of a given property/class combination exists.
Asserted Semantics	Knowledge explicitly stated by the ontology engineer.
Inferred Semantics	Knowledge derived by a reasoner through logical entailment, not explicitly written.
Open World Assumption (OWA)	The reasoning stance in which absence of a statement means "unknown," not "false."

14.10 Protégé Skills Learned (Chapter 14)

- Reading and writing OWL restriction syntax (`some` , quantifier grammar)
- Creating an existential restriction on a class in Protégé
- Running Exercise 13 to attach semantic requirements to `Pizza.owl`
- Distinguishing asserted axioms from reasoner-inferred axioms in the Protégé interface

14.11 Next Chapter (15) Preview

Chapter 15 introduces the **universal restriction** (`only`), shows why `some` and `only` must be combined to correctly define `VegetarianPizza` , and walks through implementing both restriction types as Cypher constraints in Neo4j -- giving you a concrete side-by-side comparison between OWL's Open World reasoning and a property graph's Closed World queries.

Last updated at: 2026-09-11

Chapter 15 -- Semantic Requirements: Universal Restriction, Composition, and the Neo4j Mirror

> **Editor's note:** This chapter is Part 2 of a three-chapter unit on OWL property restrictions (originally drafted as a single Chapter 14; see Chapter 14 for the foundations and existential restriction, and Chapter 16 for the EKA synthesis and common-mistakes review).

Chapter Introduction

Chapter 14 gave you the first half of the semantic requirements toolkit: the **existential restriction** (`some`), which requires that at least one qualifying relationship exists. As Chapter 14 closed, we flagged the gap that `some` leaves open -- it permits, but never forbids.

This chapter closes that gap. You will learn the **universal restriction** (`only`), see why `some` and `only` must be combined to produce a correct, reasoner-verifiable definition of `VegetarianPizza` , and then step outside Protégé entirely to see how both restriction types would be implemented as constraints in a property graph database (Neo4j) -- a comparison that sharpens your understanding of the Open World Assumption by contrasting it against the Closed World Assumption most graph databases use.

Table of Contents:

15.1 Universal Restriction -- Understanding `only`

In the previous sections, you explored:

> **existential restriction (`some`)**

and discovered one of the most important ideas in ontology engineering:

> ontology can define not only what exists,

but also:

> what **must** exist.

For example, the expression `hasTopping some CheeseTopping` allows ontology to express a meaningful semantic requirement:

> every pizza **must** contain at least one cheese topping.

This is already a major conceptual shift.

Because ontology is no longer behaving merely as a relationship repository or a connected graph structure. Instead, ontology begins acting as:

> a semantic rule system.

Meaning:

relationships are no longer interpreted only as:

> stored facts,

but also as:

> logical conditions.

At this point, you may reasonably feel:

> "I understand how ontology can require something to exist."

However, a second and equally important modeling problem soon appears.

Ontology engineers must also ask:

> **How can we control what kind of things are allowed to exist?**

This distinction may initially appear subtle.

Yet it fundamentally changes how semantic system behave.

In real-world enterprise environments, data quality problems rarely emerge only because:

> required information is missing.

More often, problems emerge because:

> incorrect relationships are introduced.

For example:

An employee may accidentally be linked to:

> a business application

instead of

> a department.

A customer may mistakenly belong to:

> a linux server infrastructure.

A business capability may incorrectly be realized by:

> a physical office location.

In all these cases:

> relationships exist,

but:

> semantic (meaning) correctness is violated.

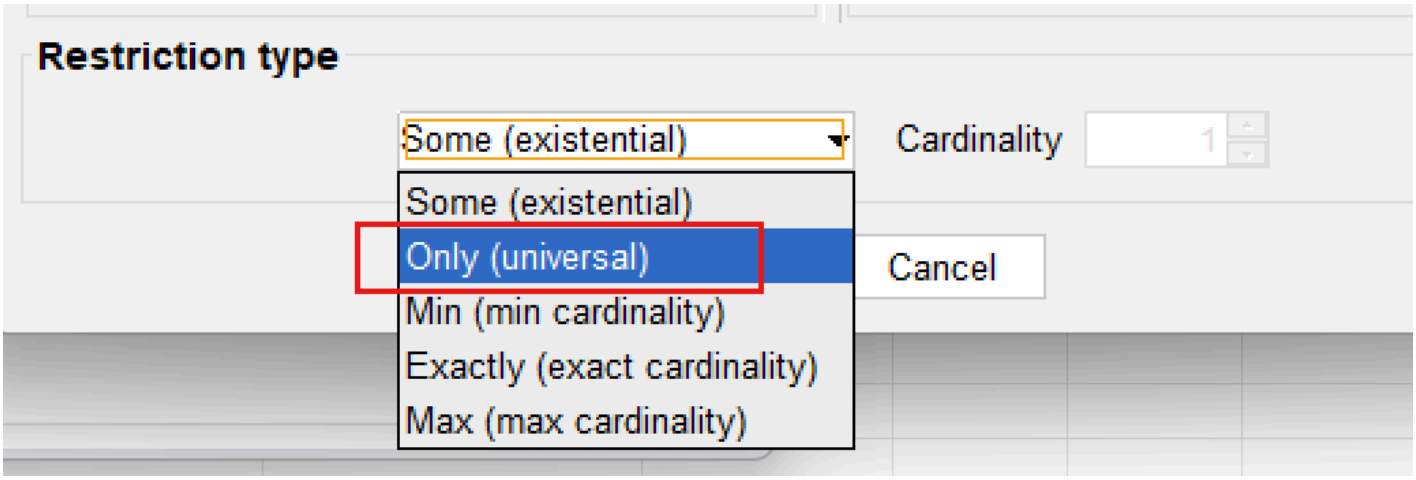
Ontology therefore requires more than **participation rules**, it also requires **semantic boundaries.**

Existential restriction helps ontology answer：

> **What kinds of things are semantically allowed?**

To solve this challenge, OWL introduces another important semantic constructor **Universal Restriction** represented in Protégé using `only` and formally implemented through `allValuesFrom` .

First, have a look that it's just under the existential (some) type in Protégé:



Unlike existential restriction, which focuses on:

> **existence** (by its naming),

universal restriction focuses on:

> **constraint.**

More specifically:

ontology moves from:

> semantic participation

toward:

> **semantic governance.**

This makes an important evolution in ontology engineering.

Because semantic systems gradually stop asking only:

> "What relationships should exist?"

and being asking:

> "Are existing relationships semantically valid?"

As ontology models scale toward:

- enterprise knowledge graphs,
- semantic integration platforms, and
- executable intelligence systems,

this distinction becomes increasingly important.

Ontology therefore evolves from:

> connected knowledge

into:

> **governed semantic knowledge.**

15.1.1 From Existence to Constraint -- Why `only` is Needed

To understand why universal restriction exists, let us first examine an important limitation of existential restriction.

Suppose we define the following semantic requirement:

```
hasTopping some VegetableTopping
```

Ontology now understands:

> a pizza must contain **at least one** vegetable topping.

At first glance: this may appear sufficient for describing a:

> vegetarian pizza.

After all: ontology can already validate:

> whether vegetable participation exists.

However, consider the following scenario.

Suppose ontology contains:

```
PizzaA hasTopping MushroomTopping
PizzaA hasTopping ChocolateTopping
```

Since `MushroomTopping` belongs to `VegetableTopping` , the existential condition succeeds.

Ontology correctly observes:

> at least one vegetable topping exists.

Yet something still feels wrong.

Because the same pizza also contains `ChocolateTopping` which clearly violates:

> common semantic expectations of vegetarian pizza.

This example reveals an important limitation.

Existential restriction successfully validates:

> **minimum participation**

But it does **NOT regulate semantic boundaries.**

Ontology can confirm:

> something acceptable exists.

But ontology still cannot answer:

> are all relationships semantically appropriate?

This distinction becomes increasingly important in enterprise modeling.

Consider an ontology according to employee.

Suppose the architect define:

```
Employee SubClassOf (belongsToDepartment some Department)
```

Ontology now ensures:

> every employee belongs to at least one department.

This is useful.

However, ontology still allows situations such as:

```
Employee_A belongsToDepartment Product_X
```

or:

```
Employee_A belongsToDepartment DataCenter_01
```

which are semantically incorrect.

The relationship exist, but the meaning becomes **polluted.**

Ontology therefore requires another mechanism.

Not merely to say:

> something valid exists.

But to say:

> **everything involved must remain semantically valid.**

This is where **universal restriction (`only`)** becomes necessary.

A useful way to think about the distinction is:

- `some` → minimum semantic participation
- `only` → semantic boundary enforcement

Or more simply:

- `some` → something valid must exist
- `only` → everything must conform

This transition represents an important maturity step in ontology thinking.

Ontology is no longer merely asking:

> "What should be present?" *-- by existential restriction*

It now also asks:

> **"What should be permitted?** *-- by universal restriction*

In enterprise terms:

- `some` ensures a **business rule**: e.g. "Every application must have an owner."
- `only` ensures **data quality**: e.g. "Every owner must belong to the class `Employee` (not `Department` , not `Supplier`)."

Both are needed.

One without the other leaves the ontology incomplete.

15.1.2 The Semantics of `only` -- "All Must Be" vs. "Only Allowed"

Universal restriction introduces a very different form of semantic logic.

Consider the OWL expression: `hasTopping only PizzaTopping`

At first glance, many learners interpret this expression incorrectly.

A common misunderstanding is:

> "This pizza must contain pizza toppings."

However, this interpretation is **semantically** inaccurate.

The keyword `only` does **NOT require existence.**

Instead, it constraints:

> the semantic type of relationship **if they exist.**

More precisely, ontology interprets the expression as:

> **IF** toppings exist, every topping must belong to the class `PizzaTopping` .

Notice the subtle difference?

Ontology is **not saying**:

> toppings MUST EXIST.

Ontology is only saying:

> whenever toppings exist, they must satisfy semantic expectations.

The hidden meaning is:

> *if toppings are not existed, I do not care since that's not triggering inference.*

This distinction is extremely important.

Because ontology engineers often **unconsciously** interpret `only` using:

> natural language intuition.

In ordinary English:

> "only pizza toppings" often sounds like **mandatory presence.**

But OWL semantics behave differently!

Ontology reasoners interpret:

> `hasTopping only PizzaTopping`

as:

> a semantic limitation,

not:

> an existence requirement.

A useful mental model is to image `only` as:

> a semantic type checker.

In programming languages, type systems help ensure:

> data conforms to expected formats.

For example:

> an integer field should NOT suddenly contain **a text string**.

Similarly:

universal restriction ensures ontology relationships remain:

> semantically consistent.

Ontology therefore uses `only` to establish:

> **relationship type safety.**

This idea connects naturally back to Chapter (13) -- Domain and Range.

At first glance, Domain/Range and universal restriction may appear similar.

Because both attempt to improve:

> semantic correctness. (kind of governance)

However, they operate at different levels.

	Domain and Range	Universal Restriction
Definition	Define "structural expectations".	Rather than describing "relationship design", it evaluates "actual semantic behavior".

Example	`hasTopping`
	`Domain: Pizza`
	`Range: PizzaTopping`
		`hasTopping only PizzaTopping`

Question Asked	"What relationship types are structurally expected?"	"Do all existing relationships remain semantically valid?"
Action	Act like "a schema-level semantic rule"	Evaluate "actual semantic behavior"
Meaning	Ontology expects "pizzas should connect to pizza toppings"	Ontology expects "whenever pizza toppings exist, the semantic expectations must be satisfied"

This distinction is especially important for:

> enterprise knowledge governance.

Because semantic quality is rarely protected by **structure alone**.

Instead, ontology must continuously evaluate **relationship meaning.**

Universal restriction therefore behaves like:

> **semantic runtime governance.**

rather than merely structural design.

15.1.3 The Critical Difference -- `some` vs. `only`

At this stage, you should clearly recognize that:

> existential restriction (`some`)

and:

> universal restriction (`only`)

are not interchangeable.

Although their syntax may initially appear similar, they represent fundamentally different forms of semantic logic.

Understanding this distinction is one of the most important milestones in ontology engineering.

Because many ontology beginners mistakenly assume `some $\approx$ only` and simply treat them as:

> different OWL keywords.

In reality, they answer **completely different** semantic questions.

Existential restriction asks:

> **"Does at least one qualifying relationship exist?"**

While universal restriction instead asks:

> **"if relationships exist, do they all satisfy semantic expectations?"**

This distinction may appear subtle.

Yet it fundamentally changes:

- reasoning behavior,
- inference outcomes, and
- semantic interpretation.

The following comparison provides a useful summary:

Characteristic	`some` (Existential Restriction)	`only` (Universal Restriction)
Core Question	"Must at least one relationship exist?"	"If relationships exist, must they all belong to a certain class?"
Semantic Meaning	Minimum existence requirement	Semantic limitation / boundary
Creates	Requirement	Constraint
Logical Role	Participation condition	Semantic governance
Reasoner Behavior	Searches for valid evidence	Searches for invalid evidence
Behavior with No Relationship	Not satisfied	Satisfied (don't care)
Example	`hasTopping some VegetableTopping`	`hasTopping only PizzaTopping`

One of the most powerful way to understand the distinction is through:

> **reasoner behavior.**

When ontology evaluates `hasTopping some CheeseTopping`, the reasoner behaves almost like:

> a semantic search engine.

Its task becomes **finding evidence.**

More specifically, the reasoner here asks:

> "Can at least one topping be found that belongs to `CheeseTopping`?"

If ontology identifies `MozzarellaTopping` or `ParmesanTopping`, then the condition succeeds.

Means ontology has found:

> qualifying semantic evidence.

Universal restriction behaves very differently.

Consider `hasTopping only PizzaTopping` , here the reasoner is not searching for **evidence of success.**

Instead, it searches for **evidence of violation!**

The reasoner effectively asks:

> "Do any topping violate semantic expectations?"

If ontology discovers `ChocolateBar` connected through `hasTopping` , while `ChocolateBar` does not belong to `PizzaTopping` , then semantic inconsistency may emerge.

A memorable mental model is:

- `some` $\longrightarrow$ `search for qualifying evidence`
- `only` $\longrightarrow$ `search for violating evidence`

This distinction becomes extremely useful when ontology models grow larger, into the enterprise scale.

Because enterprise semantic systems increasingly depend upon:

> automated validation

rather than:

> manual review.

Another important difference appears when:

> no relationship exists.

Suppose ontology contains `Pizza_B` with no toppings currently defined.

Would the following condition succeed?

```
hasTopping some CheeseTopping
```

The answer is:

> **NO.**

Because ontology cannot find evidence showing:

> at least one cheese topping exists.

Existential restriction therefore fails.

Now consider:

```
hasTopping only PizzaTopping
```

Surprisingly:

> ontology still considers the restriction satisfied.

Why?

Because ontology now finds:

> no invalid toppings.

Universal restriction asks:

> "If toppings exist, are they semantically valid?"

When no toppings exist:

> no violation can be observed.

In formal logic, this behavior is called **vacuous truth**.

Meaning:

> a universal condition remains true unless a counterexample exists.

You do not need to memorize this mathematical term immediately. (see Interesting reading for reference)

However, understanding the behavior is extremely important.

Because it reveals something fundamental:

> OWL reasoners think logically -- not intuitively.

Human may instinctively say:

> "A pizza without toppings should fail pizza rules."

But ontology behaves differently.

Ontology instead says:

> "No invalid topping exists, therefore no contradiction has been observed."

This distinction becomes increasingly important as learners transition from:

> data modeling

toward:

> logical knowledge modeling.

A useful summary is:

- `some` = `at least one valid relationship must exist`
- `only` = `all relationships must remain semantically valid`

Together, these relationships provide ontology with both:

> participation logic

and:

> semantic discipline.

==== Interesting Reading: Vacuous Truth in Formal Logic ===

The behavior of `only` when no relationship exists -- i.e., it is automatically satisfied -- is not an arbitrary design choice. It follows a fundamental principle in formal logic called **vacuous truth**.

What is Vacuous Truth?

In classical logic, a statement of the form:

> *"If P, then Q"* (written as $P \rightarrow Q$)

is considered **true** whenever the condition P is **false** -- regardless of whether Q is true or false.

This is not a convention.

It is a logical necessity to avoid contradictions and to preserve the integrity of implication.

Example in everyday reasoning:

> *"If it rains, then the ground will be wet.*

If it does **not** rain, the statement is still considered **true** (vacuously true). It makes no claim about the ground's wetness when it doesn't rain. It only promises that *whenever* it rains, wetness follows.

How Vacuous Truth Applies to "only"

Recall the semantic meaning of `only`:

> *"If a pizza has a topping, then that topping must be `PizzaTopping`.*

This is a logical implication:

> `hasTopping(pizza, topping)` $\rightarrow$ `PizzaTopping(topping)`

Scenario	Pizza has topping?	Topping is PizzaTopping?	Implication (Vacuous Truth)
`Pizza` has `MushroomTopping`	✅ True	✅ True	**True** (satisfied)
`Pizza` has `ChocolateTopping`	✅ True	❌ False	**False** (violated)
`Pizza` has **no** toppings	❌ False	(not evaluated)	**True** (vacuously satisfied)

The third row is **vacuous truth**. Because the condition "pizza has a topping" is false, the overall `only` rule is automatically satisfied. The ontology does **not** complain about missing toppings.

Why Does Logic Include Vacuous Truth?

Without vacuous truth, logical implication would break down.

Consider the statement:

> *"All humans who can fly are astronauts.*

If there are no humans who can fly, the statement is considered **true** still.

It does not falsely claim that some flying humans exist. It simply states a conditional relationship that happens to be trivially satisfied because the condition never occurs.

This same principle allows `only` to act as a **pure constraint**, not an existence requirement. It governs the *type* of relationships *if they exist*, without forcing them to exist.

Mathematical Form

In first-order logic, the universal restriction $\forall x\ (R(x) \rightarrow C(x))$ is **vacuously true** when there is no x that satisfies $R(x)$.

This is written as:

> $\forall x \in \phi,\ C(x)$ is always true.

Or equivalently:

> $\neg\exists x\ (R(x) \wedge \neg C(x))$

No counterexample exists $\rightarrow$ the statement is true still.

Why This Matters for Ontology Engineering?

- `only` **does not require existence** -- it only constrains what can exist.
- `some` **requires existence** -- it actively searches for at least one example.

Understanding vacuous truth prevents the common mistake of expecting `only` to behave like `some`.

> *"Vacuous truth is not a loophole! It is a logical feature that allow `only` to be a pure constraint, not a disguised requirement.*

==== END ====

15.1.4 Universal Restriction in Protégé -- `only` as a Semantic Guardian

Have understood the semantic meaning of `only` from both "**logical**" and "**mathematical**" perspective, you are now ready to observe how universal restriction behaves inside Protégé.

At first glance, universal restriction may appear very similar to concepts introduced earlier in Chapter (13) -- Domain and Range.

Because both mechanisms seems to control:

> semantic correctness.

However, both their behavior and purpose are fundamentally different.

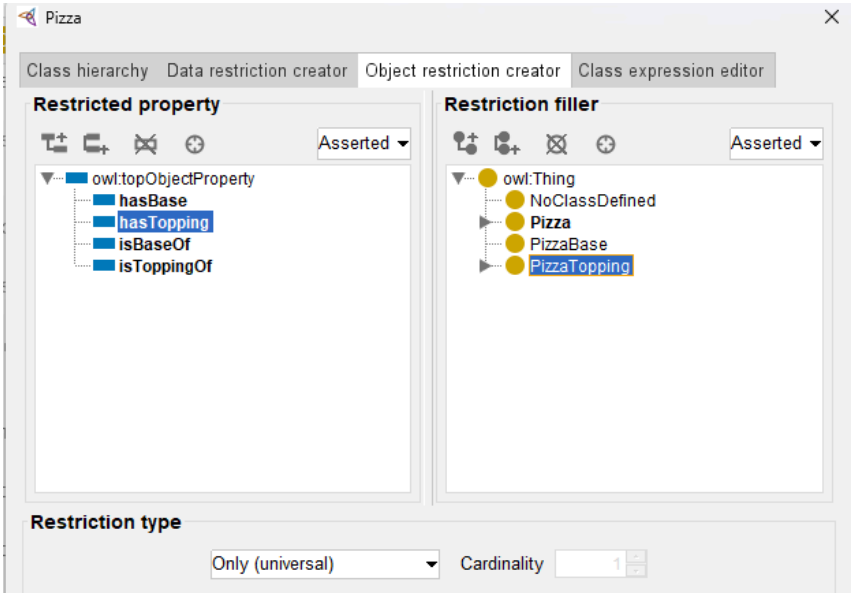
- Domain and Range primarily define **structural expectations.**
- Universal restriction instead evaluates **logical consistency.**

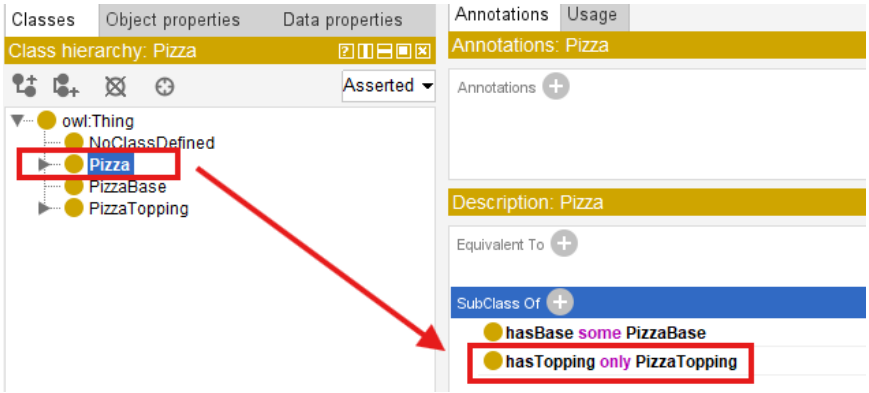
This distinction becomes easier to understand through experimentation, in our Protégé.

Suppose we define the following restriction for `Pizza` inside `SubClass Of` using:

```
hasTopping only PizzaTopping
```

Note: Since our common RDF workbook (pizza-ebook.rdf) had been already configured `some` when we practice existential constraint, to support this section without misleading, I'm clean unnecessary setting and create in one separate workbook called pizza-ebook_15.1.4.





This expression tells ontology:

> whenever a pizza contains toppings, all toppings must belong to `PizzaTopping` .

Notice again: ontology is **NOT requiring toppings to exist!**

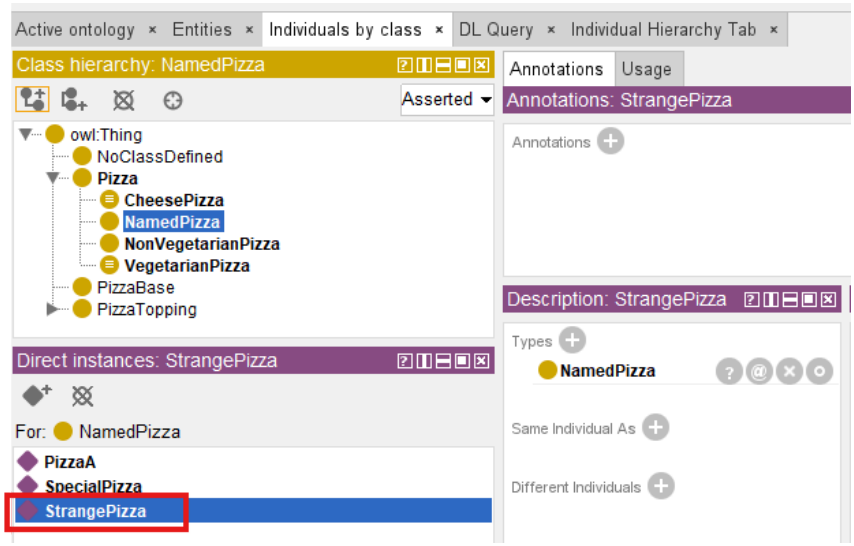
Instead, ontology governs:

> the semantic validity of toppings if they exist.

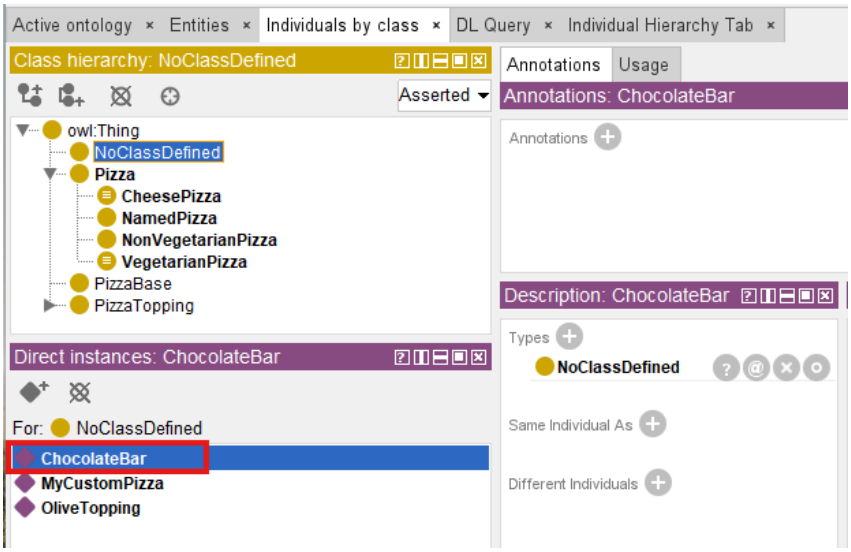
To better understand this behavior, let us intentionally construct:

> semantically problematic data.

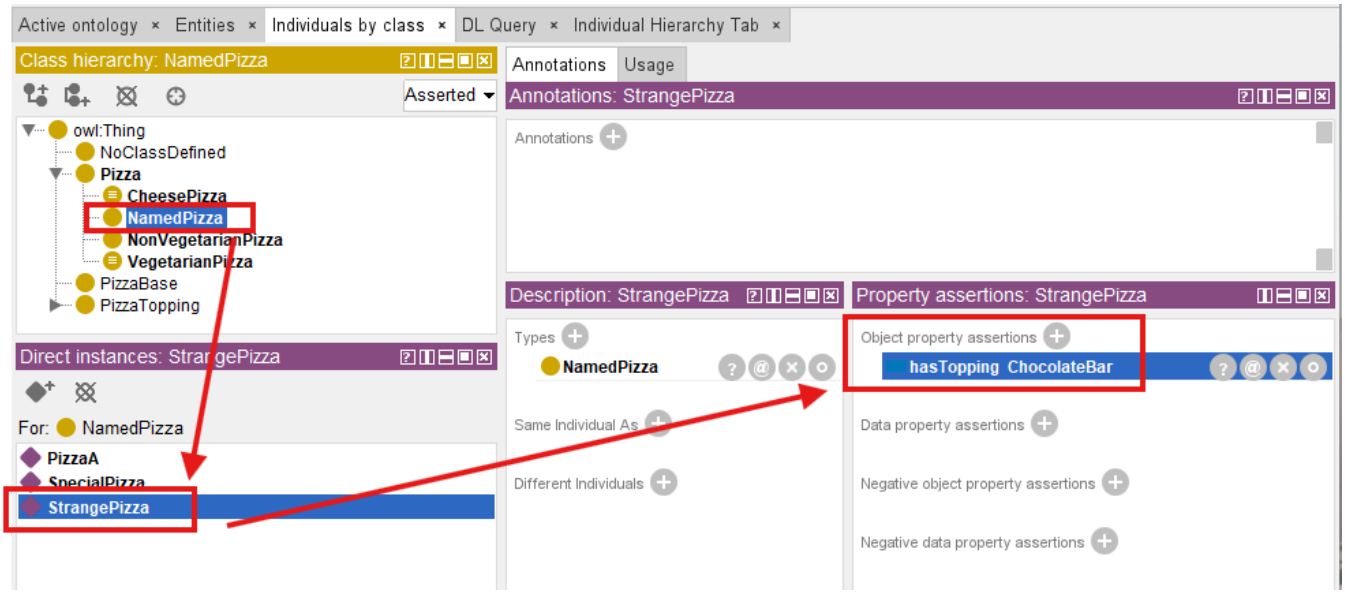
Create an individual such as `StrangePizza` under `NamedPizza` :



Then create another individual `ChocolateBar` that does **NOT** belong to `PizzaTopping` , let's create under our previous class `NoClassDefined` for simplicity:



Now connect them through `StrangePizza hasTopping ChocolateBar` relationship:



At first, Protégé may not immediately indicate a problem.

This sometimes indeed surprises beginners.

Because ontology editors often separate:

> assertion

from:

> reasoning.

Protégé first records:

> stated knowledge.

Only after synchronization does the **reasoner** begin:

> evaluating semantic consequences.

Once the reasoner executes:

> ontology starts logically evaluating the restriction.

Importantly, the reasoner does **NOT** ask:

> "Does this pizza have toppings?"

Instead, it asks:

> "Do all toppings satisfy the expected semantic type?"

This distinction matters!

Because ontology validation focuses on **meaning**, not merely **existence**.

Depending on how the ontology is modeled, several outcomes may occur.

The reasoner may identify **inconsistency**, **unsatisfied class membership**, or **classification failure**.

The exact behavior depends upon whether **disjointness**, **equivalent classes**, or **additional constraints** exist elsewhere in the ontology.

This experiment reveals an important ontology engineering principle:

> universal restriction behaves like a semantic quality monitor.

Rather than merely storing connected relationships, ontology continuously evaluates:

> whether relationships remain semantically trustworthy.

For enterprise architects, this idea is particularly important.

Because many enterprise systems fail not because:

> their relationships are absent,

but because:

> relationships become semantically **polluted**.

Let us look at one example:

an enterprise graph may technically allow following relationship:

```
Employee belongsToDepartment Server01
```

The relationship exists.

The database accepts it. (Base on the schema design.)

Yet semantically, the meaning is incorrect.

Universal restriction provides a mechanism for ontology to detect such kind of:

> **semantic drift**

before invalid meaning propagates throughout enterprise knowledge models.

A useful way to summarize the distinction is:

- `Range` $\rightarrow$ relationship expectation
- `only` $\rightarrow$ semantic validation

This is why `only` often behaves like:

> a semantic guardian.

Protecting ontology not from **missing data**, but from **"incorrect meaning**.

=== Interesting Reading: Semantic Drift in Enterprise Knowledge Models ===

Universal restriction (`only`) provides a mechanism for ontology to detect **semantic drift** before any invalid meaning propagates throughout enterprise knowledge models.

Semantic drift occurs when a relationship gradually deviates from its intended meaning over time, often through incremental, seemingly harmless assertions.

For example, an ontology initially enforces:

```
Application hostedOn only Infrastructure
```

Over months of maintenance, a modeler adds:

```
Application hostedOn VendorContract
```

Because no `only` constraint was defined for `hostedOn` , the ontology silently accepts this new relationship. Over time, the semantic boundary erodes ("poisoned"). Queries that rely on `hostedOn` to locate `infrastructure` now return `VendorContract` s, breaking impact analysis, security compliance, and also dependency tracing.

Formally, semantic drift can be defined as:

> *A gradual violation of the universal restriction $\forall R.C$ where the set of observed relationships $R(x, y)$ expands to include individual y that are not in the intended target class C, without the ontology detecting the violation.*

Universal restriction acts as a **semantic invariant**.

The reasoner continuously evaluates:

$$\forall x, y : (x, y) \in R \rightarrow y \in C$$

If a counter-example appears, the ontology flags an inconsistency or prevents the invalid relation ship from being trusted. This transforms ontology from a passive schema into an active **semantic guardian**, capable of detecting and preventing meaning erosion at scale.

In an enterprise environment, where thousands or millions of relationships may be added/updated by different teams over years, universal restriction becomes an essential tool for maintaining **semantic integrity** across the entire knowledge graph.

=== END ===

15.2 Combining `some` and `only` -- The Classic `VegetarianPizza` Pattern

At this stage, you may naturally being asking an important question:

> If existential restriction (`some`) and universal restriction (`only`) solve different semantic problems, can ontology combine them?

The answer is:

> **Yes -- and this is where ontology becomes significantly more powerful!**

In practical ontology engineering, meaningful concepts are rarely defined through **a single restriction.**

Instead, ontology begins expressing:

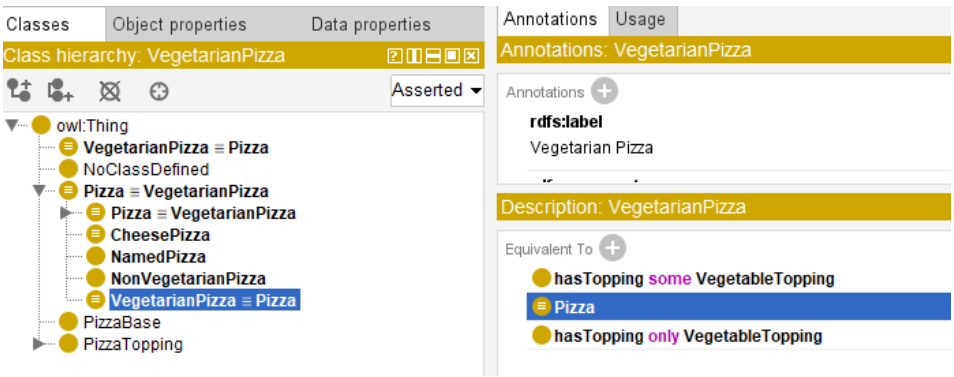
> **semantic identity through combinations of logical constraints.**

One of the most well-known examples appears in `VegetarianPizza` .

Consider the following OWL definition:

```
VegetarianPizza EquivalentTo

Pizza

and (hasTopping some VegetableTopping)

and (hasTopping only VegetableTopping)
```

Reflect into Protégé, as below setting:



At first glance, this expression may appear repetitive.

After all, both restrictions reference `VegetableTopping` .

However, each restriction performs a completely different semantic role.

- `hasTopping some VegetableTopping` ensures that **at least one** vegetable topping exists.
- `hasTopping only VegetableTopping` ensures that **every** topping must be a vegetable.

Neither alone is sufficient. Only the combination of `some` and `only` creates a complete semantic definition.

To understand why both are necessary, it would be helpful to first examine:

> why isolated restrictions are incomplete.

15.2.1 Why `some` Alone Is Not Enough?

Suppose ontology only defines `hasTopping some VegetableTopping` , this means:

> at least one vegetable topping must exist.

At first glance, this may seem sufficient to describing:

> a vegetarian pizza.

However, consider the following situation:

```
PizzaA hasTopping MushroomTopping
```

```
PizzaA hasTopping PepperoniTopping
```

Ontology evaluates `hasTopping some VegetableTopping` and immediately **succeeds.**

Why?

Because ontology finds:

> qualifying evidence.

The pizza contains `MushroomTopping` .

Then the existential condition therefore succeeds.

However, the pizza still contains `PepperoniTopping` , which clearly violates **vegetarian meaning.**

This illustrates an important ontology engineering lesson:

> existential restriction validates **participation** -- NOT semantic purity.

Or more simply, `some` ensures:

> something acceptable exists.

But it does **NOT** ensure:

> everything remains acceptable.

The reasoner asks:

> "Can qualifying evidence be found?"

It does **NOT** ask:

> "Are there any violations?"

Semantic meaning therefore remains incomplete.

15.2.2 Why `only` Alone Is Not Enough

Now consider the opposite situation.

Suppose ontology defines `hasTopping only VegetableTopping` .

This restriction states:

> if toppings exists, they must all belong to `VegetableTopping` .

At first glance, this appears semantically stronger.

However, a subtle logical problem emerges.

Image `PizzaB` with no toppings at all.

Surprisingly, ontology still consider `hasTopping only VegetableTopping` to be **satisfied.**

Why?

Because ontology searches for:

> violating evidence.

and in this case:

> no violating topping exists.

This follows the mathematical principle introduced earlier:

> **vacuous truth.**

In formal logic, a universal statement remains **true** unless:

> a counterexample exists.

Ontology therefore concludes:

> nothing violates the rule.

However, from a human perspective, an empty pizza may not intuitively feel like:

> a valid vegetarian pizza.

This highlights an important distinction:

> reasoners think **logically** -- **NOT** intuitively.

Universal restriction provides:

> semantic discipline.

But it does **NOT require participation.**

15.2.3 Why Logical Composition Matters

Ontology becomes significantly more expressive when both restrictions are combined.

Consider again:

```
VegetarianPizaa EquivalentTo

Pizza

and (hasTopping some VegetableTopping)

and (hasTopping only VegetableTopping)
```

Together, the restrictions now express two independent requirements:

1. **At least one vegetable topping must exist**, and
2. **All toppings must remain vegetable toppings**.

The first restriction guarantees:

> semantic participation.

The second restriction guarantees:

> semantic consistency.

Only together do they express:

> authentic vegetarian meaning.

Note

While, here I'd like to express the actual definition - from tutorial - for `VegetarianPizza` as `VegetarianPizza EquivalentTo Pizza and (hasTopping only (CheeseTopping or VegetableTopping))`

This reveals one of ontology engineering's most important principles:

> **meaning rarely emerges from a single rule.**

Instead:

> **meaning emerges from the interaction of multiple logical restrictions.**

A useful mental model can be thought as:

- `some` → participation
- `only` → discipline

Together, they produce:

> **semantic identity.**

Ontology therefore moves beyond:

> relationship storage

toward:

> governed conceptual meaning.

This is one of the reasons ontology differs fundamentally from traditional databases.

Databases typically store **facts**.

Ontology additionally governs **what facts mean**.

15.3 Implementing `some` and `only` in Graph Database - A Neo4j Perspective

At this point, you may reasonably ask:

> If existential and universal restrictions are so powerful, can similar logic be implemented in a graph database such as Neo4j?

The answer is:

> **Yes -- but with important limitations.**

This distinction is particularly important because many enterprise practitioners initially assume:

> **Neo4j = Ontology**

or:

> **Knowledge Graph automatically means semantic intelligence.**

In reality, graph databases and ontologies solve fundamentally different problems.

Neo4j primarily manages **connected data.**

Ontology additionally governs **logical meaning**. (note the "additionally" wording)

A graph database excels at:

- relationship storage,
- graph traversal, and
- connected querying.

However, Neo4j does not natively understand $\exists R.C$ or $\forall R.C$, nor does it automatically infer:

- class membership,
- semantic consistency, or,
- logical identity.

Instead, engineers must explicitly implement semantic behavior through:

> **Cypher queries.**

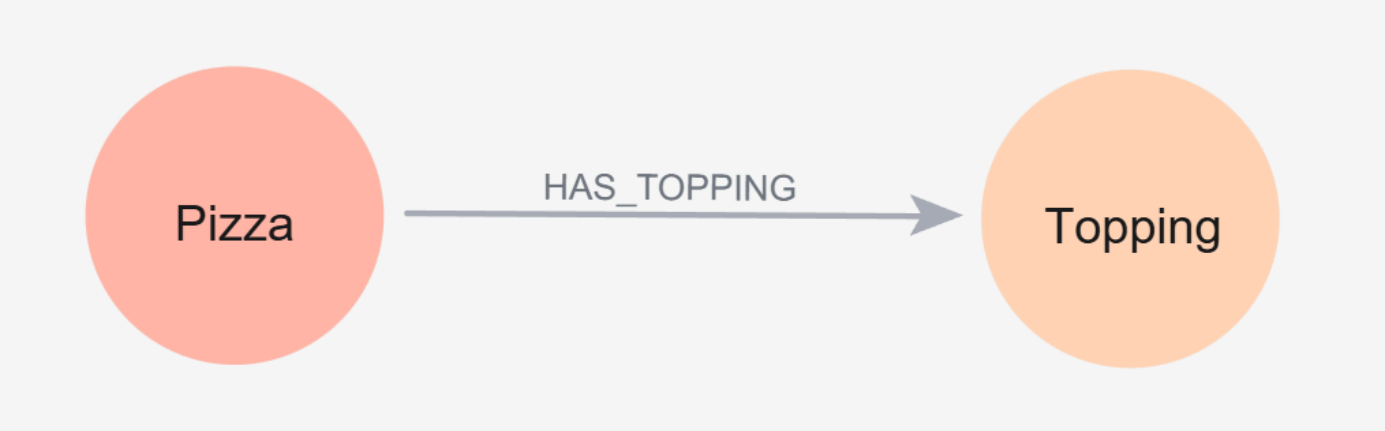
This becomes easier to understand using `VegetarianPizza`.

Suppose Neo4j contains below relationship:

```
(:Pizza)-[:HAS_TOPPING]->(:Topping)
```

You can use below Cypher to create these 2 nodes and 1 relationship (note: this is just for demo purpose, using below query you'll create two actual instances as well):

```
CREATE (:Pizza)-[:HAS_TOPPING]->(:Topping)
```

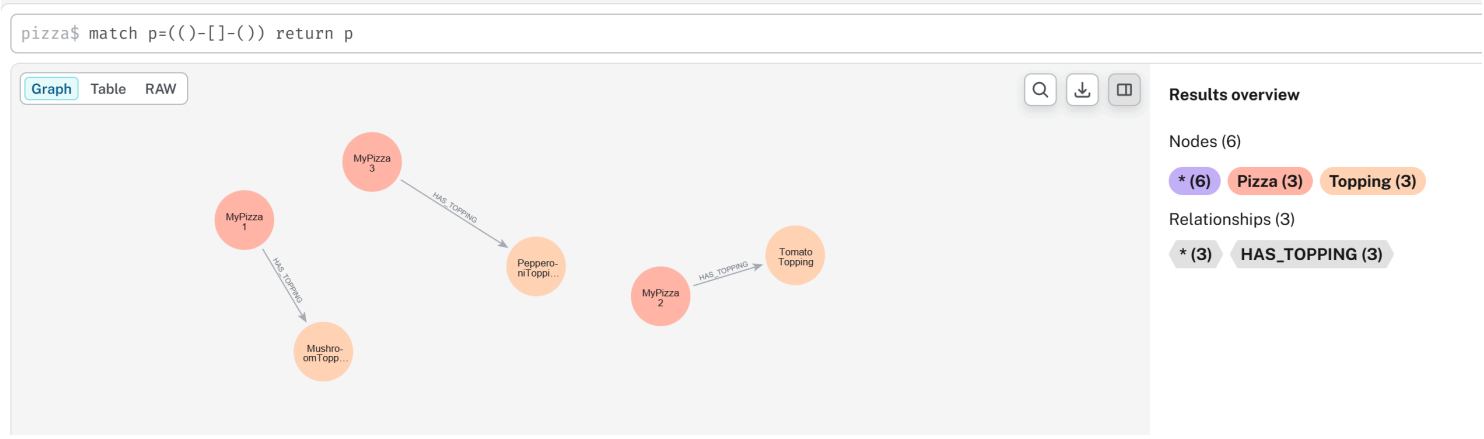


A pizza may connect to `MushroomTopping`, `TomatoTopping`, or `PepperoniTopping`.

Following Cypher queries are used to create these new relationships:

```
MERGE (p1:Pizza {name:"MyPizza1"})
MERGE (p2:Pizza {name:"MyPizza2"})
MERGE (p3:Pizza {name:"MyPizza3"})
MERGE (t1:Topping {name:"MushroomTopping"})
MERGE (t2:Topping {name:"TomatoTopping"})
MERGE (t3:Topping {name:"PepperoniTopping"})
MERGE (p1)-[:HAS_TOPPING]->(t1)
MERGE (p2)-[:HAS_TOPPING]->(t2)
MERGE (p3)-[:HAS_TOPPING]->(t3)
```

Then you may extract them as below:



Neo4j itself does not automatically determine:

> whether the pizza qualifies as vegetarian.

Instead, semantic interpretation must be manually implemented.

> Tip: You may find the working Neo4j dump database in ebook's *neo4j* subfolder.

15.3.1 Implementing Existential Restriction (`some`) in Neo4j

Recall earlier `hasTopping some VegetableTopping` means:

> at least one vegetable topping **must** exist.

In mathematical notation:

$$\exists \, \text{hasTopping}.\text{VegetableTopping}$$

This expression asks:

> "Can qualifying evidence be found?"

Inside Neo4j, this behavior can be approximated using:

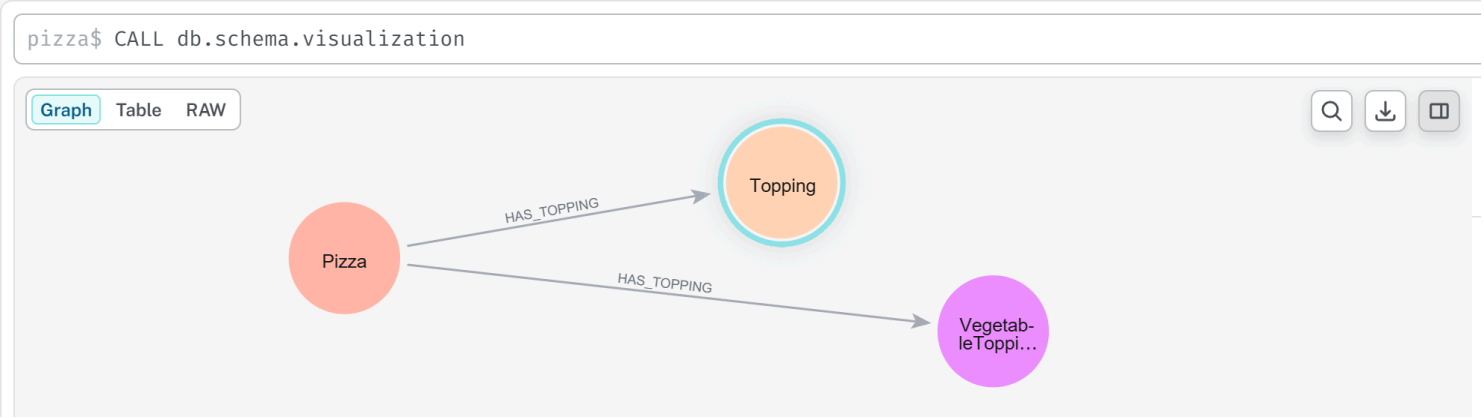
```
MATCH (p:Pizza)
WHERE EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(:VegetableTopping)
}
RETURN p.name AS Pizza
```

To practice above query, we need to add `VegetableTopping` as another node in Neo4j, then double assign `MushroomTopping` and `TomatoTopping` to this new node:

```
MATCH (n:Topping {name:"MushroomTopping"}), (m:Topping {name:"TomatoTopping"})
SET n:VegetableTopping
SET m:VegetableTopping
RETURN m, n LIMIT 25;
```

After above preparation, using `CALL db.schema.visualization` , you can see below schema:

```
pizza$ CALL db.schema.visualization
```



Now, you may execute previous Cypher query, and get result `MyPizza` :

```
1 MATCH (p:Pizza)
2 WHERE EXISTS {
3   MATCH (p)-[:HAS_TOPPING]->(:VegetableTopping)
4 }
5 RETURN p.name AS Pizza
```

Table RAW

Pizza
1 "MyPizza1"
2 "MyPizza2"

Notice the similarity.

The Cypher keyword `EXISTS` functions conceptually like:

> existential search.

Neo4j asks:

> "Does there exist at least one path to `VegetableTopping` ?"

If answer is Yes, the pizza satisfies the condition.

However, an important difference remains.

Neo4j executes **an explicit query.**

While ontology instead performs **automatic reasoning.**

Once semantic restrictions are defined, the reasoner continuously evaluates:

> logical consequences.

Not require engineers repeatedly writing validation logic like in Neo4j.

15.3.2 Implementing Universal Restriction (`only`) in Neo4j

Now consider: `hasTopping only VegetableTopping` .

In mathematical notation: $\forall\, hasTopping.VegetableTopping$.

Unlike existential restriction, universal restriction asks:

> "Are all relationships semantically valid?"

This creates an interesting challenge.

Neo4j has no direct equivalent for $\forall$, instead, universal semantic are typically approximated through:

> **violation detection.**

Rather than proving **everything is valid.**

Neo4j attempts to prove **nothing invalid exists** (or say "nothing is not invalid")

For example below Cypher query:

```
MATCH (p:Pizza)
WHERE NOT EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(t)
  WHERE NOT t:VegetableTopping
}
RETURN p.name AS VegetarianCandidate
```

You may see below result:

```
1 MATCH (p:Pizza)
2 WHERE NOT EXISTS {
3   MATCH (p)-[:HAS_TOPPING]->(t)
4   WHERE NOT t:VegetableTopping
5 }
6 RETURN p.name AS VegetarianCandidate
```

Table RAW

VegetarianCandidate
1 "MyPizza1"
2 "MyPizza2"

Why here still `MyPizza1` and `MyPizza2` ?

This query asks:

> "Can any invalid topping be found?"

If no violating topping exists, the pizza passes validation.

This logic closely resembles $\forall\ hasTopping.VegetableTopping$ where semantic correctness is evaluated through:

> absence of contradiction.

Notice how this aligns perfectly with the reasoning distinction discussed earlier:

- `some` $\longrightarrow$ for qualifying evidence
- `only` $\longrightarrow$ for violating evidence

Ontology and Cypher therefore share:

> similar logical goals.

But they implement them differently.

15.3.3 Combining `some` and `only` in Neo4j

To simulate:

```
VegetarianPizza EquivalentTo

Pizza

and (hasTopping some VegetableTopping)

and (hasTopping only VegetableTopping)
```

Neo4j must manually combine **both logical conditions.**

Below is the Cypher query:

```
MATCH (p:Pizza)
WHERE EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(:VegetableTopping)
}
AND NOT EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(t)
  WHERE NOT t:VegetableTopping
}
RETURN p.name AS VegetarianPizza
```

Result as below:

```
MATCH (p:Pizza)
WHERE EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(:VegetableTopping)
}
AND NOT EXISTS {
  MATCH (p)-[:HAS_TOPPING]->(t)
  WHERE NOT t:VegetableTopping
}
RETURN p.name AS VegetarianPizza
```

Table RAW

VegetarianPizza
"MyPizza1"
"MyPizza2"

This query approximates ontology semantics by enforcing:

1. At least one vegetable topping exists, and
2. No invalid topping exists

The result may appear similar to OWL reasoning.

However, the implementation philosophy remains fundamentally different.

Neo4j performs **procedural semantic checking.**

Ontology performs **declarative semantic reasoning**.

In Neo4j, engineers must repeatedly write logic describing:

> how validation should happen.

In ontology, meaning becomes:

> part of the model itself.

The reasoner automatically performs **inference**, **validation** and **classification**.

This reveals an important architectural distinction:

- Neo4j → relationship intelligence
- OWL → semantic intelligence

Or more precisely:

- Graph Database → connected facts
- Ontology → governed meaning

This distinction becomes increasingly important as enterprise knowledge graphs scale.

Without ontology, semantic logic often becomes scattered across:

- Cypher queries,
- applications,
- APIs, and
- business rules.

Over time, semantic governance becomes fragmented.

Ontology instead centralizes:

> meaning itself.

Meaning no longer lives only inside:

> application code.

It becomes:

> an explicit enterprise asset.

15.3.4 Cypher Query vs. Pellet Reasoner vs. Open World Assumption

Sections 15.3.1 through 15.3.3 established a practical bridge between OWL restrictions and Cypher queries in Noe4j:

- Section 15.3.1 demonstrated how to implement existential restriction (`some`) using pattern matching with `MATCH` and `WHERE` clauses.
- Section 15.3.2 showed the implementation of universal restriction (`only`) using `NOT EXISTS` to detect invalid relationships.
- Section 15.3.3 then combined both restrictions, illustrating how complex semantic requirements -- such as "a pizza must have at least one cheese topping and no non-pizza toppings" -- can be expressed as graph patterns.

However, this pedagogical bridge -- while valuable for initial understanding and hands-on experimentation -- carries a significant risk:

> it may obscure two foundational semantic differences that distinguish OWL reasoning from graph database querying.

First, as discussed in Chapter 14 (section 14.7.4), OWL reasoners operate under the **Open World Assumption (OWA)**, while Neo4j (like most property graph databases) operates under the **Closed World Assumption (CWA)**.

Second, and more critically, the Cypher implementations in 15.3.1~15.3.3 treat `some` and `only` as *query patterns* that return results, whereas an OWL reasoner treats them as *logical axioms* that derive new knowledge.

To build truly robust executable knowledge architectures, an ontology engineer must understand how *both* restriction types behave under *both* world assumptions, and how they contrast with the property graph querying paradigm demonstrated in the previous three sections.

This section provides a two-dimension comparative analysis using a unified scenario. We keep extending our `Pizza.owl` knowledge base to evaluate both:

- **Existential Restriction (`some`)**: `hasTopping some CheeseTopping` -- "At least one cheese topping must exist."
- **Universal Restriction (`only`)**: `hasTopping only CheeseTopping` -- "If toppings exist, they must all be PizzaToppings."

Example Scenario Definition

Ontology Definition:

```
Existential Rule (some):
  CheesePizza EquivalentTo: Pizza and (hasTopping some CheeseTopping)

Universal Rule (only):
  CleanPizza EquivalentTo: Pizza and (hasTopping only PizzaTopping)
```

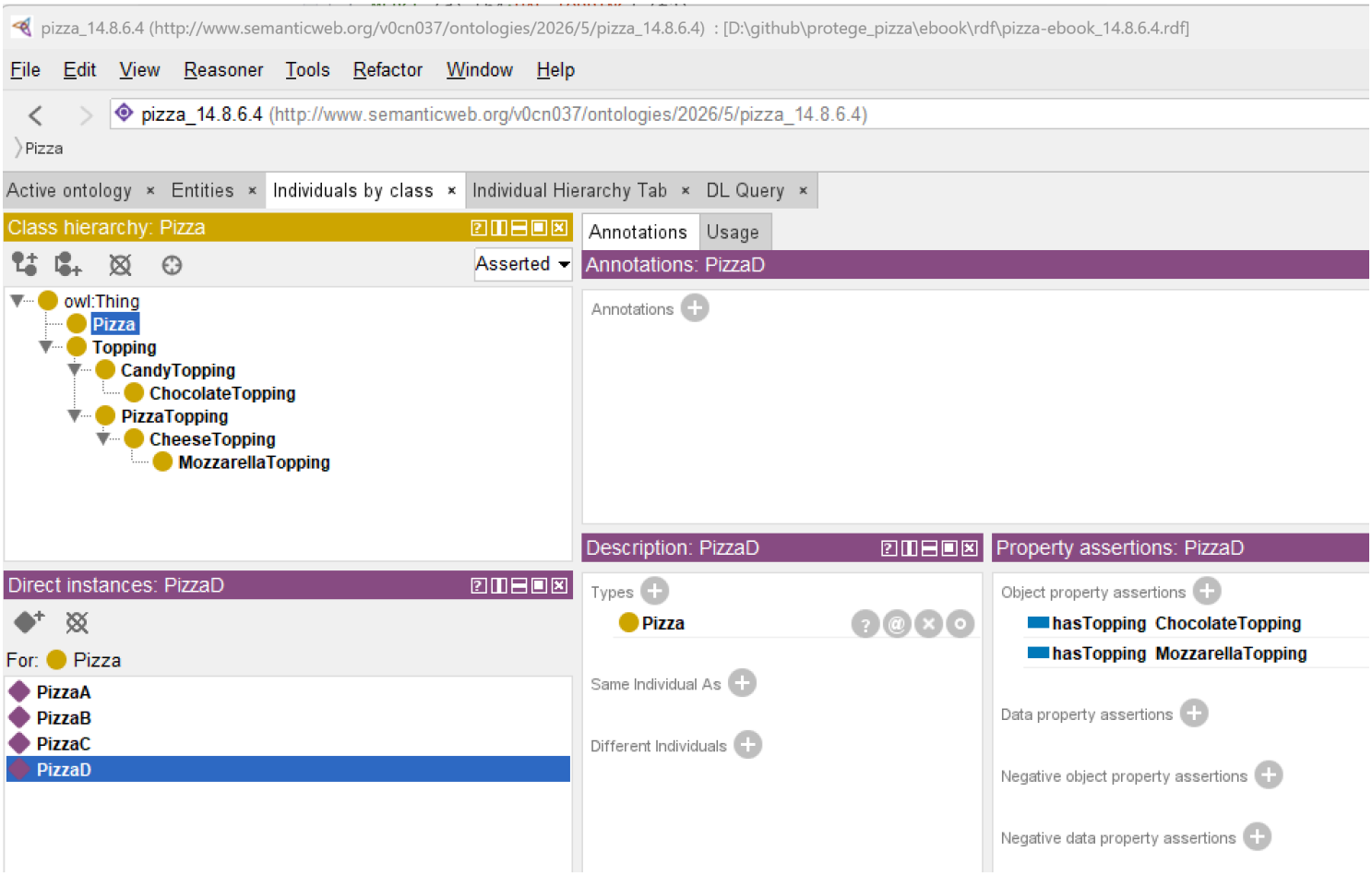
Knowledge Base (Asserted Facts):

Individual	Asserted Type	Asserted Relationships	Class Hierarchy
PizzaA	`Pizza`	`hasTopping MozzarellaTopping`	`MozzarellaTopping` ⊑ `CheeseTopping` ⊑ `PizzaTopping`
PizzaB	`Pizza`	*(no `hasTopping` assertions)*	---
PizzaC	`Pizza`	`hasTopping ChocolateTopping`	`ChocolateTopping` ⊑ `CandyTopping` (`CandyTopping` ⊓ `PizzaTopping` = `owl:Nothing`)
PizzaD	`Pizza`	• `hasTopping MozzarellaTopping` • `hasTopping ChocolateTopping`	(As above)

Initialize ontology in Protégé:

New working ontology file: `/ebook/rdf/pizza-ebook_15.3.4.rdf` .

Base on above asserted facts, you may find following initialized knowledge:

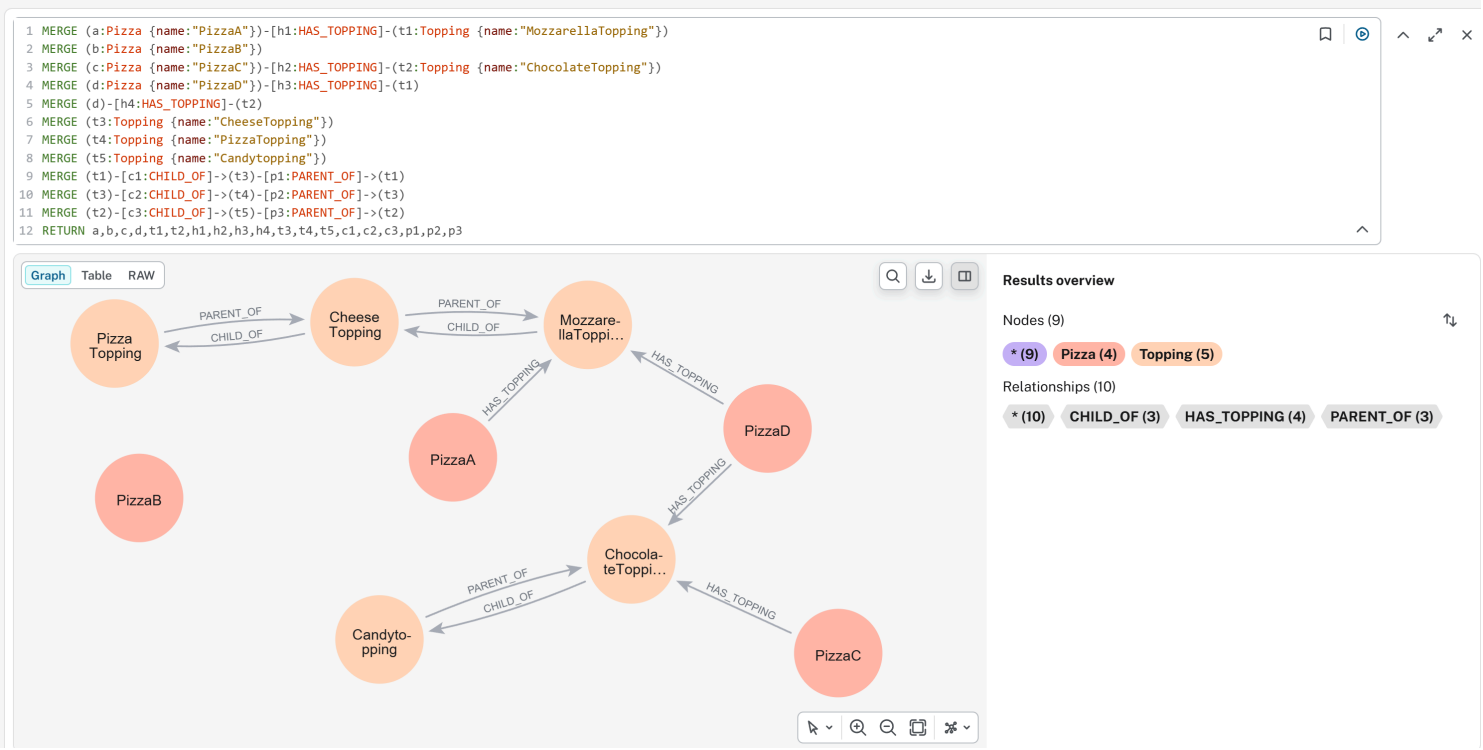


- Set `Disjoint With` between `PizzaTopping` and `CandyTopping`
- Set proper Domain and Range to relationship `hasTopping`
- Create necessary instances for building up the relationships.

Initialize graph database in Neo4j:

Using following Cypher query to create all asserted facts:

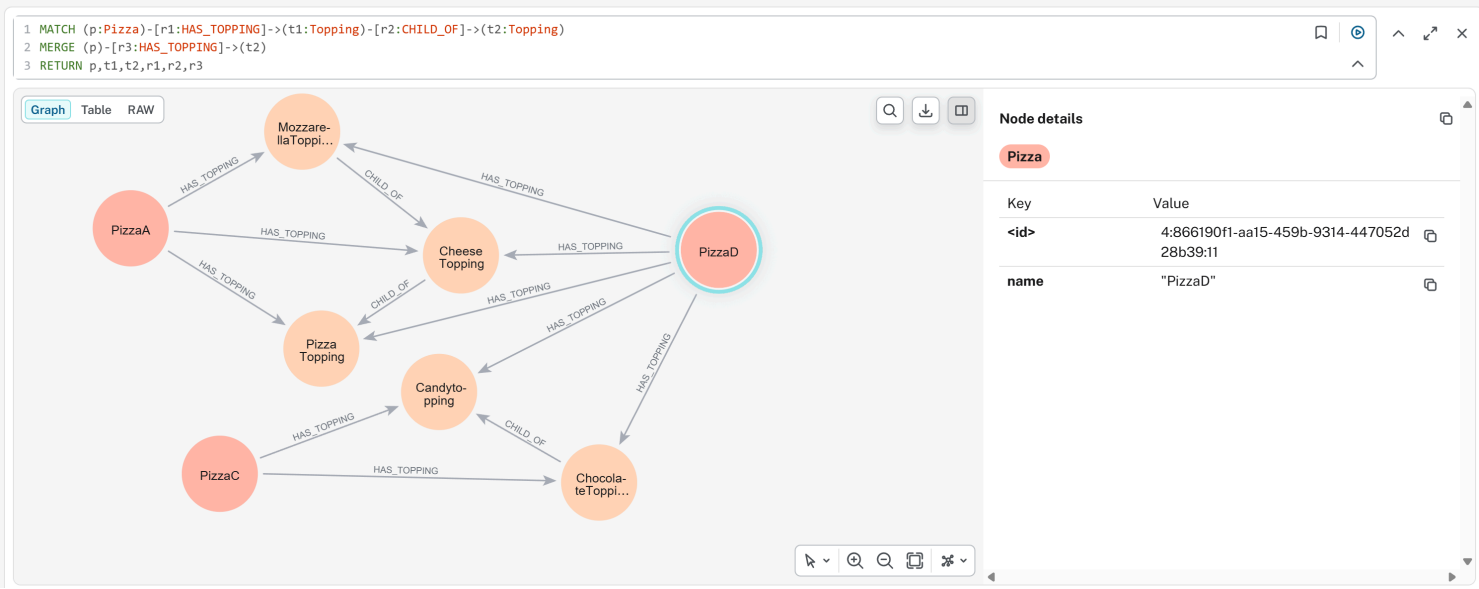
```
MERGE (a:Pizza {name:"PizzaA"})-[h1:HAS_TOPPING]-(t1:Topping {name:"MozzarellaTopping"})
MERGE (b:Pizza {name:"PizzaB"})
MERGE (c:Pizza {name:"PizzaC"})-[h2:HAS_TOPPING]-(t2:Topping {name:"ChocolateTopping"})
MERGE (d:Pizza {name:"PizzaD"})-[h3:HAS_TOPPING]-(t1)
MERGE (d)-[h4:HAS_TOPPING]-(t2)
MERGE (t3:Topping {name:"CheeseTopping"})
MERGE (t4:Topping {name:"PizzaTopping"})
MERGE (t5:Topping {name:"CandyTopping"})
MERGE (t1)-[c1:CHILD_OF]->(t3)-[p1:PARENT_OF]->(t1)
MERGE (t3)-[c2:CHILD_OF]->(t4)-[p2:PARENT_OF]->(t3)
MERGE (t2)-[c3:CHILD_OF]->(t5)-[p3:PARENT_OF]->(t2)
RETURN a,b,c,d,t1,t2,h1,h2,h3,h4,t3,t4,t5,c1,c2,c3,p1,p2,p3
```



- Use two direction `PARENT_OF` and `CHILD_OF` for inverse class hierarchical structure

Since we have maximum two levels class hierarchy structure, to simulate the Protégé's Parent/Child inferencing behavior into Neo4j, run below Cypher query twice:

```
MATCH (p:Pizza)-[r1:HAS_TOPPING]->(t1:Topping)-[r2:CHILD_OF]->(t2:Topping)
MERGE (p)-[r3:HAS_TOPPING]->(t2)
RETURN p,t1,t2,r1,r2,r3
```



Note: here we don't see `PizzaB` since it has no topping relationship.

Part 1: Existential Restriction (`some`) -- "At Least One Must Exist"

Goal: Determine whether each individual satisfies `hasTopping some CheeseTopping` (i.e., is a `CheesePizza`).

Recall from 15.3.1 that the Cypher implementation of `some` uses a single pattern match:

```
MATCH (p:Pizza)-[:HAS_TOPPING]->(t1:Topping)
WHERE toLower(t1.name) CONTAINS "cheese"
RETURN p
```

You get `PizzaA` and `PizzaD`

Note: we don't create `CheeseTopping` as another node in Neo4j, it's one instance of `Topping` node and we use `WHERE` clause to filter that.

Now compare this closed-world query against OWL open-world reasoning:

Individual	Cypher (CWA) - per 15.3.1	Pellet Reasoner (OWA)	OWA Formal Conclusion
PizzaA	✅ **Returned** - Pattern matches	✅ **Inferred as** `CheesePizza` - Evidence found	**True** - Sufficient evidence exists
PizzaB	❌ **Not returned** - No matching relationship	❌ **Not inferred** - Also not inferred as $\neg CheesePizza$	**Unknown** - Missing information; future facts could change this
PizzaC	❌ **Not returned** - Target is not `CheeseTopping`	❌ **Not inferred as** `CheesePizza` - Condition false for known relationship	**False** - Evidence exists and definitively fails the condition
PizzaD	✅ **Returned** - At least one relationship satisfies pattern	✅ **Inferred as** `CheesePizza` - `MozzarellaTopping` satisfies condition	**True** - One valid relationship is sufficient; invalid relationships are irrelevant

Key Insight for `some` :

- Under OWA, the existential restriction is *monotonic*. Once satisfied (`PizzaD`), it returns satisfied regardless of additional assertions. The reasoner searches for *at least one piece of validating evidence.*
- The Cypher query from 15.1.6.1 and customized here correctly identifies `PizzaA` and `PizzaD` , but it cannot distinguish `PizzaB` (unknown) from `PizzaC` (false) -- both simply "not returned". This conflation is the primary limitation of the closed-world emulation.

Part 2: Universal Restriction (`only`) -- "Everything Must Conform"

Goal: Determine whether each individual satisfies `hasTopping only PizzaTopping` (i.e., has no topping outside the `PizzaTopping` class).

Recall from 15.3.2 that the Cypher implementation of `only` uses `NOT EXISTS` to detect violations:

```
MATCH (p:Pizza)
WHERE NOT EXISTS {
  (p)-[:HAS_TOPPING]->(t:Topping)
  WHERE NOT toLower(t.name) CONTAINS "pizza"
}
RETURN p
```

Result is `PizzaB` in the Neo4j graph; however, our testing graph doesn't return `PizzaA` due to we don't complicated query to check Parent/Child context here, in reality, `PizzaA` si also fulfill the CWA.

Now compare this against OWL open-world reasoning:

Individual	Cypher (CWA) - per 15.3.2	Pellet Reasoner (OWA)	OWA Formal Conclusion
PizzaA	✅ **Returned** - No invalid topping found	✅ **Satisfies restriction** - All relationships valid	**True** - All asserted relationships are valid
PizzaB	✅ **Returned (vacuously)** - See vacuous truth discussion in 15.1.3	**True (vacuously)** - No counterexample exists	
PizzaC	❌ **Not returned** - Invalid topping detected	❌ **Violates restriction** - Counterexample found	**False** - Definitive counterexample exists
PizzaD	❌ **Not returned** - The `ChocolateTopping` is invalid, even though `MozzarellaTopping` is valid	❌ **Violates restriction** - Universal quantifier requires *all* relationships to be valid	**False** - A single counterexample falsifies universal condition

Key Insight for `only`:

- Under OWA, the universal restriction is *falsified by a single counterexample*. The reasoner searches for *any violating evidence*.
- The Cypher query from 15.3.2 correctly identifies valid pizzas (A and B) -- has explained result of A -- and excludes invalid ones (C and D). However, like the `some` case, it conflates `PizzaB` (vacuously true) with `PizzaA` (genuinely true), and cannot represent the logical distinction between "no toppings yet" and "has only valid toppings."

Part 3: Combining `some` and `only` -- The Composite Pattern

Section 15.3.3 demonstrated how to combine both restrictions in a single Cypher query, identifying pizzas that satisfy *both* conditions: at least one cheese topping AND no non-pizza toppings.

```
MATCH (p:Pizza)
WHERE (p)-[:HAS_TOPPING]->(:CheeseTopping)
AND NOT EXISTS {
  (p)-[:HAS_TOPPING]->(t:Topping)
  WHERE NOT toLower(t.name) CONTAINS "pizza"
}
RETURN p
```

Note: it should have `PizzaA` returned.

The following table shows the integrated results across all four individuals, comparing the Cypher composite query against OWL reasoning for the combined definition:

Combined Definition:

```
GoodPizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)

and (hasTopping only PizzaTopping)
```

Individual	Cypher (CWA) - per 15.3.3	Pellet Reasoner (OWA)	OWA Formal Conclusion
PizzaA	✅ **Returned** - Has cheese topping AND all toppings valid	✅ **Inferred as GoodPizza** - Both conditions satisfied	**True** - Satisfies both existential and universal requirements
PizzaB	❌ **Not returned** - Fails `some` condition (no cheese topping)	❌ **Not inferred** - `some` unknown; `only` vacuous but insufficient	**Unknown** - Mising information for condition
PizzaC	❌ **Not returned** - Fails `some` (no cheese topping) AND fails `only` (invalid topping)	❌ **Not inferred** - `some` false, `only` false	**False** - Definitive failure on both dimensions
PizzaD	❌ **Not returned** - Has cheese topping (passes `some`) BUT fails `only` (invalid topping exists)	❌ **Not inferred as GoodPizza** - `only` condition violated	**False** - Universal restriction fails despite existential satisfaction

Critical Observation - `PizzaD`:

- This individual reveals the non-complementary nature of `some` and `only`.
- A pizza can simultaneously satisfy an existential restriction (it has at least one cheese topping) *and* violate a universal restriction (it also has a non-pizza topping).
- In the composite query from 15.3.3, `PizzaD` is correctly excluded.
- However, an OWL reasoners provides additional insight: `PizzaD` *is* a `CheesePizza` (from Part 1) but is *NOT* a `GoodPizza` (from Part 3).
- This distinction -- that an individual can belong to one inferred class but not another -- is a form of logical nuance that Cypher's flat pattern matching CANNOT express without explicitly modeling both classifications.

Part 4: Integrated Comparison - `some` vs `only` Across Both Worlds

The following table integrates both restriction types, showing the classification outcomes for each individual under each condition, comparing Cypher (from 15.3.1~15.3.3) against OWL reasoning:

Individual	`some CheeseTopping` (Cypher/CWA)	`some CheeseTopping` (Pellet/OWA)	`only PizzaTopping` (Cypher/CWA)	`only PizzaTopping` (Pellet/OWA)
PizzaA	✅ True	✅ True	✅ True	✅ True
PizzaB	❌ False	❓ Unknown	✅ True	✅ True (vacuous)
PizzaC	❌ False	❌ False	❌ False	❌ False
PizzaD	✅ True	✅ True	❌ False	❌ False

Critical Observation -- the Unknown Column:

- The most significant difference between the Cypher columns and the OWA columns appears in `PizzaB` for the `some` condition.
- Cypher returns `False` (the individual is excluded from results), while OWL returns `unknown` (the individual cannot be classified yet).
- The difference is not a bug or an implementation detail -- it is the direct consequence of the Open World Assumption.
- In enterprise scenarios where data arrives incrementally (e.g., a pizza order system where toppings are added over time), `Unknown` is often the correct and more useful answer than `False`.

Part 5: Why the Difference Matters - From Query to Reasoning

The Cypher implementations in 15.3.1~15.3.3 are *operational filters*. They answer the questions:

> "Given the data currently in the graph, which individuals match my explicit pattern?"

OWL reasoning, by contrast, answers a deeper question:

> "Given the logical axioms defined in the ontology and the data currently available, what class memberships can be *necessarily* inferred -- and which remain *possibly* true?"

Dimension	Cypher in Neo4j (15.3.1~15.3.3)	OWL Reasoner (e.g. Pellet)
Work Assumption	Closed World: Missing = False	Open World: Missing = Unknown

Dimension	Cypher in Neo4j (15.3.1~15.3.3)	OWL Reasoner (e.g. Pellet)
Restriction Type	Written as explicit `MATCH` or `NOT EXISTS` patterns	Declared as logical axioms in the ontology
Output	Query result set (individuals matching pattern)	Inferred class memberships (new `rdf:type` assertions)
Handling of `PizzaB`	Excluded from `some` query (False)	Not classified as `CheesePizza`, but not classified as ¬ `CheesePizza` either (Unknown)
Composability	Manual composition via `AND` / `OR` in query	Automatic composition via logical conjunction in class expression
Maintenance	Query logic lives in application code	Logical axioms live in ontology; all queries benefit automatically

Part 6: Practical Guidance - When to Use Each Approach

Scenario	Recommended Approach	Rationale
Real-time filtering on stable complete data	Use Cypher patterns (15.3.1~15.3.3) directly on Neo4j	• High performance • Closed world assumption is valid when data is known to be complete
Incremental data integration from multiple sources	Use OWL reasoner + materialize inferred types to Neo4j	• Open world prevents premature false negatives • Reasoner handles missing data gracefully
Data quality validation (missing required fields)	• Use `some` restriction with OWL reasoner • Query for individuals *not* satisfying the restriction	Under OWA, missing data flags as "incomplete" (Unknown), not "invalid" (false) -- more actionable for data stewardship
Compliance checking (forbidden relationships)	• Use `only` restriction with OWL reasoner • Any violation yields definitive `False`	• Universal restriction catch type violations directly • No ambiguity
Detecting "partially correct" data (like `PizzaD`)	• Combine both restrictions • Query for individuals satisfying `some` but failing `only`	Identifies data that has required components but also contains invalid elements -- often signals need for human review
Production dashboards on stable data	Run Cypher queries on a materialized inference graph	First use reasoner to compute all inferred types, then export to Neo4j for high-performance operational queries

Summary: The Logical Bridge Between Sections 15.3.1~15.3.3 and True OWL Reasoning

Let's take a breathe before moving to next section.

> The Cypher patterns in 15.3.1~15.3.3 answers:
> "What matches my explicit patter *right now* in this **closed** graph?"

> An OWL reasoner answers:
> "What *must be true* given my logical axioms and available data -- and what *remains possibly true* -- under open-world semantics?"

The implementations in 15.3.1 (`some` as `MATCH`), 15.3.2 (`only` as `NOT EXISTS`), and 15.3.3 (composite patters) are excellent pedagogical (and self-learning) tools.

They allow developers familiar with graph databases to experience the *filtering effect* of OWL restrictions without immediately learning description logic.

However, as this section has demonstrated, these Cypher patterns are *simulations*, not equivalents.

- They cannot represent the open-world distinction between `False` and `Unknown` (`PizzaB` for `some`),
- They cannot automatically derive new classifications without being explicitly written for each query, and
- They cannot capture the logical nuance that an individual like `PizzaD` can be a `CheesePizza` (satisfying `some`) while not being a `GoodPizza` (failing `only`).

Understanding these differences -- mastering the distinct behaviors of `some` and `only` under both closed-world and open-world assumptions -- is a fundamental maturity milestone in ontology engineering.

As you move from writing Cypher queries (sections 15.3.1~15.3.3) to deploying OWL reasoners in executable knowledge architectures (EKA), this distinction will determine whether your semantic system merely filters data or truly understands it.

Congratulations for you to read through this chapter until here, this is a big long journey for myself as well to make existential and universal restriction clearer. If you feel any parts uncertain, read again and hands-on practice with your Protégé and Neo4j.

15.4 Chapter 15 -> 16 Bridge

You can now build both halves of a semantic requirement, combine them into the classic `some` + `only` pattern that correctly and reasoner-verifiably defines `VegetarianPizza`, and you have seen the same requirement implemented twice -- once as OWL restrictions under the Open World Assumption, once as Cypher constraints under Neo4j's Closed World Assumption.

What remains is to step back and ask what all of this means for the larger architecture. Chapter 16 places property restrictions inside the full EKA five-layer model, walks through a milestone self-assessment, catalogs the most common mistakes learners make with restrictions, and closes out this three-chapter unit before the book moves on to subclass creation.

15.5 Key Concepts (Chapter 15)

Concept	Description
Universal Restriction (`only`)	Constrains a class so that *all* values of a given property must belong to a specified class.
Vacuous Truth	The logical principle by which `only` restrictions are automatically satisfied when no relationship exists at all.
Composite Restriction (`some` + `only`)	The combination pattern required to both require and bound a relationship -- the correct way to define `VegetarianPizza`.
Closed World Assumption (CWA)	The reasoning stance (used by Neo4j/Cypher) in which absence of a statement means "false."
OWA vs. CWA	The core philosophical divide between OWL reasoning and typical property-graph querying.

15.6 Protégé Skills Learned (Chapter 15)

- Creating a universal restriction (`only`) on a class in Protégé
- Combining `some` and `only` restrictions into a single composite class definition
- Recognizing vacuous truth in reasoner output
- Translating OWL restriction semantics into equivalent Cypher query logic for Neo4j

15.7 Next Chapter (16) Preview

Chapter 16 maps everything learned across Chapters 14 and 15 onto the EKA five-layer model (K, R, Θ, Φ, Γ), offers a self-assessment checklist, reviews the most common restriction-modeling mistakes (including the closed-world trap and `EquivalentTo` vs. `SubClassOf` confusion), and closes this unit before moving on to subclass creation.

Last updated at: 2026-09-11

Chapter 16 -- Restriction as Governance: EKA Synthesis, Common Mistakes, and Unit Summary

> **Editor's note:** This chapter is Part 3 of a three-chapter unit on OWL property restrictions (originally drafted as a single Chapter 14; see Chapter 14 for existential restriction and Chapter 15 for universal restriction, composition, and the Neo4j comparison).

Chapter Introduction

Chapters 14 and 15 gave you the mechanics: existential restriction (`some`), universal restriction (`only`), their composition, and a working comparison against Neo4j's Closed World query model. This chapter steps back from mechanics to meaning.

You will place property restrictions inside the full EKA five-layer model, take a milestone self-assessment to confirm the ideas have actually landed, review the most common mistakes learners make when modeling restrictions, and then close out this three-chapter unit with a consolidated summary, key concepts list, and skills checklist covering all of Chapters 14-16.

Table of Contents:

16.1 EKA Perspective -- Restriction as Semantic Governance

Throughout this chapter, you explored one of the most important capabilities in OWL ontology engineering:

> **Property Restriction.**

At first glance, restrictions may appear to be merely:

> OWL syntax,

or:

> Protégé modeling techniques.

Expressions such as:

`hasTopping some CheeseTopping` or `hasTopping only VegetableTopping`

may initially feel like technical notation designed only to support:

> ontology classification.

However, from the perspective of:

> **Executable Knowledge Architecture (EKA)**,

property restrictions represent something far more significant.

They transform ontology from:

> a passive relationship model

into:

> **governed executable semantic logic.**

This distinction matters.

Because knowledge graphs alone rarely produce:

> meaningful and trustworthy intelligence.

A graph can successfully connect:

`Pizza` $\longrightarrow$ `hasTopping` $\longrightarrow$ `Mushroom` ,

yet, this relationship alone cannot explain:

- whether the pizza should be classified as vegetarian,
- whether semantic expectations are satisfied,
- whether governance policies/rules are violated, or
- whether downstream actions should occur.

In other words, relationships create:

> connected knowledge.

Restrictions create:

> **meaningful semantic behavior.**

Seen through the EKA perspective, property restrictions become an essential bridge between:

> semantic knowledge

and:

> executable intelligence.

Recall the formal EKA tuple introduced in Chapter (00):

$$EKA = (K, R, \Theta, \Phi, \Gamma),$$

where:

- K = Knowledge Graph layer
- R = Reasoning & Rules layer
- Θ = Trigger layer
- Φ = Execution layer
- Γ = Governance layer

Property restrictions contributes directly across every one of these layers.

16.1.1 Restriction and the Knowledge Graph Layer (K)

The first contribution appears in:

> **K - Knowledge Graph layer.**

Recall from Chapter (00):

> K represents a set of entities (nodes) and semantic relationships (edges) conforming to an ontology.

Without restrictions, knowledge graphs primarily describe:

> connected facts.

For example:

```
PizzaA hasTopping MozzarellaTopping
```

simply tell us **a relationship exists.**

This is valuable, but semantically incomplete.

The graph itself cannot determine:

- whether this pizza qualifies as a `CheesePizza`,
- whether toppings violate semantic expectations, or
- whether business meaning should change.

Property restrictions enrich the knowledge graph by adding:

> semantic interpretation.

The graph stops being merely:

> connected information.

Instead, it becomes:

> semantically meaningful knowledge.

For example:

```
CheesePizza EquivalentTo Pizza and (hasTopping some CheeseTopping)
```

adds semantic significance to the graph.

Now, the presence of `MozzarellaTopping` may imply `CheesePizza`.

This marks an important shift.

Knowledge graph becomes **inferable.**

Rather than **manually asserted.**

Within EKA, restrictions therefore improve:

> semantic richness inside the Knowledge Graph layer.

16.1.2 Restriction and the Reasoning & Rules Layer (R)

The **strongest contribution** of Chapters 14–16 appears in:

> R **- Reasoning & Rules layer.**

As defined in Chapter (00):

> R **represents a set of inference rules (OWL, SWRL, or custom rules) that derive new facts from** K**.**

This chapter introduced one of the most important mechanisms for enabling reasoning:

> **property restriction logic.**

Existential restriction `hasTopping some CheeseTopping` behaves as:

> a reasoning rule.

It establishes a logical condition:

> if evidence exists, then semantic consequences may follow.

Similarly,

universal restriction `hasTopping only VegetableTopping` acts as:

> a semantic rule boundary.

It contains:

> what relationships remain logical acceptable.

Together these two restrictions become:

> executable semantic rules.

Unlike procedural code, restrictions express *what* must be true, not *how* to check it.

The reasoner no longer simply stores information.

Now it evaluates:

> logical meaning!

Let's see an example, suppose ontology contains:

```
CheesePizza EquivalentTo

Pizza

and (hasTopping some CheeseTopping)
```

and the graph contains `PizzaA hasTopping MozzarellaTopping`

with:

```
MozzarellaTopping SubClassOf CheeseTopping
```

The reasoner may automatically infer `PizzaA rdf:type CheesePizza` without manual classification.

This is an important maturity shift.

Ontology transitions from:

> descriptive modeling

toward:

> executable semantic logic.

This is precisely the purpose of:

> R **- Reasoning & Rules**

Knowledge begins producing:

> **NEW** knowledge.

16.1.3 Restriction and the Trigger Layer (Θ)

A less obvious, but highly important contribution appears in:

> Θ **- Trigger layer.**

Chapter (00) defines:

> Θ **as a set of conditions (semantic events, queries, or time-based triggers) that initiate execution.**

This perspective becomes particularly interesting when viewed through:

> ontology restrictions.

Restrictions can behave as **semantic trigger conditions.**

Restrictions enable the execution layer to distinguish between ‘data missing’ (treatable) vs. ‘semantic violation’ (escalatable).

Recall different Pizzas we have modeled with restrictions in previous sections, putting them in below execution decision matrix bases on reasoning outcomes:

Scenario	Data State (K)	Reasoning Conclusion (R)	Execution Action (Θ)
`PizzaA`	Has cheese topping	`CheesePizza` (True)	Normal processing
`PizzaB`	No topping information	Unknown	Trigger data quality workflow, request completion automatically

Scenario	Data State (K)	Reasoning Conclusion (R)	Execution Action (Θ)
`PizzaC`	Has non-cheese topping	`Not CheesePizza` (False)	Reject / Alert / Route to manual review

Note: The Θ (Trigger) layer is responsible for emitting signals based on reasoning outcomes, while the Φ (Execution) layer carries out the corresponding actions. (Or in short: Θ signals, Φ executes)

Furthermore, let's switch to enterprise environment, suppose an enterprise ontology defines:

```
HighRiskServer EquivalentTo

Server

and (hasExposure some CriticalVulnerability)
```

The moment semantic conditions becomes satisfied:

> the trigger activates.

In our `Pizza.owl` ontology terms, the reasoner may infer `PizzaA rdf:type VegetarianPizza` once semantic conditions are met.

In enterprise architecture, the same pattern becomes significantly more powerful.

A semantic trigger might initiate when:

- a customer becomes high-value,
- a server becomes high-risk,
- a process violates governance rules, or
- a compliance breach is detected.

The important idea here is:

> restrictions create semantic conditions.

These semantic conditions later become **executable triggers.**

Ontology therefore move beyond:

> classification,

toward:

> event-driven semantic behavior.

16.1.4 Restriction and the Execution Layer (Φ)

Once triggers activate, knowledge may begin producing:

> actions.

This connects directly to:

> **Φ - Execution layer.**

Chapter (00) defines:

> Φ as a set of actions (API calls, process invocations, alerts, graph updates) that change the real world or the knowledge graph.

This is where ontology begins becoming:

> executable.

For example, suppose ontology infer `HighRiskServer` through property restrictions (see 16.1.2).

This classification may automatically trigger:

- a security alert,
- a ServiceNOW (or other ticketing system) incident registration,
- an API invocation, or
- an automated remediation workflow.

In Neo4j and enterprise knowledge graphs, ontology reasoning may therefore move beyond:

> semantic understanding

toward:

> operational execution.

Even in the `Pizza.owl` ontology example, we can imagine semantic outcomes.

A food recommendation engine might automatically suggest:

> vegetarian menu options

after ontology infer `VegetarianPizza` .

This may sound simple.

Yet the architectural implication is significant.

Restrictions are no longer merely:

> logic constraints.

They become:

> executable decision conditions.

Ontology therefore evolves from:

> passive knowledge

toward:

> actionable (proactive) intelligence.

16.1.5 Restriction and the Governance Layer (Γ)

Finally, property restriction contribute strongly to:

> Γ **- Governance layer.**

Chapter (00) defines:

> Γ **as constraints, validation rules (SHACL), and access policies that ensure semantic integrity.**

This is one of the most important contributions of:

> universal restriction (`only`).

In earlier sections of this chapter, we observed `hasTopping only VegetableTopping` acting as:

> a semantic boundary.

Rather than merely storing relationships, ontology begin governing:

> acceptable meaning!

This becomes increasingly important as enterprise knowledge systems scale.

Because semantic drift naturally emerges.

Relationship continue existing, yet meaning gradually becomes corrupted.

For example, an enterprise graph may accidentally contain:

```
Employee belongsToDepartment Server01
```

Technically, the relationship exists.

The database accept it and queries still function.

Yet, **semantically**, the meaning becomes invalid.

Restrictions help prevent **semantic pollution**, before inconsistent meaning propagation into:

- reporting,
- automation,
- analytics, or
- decision-making systems.

This emphasis on semantic governance is increasingly recognized beyond ontology engineering communities.

For example, the emerging idea of the:

> **Semantic Data Charter (SDC)**

highlights the importance of:

- trustworthy semantics,
- explainable knowledge,
- embedded semantic consistency, and
- responsible incremental governance of enterprise semantic ecosystems.

As discussed in the foreword by Timothy Cook, organizations must move beyond:

> merely connected data

toward:

> governed and trustworthy meaning.

This aligns closely with Γ **- Governance** within EKA.

Readers interested in broader semantic governance practices may explore Semantic Data Charter for additional perspectives.

Ultimately, governance ensures semantic systems remain:

- trustworthy,
- explainable, and
- operationally reliable.

16.1.6 Restriction as the Foundation of Executable Intelligence

When viewed across the complete EKA perspective:

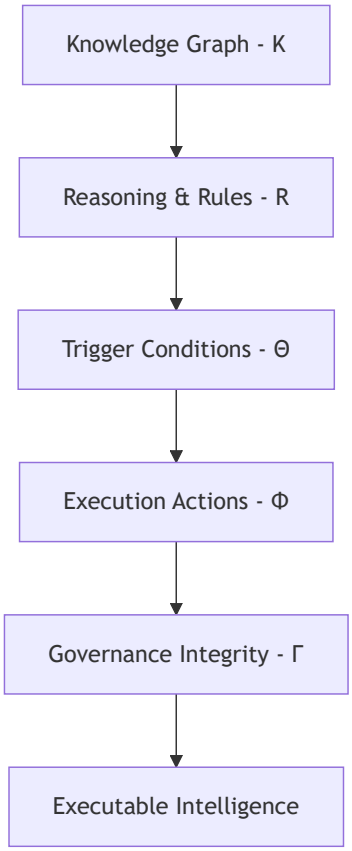
property restrictions (existential, universal as discussed) become much more than:

> OWL syntax.

They actually become:

> **semantic execution logic.**

The progression introduced throughout this chapter may be summarized as below end-to-end EKA implementation roadmap:



Chapters 14–16 represents an important maturity milestone in ontology engineering.

Earlier chapters focused on:

- concepts,
- classes,
- properties
- relationships, and
- semantic structure.

This chapter introduced something fundamentally different:

> **behavioral semantics.**

Ontology no longer merely describes:

> what exists,

it increasingly governs:

> what should happen.

This transition marks the beginning of:

> executable knowledge.

By this stage, you are no longer simply modeling ontology.

You are beginning to engineer:

> **governed executable semantics.**

16.2 EKA Learning Progression Across Chapters

16.2.1 A Milestone Worth Recognition

Before continuing to the final sections of this chapter, it is worth recognizing something important:

> **Chapters 14–16 are intentionally more extensive than previous chapters.**

By now, you have likely noticed that this chapter introduced:

- significantly more logical notation,
- more conceptual explanation,
- deeper semantic discussion, and
- more detailed reasoning examples.

This was deliberate.

Because Chapters 14–16 represents one of the most important conceptual transitions in ontology engineering.

Earlier chapters primarily focused on:

- concepts,
- classes,
- relationships,
- hierarchy,
- individuals, and
- semantic structure.

These ideas are foundational.

However, property restrictions introduce something fundamentally different:

> **formal semantic logic.**

For the first time, ontology modeling moves beyond:

> describing what exists.

Instead, you are beginning to model:

- what must exist,
- what is allowed to exist,
- what can be inferred, and
- what logical conditions govern semantic meaning.

This is also where ontology starts becoming:

> executable!

In many ways, Chapters 14–16 marks the transition from:

> "Learning Protégé"

toward:

> **thinking like an ontology engineer.**

Concepts such as:

- existential restriction (`some`),
- universal restriction (`only`),
- open world assumption,
- semantic constraints, and
- reasoning behavior

form part of the intellectual foundation for nearly all advanced ontology modeling.

Without a strong understanding of these concepts, later topics often become:

- difficult,
- confusing, or
- semantically disconnected.

When I learned `Pizza.owl` tutorial the first time, I still remember how hard to me to graph the real theory behind those property restrictions.

Because follow the tutorial to play around in Protégé is one thing, but understanding the inter-relationships among those restrictions are the another layer of knowledge.

This is one of the reasons why Chapters 14–16 now intentionally contains:

- more explanation,
- more examples, and
- deeper reflection (especially I found from mathematic aspects).

The additional depth is not accidental.

It is

> **architecturally important.**

Equally important: if you have reached this point in the book, you have already completed a substantial semantic journey.

You have learned how to:

- model classes,
- design hierarchies,
- define relationships,
- create individuals,
- distinguish types of property,
- apply reasoning,
- govern semantic meaning, and now:
- **express formal logic through restrictions**.

This is already significantly beyond what many introductory ontology learners experience.

So before moving forward, this may actually be a good moment to:

- take a short break,
- revisit earlier chapters,
- experiment again inside Protégé,
- reflect on what has been learned, or even simply
- **celebrate how far you have already come!**

Ontology engineering is rarely mastered in a single reading.

It develops gradually through:

- experimentation,
- reflection,
- mistakes, and
- repeated semantic refinement.

By reaching Chapters 14–16, you are already beginning to think more like:

> **a professional semantic architect.**

With that milestone recognized, this is a good moment to step back and examine:

> how all previous chapters collectively contributed toward building on:

> **Executable Knowledge Architecture (EKA)**

16.2.2 Before Your Move Forward: A Quick Self-Assessment

Before continuing to the next section, take a moment to check your understanding of the core concepts introduced in this chapter.

If you can answer these four questions confidently, you have successfully internalized the most important ideas.

If not, no worry, I myself learned this chapter in tutorial for several times, so just consider revisiting the relevant sections before proceeding.

Question 1: What is the Core Difference between `some` (Existential Restriction) and `only` (Universal Restriction)?

Can you explain it in your own words, not just by reading the syntax?

Focus on what each restriction asks, what kind of evidence each searches for, and how each behaves when no relationship exists.

Related sections: 14.4 (Chapter 14), 15.1.3 (Chapter 15)

Question 2: Why is `only` Automatically Satisfied (Vacuously True) When No Relationship Exists?

This is one of the most counter-intuitive aspect of OWL reasoning for beginners.

Can you explain the logical principle of vacuous truth using a simple everyday example, and then connect it back to the `Pizza.owl` scenario?

Related sections: 15.1.3 (Chapter 15), Interesting Reading: Vacuous Truth in Formal Logic

Question 3: Why does `PizzaB` Return `False` in Cypher (Closed World) but `Unknown` in OWL Reasoning (Open World)?

`PizzaB` has no topping assertions.

- Under closed world assumption, missing information means false.
- Under open world assumption, missing information means unknown.

Can you explain why this distinction matters in enterprise environments where data arrives incrementally?

Related sections: 14.7.4 (Chapter 14), 15.3.4 Part 1, Part 4 (Chapter 15)

Question 4: What is the Difference between `EquivalentTo` and `SubClassOf` in terms of Reasoning Behavior?

One defines a one-way inheritance rule.

The other defines a two-way logical equivalence.

Can you explain why using `SubClassOf` instead of `EquivalentTo` would prevent the reasoner from automatically classifying `PizzaA` as a `CheesePizza`?

Related sections: 14.7.2 (Chapter 14), Interesting Read: Necessary vs. Sufficient Conditions

=== How to use this checklist: ===

- If you can answer all four questions clearly and confidently, you are ready to continue to 16.2.3.
- If any question feels unclear, take a short break, revisit the indicated related sections, and experiment again in Protégé or Neo4j.
- If you find a mistake in your understanding while reviewing, this is not a failure. It is a learning opportunity! Ontology engineering is refined through iteration.

16.2.3 From Ontology Learning to Executable Knowledge Architecture

Let's move, at first glance, the previous chapters of this book may appear to teach:

> isolated Protégé features,

One chapter introduced:

> class hierarchies.

Another focused on:

> inverse properties,

> domain and range,

or:

> property characteristics.

Yet, from the perspective of:

> **Executable Knowledge Architecture (EKA)**,

an important pattern begins to emerge.

The book has never been merely teaching:

> ontology syntax.

Instead, each chapter has progressively contributed toward constructing:

> **an executable semantic system.**

Recall the formal EKA definition introduced in Chapter (00):

$$EKA = (K, R, \Theta, \Phi, \Gamma)$$

Where:

- K = Knowledge Graph Layer
- R = Reasoning & Rules Layer
- Θ = Trigger Layer
- Φ = Execution Layer
- Γ = Governance Layer

Together, these components describe how semantic knowledge evolves from:

> connected information

toward:

> **executable intelligence.**

Seen through this perspective:

the `Pizza` ontology journey can be understood not simply as:

> learning ontology modeling,

but rather:

> progressively constructing the foundations of executable semantics.

16.2.4 EKA Tuple Mapping Across Chapters

The following table trying to summarize how each chapter contributed to the different layers of:

> **Executable Knowledge Architecture (EKA)**

The contribution levels are interpreted as:

- ● ● ●: Strong contribution
- ●●: Moderate contribution
- ●: Light contribution
- —: Minimal or not yet introduced

Chapter	Main Focus	K Knowledge Graph	R Reasoning & Rules	Θ Trigger	Φ Execution	Γ Governance
Ch00	EKA Foundation	●●	●●	●●	●●	●●
Ch01	Entering Ontology Engineering	●	—	—	—	—
Ch02	First Ontology in Protégé	●●	●	—	—	—
Ch03	Protégé Workspace	●	—	—	—	—
Ch04	Classes & Ontology Skeleton	● ● ●	●	—	—	●
Ch05	Named Classes	● ● ●	●	—	—	●
Ch06	Applying a Reasoner	●●	● ● ●	—	—	●
Ch07	Disjoint Classes	●●	●●	—	—	● ● ●
Ch08	RDF File Structure	● ● ●	●	—	—	●
Ch09	Class Hierarchy	● ● ●	●●	—	—	●
Ch10	Object Properties	● ● ●	●●	—	—	●
Ch11	Inverse Properties	● ● ●	● ● ●	—	—	●●
Ch12	Object Property Characteristics	● ● ●	● ● ●	—	—	● ● ●
Ch13	Domain & Range	● ● ●	● ● ●	—	—	● ● ●
Ch14	Property Restrictions	● ● ●	● ● ●	●	●	● ● ●

At this stage of the learning journey, you may already begin noticing an important reality:

> ontology engineering is becoming increasingly different from simply modeling data.

Earlier chapters primarily focused on:

- concepts,
- classes,
- hierarchies,
- relationships, and
- semantic structures.

The intellectual challenge was largely centered on:

> *how knowledge is represented.*

Chapters 14–16 introduced something considerably deeper.

The challenge is no longer merely:

> representing knowledge.

Instead, you are increasingly required to think about:

- semantic requirements,
- logical boundaries,
- reasoning behavior,
- incomplete information, and
- machine-interpretable meaning.

In other words, ontology is beginning to shift from:

> **structural modeling**

toward:

> **semantic logic engineering.**

This transition is both **powerful** and at times **intellectually demanding.**

Because formal semantic logic does not always behave according to:

> everyday intuition.

What appears obvious to humans may not be logically inferable to a reasoner.

Likewise, what seems semantically restrictive may still remain logically valid under:

> the Open World Assumption (OWA).

This is especially true when working with:

- existential restrictions (`some`),
- universal restrictions (`only`),
- necessary and sufficient conditions, and
- automated reasoning behavior.

As ontology models become increasingly expressive:

small misunderstanding may produce disproportionately large consequences.

A seemingly minor modeling choice may result in:

- unexpected inferences,
- missing classifications,
- semantic inconsistency, or
- unintended meaning propagation throughout a knowledge graph.

For enterprise semantic systems, such misunderstandings may eventually influence:

- governance quality,
- reasoning reliability,
- automation behavior, and ultimately
- trust in semantic intelligence.

For this reason, before concluding Chapters 14–16, it is valuable to step briefly away from:

> what restrictions are,

and instead examine:

> where ontology engineers most commonly make mistakes.

Understanding these modeling pitfalls is not simply about avoiding errors.

It is about strengthening:

- semantic precision,
- reasoning confidence, and
- professional ontology engineering practice.

With this perspective established:

we can now examine several of the most common misunderstandings encountered when working with:

> **OWL Property Restrictions.**

16.3 Common Mistakes in Restriction Modeling

By this point in Chapters 14–16, you have explored one of the most powerful capabilities in OWL ontology engineering:

> **Property Restrictions.**

You have learned:

- existential restriction (`some`),
- universal restriction (`only`),
- reasoning behavior,
- open world assumption (OWA), and
- semantic governance.

At first glance, restrictions may appear relatively straightforward.

After all, the syntax seems simple.

Expressions such as:

`hasTopping some CheeseTopping` or `hasTopping only VegetableTopping`

are visually concise and relatively easy to enter inside Protégé.

However, semantic simplicity in notation does not necessarily mean:

> conceptual simplicity.

In practice, property restrictions represent one of the area where ontology learners most frequently encounter:

- misunderstanding,
- unexpected reasoning results, and
- semantic modeling mistakes.

Interestingly, many ontology problems do not originate from:

> syntax errors.

Instead, they arise from:

> misunderstanding semantic meaning.

The ontology technically works.

The reasoners runs successfully.

No visible errors appear

Yet, the resulting semantic behavior becomes:

- incomplete,
- misleading, or sometimes,
- logically unintended.

This is particularly important because ontology engineering increasingly influences:

- enterprise knowledge systems,
- graph intelligence,
- governance frameworks, and
- executable reasoning.

A small misunderstanding in restriction modeling may eventually affect:

- classification quality,
- semantic trust,
- automation behavior, or
- reasoning accuracy.

For this reason, before concluding Chapters 14–16, it is valuable to review several of the most common mistakes learners encounter when working with:

> OWL property restrictions.

Understanding these pitfalls early will significantly strengthen:

- ontology engineering practice,
- semantic precision, and
- professional reasoning confidence.

16.3.1 Confusing `some` with `only

Perhaps the single most common mistake in ontology learning is confusing:

> **existential restriction (`some`)**

with:

> **universal restriction (`only`)**.

At first glance: the two may appear similar.

- Both involve **object properties.**
- Both reference **classes**.
- Both influence **reasoning behavior**, and
- Both appear inside **restriction expressions**.

Yet semantically: they express fundamentally different logical ideas.

Recall the existential restriction: `hasTopping some CheeseTopping`

Semantically, this means:

> **at least one `CheeseTopping` must exist.**

This creates:

> a requirement (to the ontology).

It answers the logical question:

> *Must something exist?*

By contrast:

the universal restriction: `hasTopping only VegetableTopping`

means:

> **if topping exist, all of them must belong to `VegetableTopping` .**

This creates:

> a boundary.

It answers the question:

> *What is allowed to exist?*

This distinction is critically important.

Many beginners incorrectly interpret: `hasTopping only VegetableTopping` as meaning:

> the pizza must contain vegetables.

However, this interpretation is logically incorrect.

A pizza with **no toppings at all** still satisfies the restriction.

Why?

Because OWL evaluates the statement logically as:

> *if toppings exist, they must all be vegetables.*

While, if NO toppings exist, **there is no violation.**

This behavior often surprises learners because:

> human intuition

and:

> formal semantic logic

do not always behave identically.

Humans tend to think:

> "No vegetables means not vegetarian."

The reasoner thinks differently:

> "No evidence violates the restriction (boundary)."

A useful memory aid is:

- `some` = requirement
- `only` = limitation

Or more intuitively:

- `some` asks: "What must exist?"
- `only` asks: "What is allowed to exist?"

Understanding this distinction is one of the most important milestones in mastering:

> OWL restriction semantics.

16.3.2 Assuming Closed World Behavior

Another major misunderstanding occurs when learners unconsciously expect ontology to behave like:

> a traditional database.

This expectation is understandable.

Most software systems, relational databases, and business applications implicitly operate under what is called:

> **Closed World Assumption (CWA)**.

Under CWA, missing information is typically interpreted as:

> false.

For example, if a customer database contains no email address, the system may assume:

> the customer does not have one.

OWL ontology behave differently.

As discussed earlier in this chapter, OWL follows the:

> **Open World Assumption (OWA)**.

Under OWA, missing information means:

> **unknown.**

Not:

> false.

This difference becomes especially important when working with:

> restrictions.

Consider the following definition:

```
CheesePizza EquivalentTo

Pizza and

(hasTopping some CheeseTopping)
```

Now imaging `PizzaB` exists.

But no topping information has been added.

Then many learners instinctively conclude:

> `PizzaB` is not `CheesePizza` .

The reasoner does not make this conclusion.

Instead, the reasoner says:

> **I do not know yet.**

No evidence currently proves:

```
hasTopping some CheeseTopping
```

But equally, no evidence disproves it.

This distinction is subtle, yet extremely important!

OWL reasoners think:

> logically.

Not:

> intuitively.

A useful principle to remember is:

$$\texttt{No evidence} \neq \texttt{Evidence of absence}$$

This idea becomes increasingly important in:

> enterprise semantic systems.

Because enterprise knowledge is often:

- incomplete,
- delayed, or
- continuously evolving.

Ontology engineering therefore learns to tolerate:

> uncertainty.

Rather than prematurely forcing conclusions.

16.3.3 Forgetting the Difference Between `EquivalentTo` and `SubClassOf`

Another very common mistake appears when learners misunderstand the difference between:

> `EquivalentTo`

and:

> `SubClassOf` .

This distinction becomes especially important when expecting:

> automatic reasoning.

Earlier examples in this chapter used definitions such as:

```
CheesePizza EquivalentTo

Pizza and

(hasTopping some CheeseTopping)
```

Why use `EquivalentTo` instead of `SubClassOf` ?

Because `EquivalentTo` establishes:

> **necessary and sufficient conditions.**

Reasoning therefore works **in both directions.**

This means:

If something is `CheesePizza` , then `hasTopping some CheeseTopping` MUST hold true.

But equally important:

If a `Pizza` satisfies `hasTopping some CheeseTopping` , the reasoner may infer `CheesePizza` .

This enables:

> automatic classification.

By contrast:

```
CheesePizza SubClassOf

(hasTopping some CheeseTopping)
```

only establishes **a one-way semantic rule.**

It means:

> every `CheesePizza` must satisfy the condition.

However, the reasoner will **not automatically infer** `CheesePizza` simply because cheese toppings exist.

This misunderstanding causes many learners to think:

> "The ontology is not working!"

In reality, the ontology behaves exactly as designed.

The issue lies in:

> semantic expectation.

A useful mental shortcut could be:

- `EquivalentTo` = `necessary + sufficient`
- `SubClassOf` = `necessary only`

Whenever automatic classification matters, carefully consider:

> whether bi-directional reasoning is required.

Because this modeling decision strongly influences:

> reasoning outcomes.

16.3.4 Misunderstanding Governance vs Validation -- OWL vs SHACL

One of the most important distinctions for enterprise architects is **understanding the difference between:**

> **semantic governance**

and:

> **data validation.**

Many learners mistakenly assume restrictions behave like:

> validation rules.

This is only partially true.

OWL restrictions do not behave exactly like:

> database constraints.

Nor are they identical to:

> validation frameworks such as SHACL.

For example: consider `hasTopping only VegetableTopping` , OWL interprets this semantically.

The reasoner evaluates **logical meaning**.

It may:

- infer knowledge,
- identify inconsistency, or
- preserve semantic coherence.

However, OWL does NOT necessarily reject incomplete information.

This reflects:

> Open World Assumption.

By contrast, validation frameworks such as **SHACL** operate differently.

SHACL explicitly asks:

> *Does the data violate defined constraints?*

This resembles:

> governance enforcement.

Rather than:

> semantic reasoning.

A simplified distinction is:

- `OWL` = `meaning and inference`
- `SHACL` = `validation and enforcement`

Within:

> Γ **- Governance layer**

of EKA: both are valuable.

OWL restrictions help govern:

> semantic meaning.

SHACL helps validate:

> operational correctness.

Together, they strengthen:

- trustworthiness,
- semantic integrity, and
- governed enterprise knowledge.

This distinction becomes increasingly important when ontology evolves toward:

> production-grade semantic systems.

=== Interesting Reading: A First Look at SHACL ===

At this point, curious readers may wonder:

> If OWL focuses on semantic reasoning, is there a formal language specifically designed for validating graph data?

The answer is **Yes - SHACL**.

SHACL stands for:

> **SHApes Constraint Language**

and is a W3C recommendation designed for validating RDF graphs against explicitly defined structural constraints.

Conceptually: SHACL introduces the idea of a **shape**.

A shape describes:

> what a valid node in a graph should look like.

For example, a SHACL shape may specify rules such as:

- every `Pizza` must have at least one topping,
- every topping must belong to `PizzaTopping` , or
- a `VegetarianPizza` cannot contain `MeatTopping` .

Unlike OWL, which primarily asks:

> *What semantic meaning can be inferred?*

SHACL asks a different question:

> *Does this graph instance satisfy required constraints?*

This difference is subtle but important!

- OWL is fundamentally **inferential and descriptive.**
- SHACL is fundamentally **constraint-oriented and validation-driven.**

From a mathematical perspective, SHACL can be viewed as form of:

> predicate-based constraint evaluation over graph structures.

Given a graph: $G = (V, E)$, where

- $V =$ `vertices (nodes)`
- $E =$ `edges (relationships)`

a SHACL shape can be interpreted as a constraint function:

$$S(v) \rightarrow \{true, false\}$$

where a node: $v \in V$ either **satisfies the constraint** or **violates it**.

In simplified mathematical terms:

> SHACL behaves somewhat like a set of logical predicates evaluated over graph topology.

This makes SHACL especially useful in enterprise environments where semantic governance requires not only:

> reasoning,

but also:

> strict data quality validation.

Michael's original `Pizza` tutorial formally introduces SHACL later in his Chapter 11 of the PDF.

In this book, we will revisit SHACL in much greater practical detail, including additional examples beyond the original tutorial to show how SHACL can complement OWL inside modern semantic architectures.

For now, the key takeaway is simple:

- `OWL helps machines infer meaning.`
- `SHACL helps engineers validate quality.`

Both play important roles in building trustworthy semantic systems.

There is no need to master SHACL yet.

For now, simply remember:

> **semantic intelligence requires both reasoning and validation.**

=== END Interesting Reading ===

16.3.5 Overusing Restrictions

After learners discover the expressive power of restrictions, another common mistake often appears:

> **over-modeling.**

Restrictions are powerful.

They allow ontology engineers to express:

- requirements,
- boundaries,
- logical expectations, and
- inferential conditions.

Because of this, beginners sometimes attempt to model **everything.**

Every class becomes filled with **nested restrictions**, **multiple logical conditions** and **excessive semantic detail**.

This may initially appear:

> sophisticated.

However, in practice, it often produces:

> unnecessary complexity.

Large numbers of restrictions may eventually cause:

- reasoning slowdown,
- maintenance difficulty,
- conceptual confusion, or
- semantic rigidity.

A useful engineering principle is:

> **model only semantics that create VALUE**.

Before introducing a restriction, ask:

> Does this improve reasoning?

> Does it strengthen governance?

> Does it clarify business meaning?

> Does it improve semantic precision?

> Does it contribute to executable intelligence?

If the answer is unclear, then the restriction may NOT be necessary.

In enterprise architecture, this problem resembles:

> technical debt.

Poorly designed semantic logic creates:

> **semantic debt**.

Meaning, the ontology becomes:

- difficult to explain,
- difficult to maintain, or
- difficult to trust.

Professional ontology engineering values:

> semantic clarity

over:

> semantic complexity.

A good ontology is not necessarily:

> the most expressive.

It is often:

> the most understandable.

16.3.6 Enterprise Architecture Perspective -- Why These Mistakes Matter

At this point, a reasonable question may emerge:

> *Do these mistakes really matter?*

In small learning exercises: perhaps NOT.

In a `Pizza` ontology, the consequences remain limited.

However:

inside real enterprise semantic systems, these misunderstandings can become:

> expensive!

Consider a semantic knowledge graph used for:

- financial risk,
- cybersecurity,
- healthcare governance, or
- enterprise architecture.

A misunderstanding of `some` vs. `only` may produce:

> incorrect classification.

A misunderstanding of `EquivalentTo` may cause:

> missing inference.

Incorrect expectations about "**Open World Assumption**" may lead to:

> flawed analytics.

Weak governance design may create:

> semantic inconsistency.

Over time, trust in semantic systems may decline.

This is one of the reasons why **ontology engineering** requires careful thinking.

Ontology is merely:

> data modeling.

It increasingly becomes:

> **semantic engineering**.

Viewed through:

> **Executable Knowledge Architecture (EKA)**,

the restriction mistakes may impact multiple layers simultaneously:

- $K \rightarrow$ `incorrect semantic relationships`
- $R \rightarrow$ `incorrect reasoning outcomes`
- $\Theta \rightarrow$ `incorrect trigger conditions`
- $\Phi \rightarrow$ `unreliable execution behavior`
- $\Gamma \rightarrow$ `weakened semantic governance`

This demonstrates why semantic precision matters.

Because semantic systems increasingly influence:

- automation,
- trust,
- decision, and
- executable intelligence.

By understanding these common mistakes early, learners become better prepared to design ontology that is not only **logically correct**, but also **explainable**, **governable** and **operationally trustworthy**.

16.4 Chapter Summary

16.4.1 What We Learned

Chapters 14–16 marked one of the most important conceptual milestones in this ebook.

In earlier chapters, ontology engineering primarily focused on:

> semantic representation.

You built knowledge structures through:

- classes,
- hierarchies,
- object properties,
- inverse properties, and
- semantic relationship constraints.

Those foundational components taught us how ontology represents:

> structured meaning.

In this chapter (14), however, ontology became significantly more expressive.

We moved beyond simple structural modeling and introduced:

> semantic requirements.

This shift occurred through the study of **Property Restrictions.**

Property restrictions allowed us to express **logical** conditions over relationships, enabling ontology to describe not only:

- *what entities are,*

but also:

- *what relationships must exist,*
- *what relationships are allowed,* and
- *what semantic boundaries must be respected.*

This marks a major conceptual transition in OWL modeling.

Ontology is no longer merely descriptive.

It increasingly becomes:

> **logical!**

16.4.2 Why Restrictions Matter

Throughout this chapter (14), two major forms of restrictions were introduced.

First:

> **Existential Restriction (`some`)**

This restriction answers the question:

> *Must at least one relationship exist?*

For example: `hasTopping some CheeseTopping`

This expresses a semantic requirement that at least one cheese topping must exist.

Second:

> **Universal Restriction (`only`)**

This restriction answers:

> *If relationships exist, what are they allowed to connect to?*

For example: `hasTopping only VegetableTopping`

This creates a semantic boundary.

Together, these two restrictions form one of the most powerful modeling tool in OWL.

They enable ontology engineers to define concepts with far greater precision than traditional data models typically allow.

Restrictions therefore contribute to:

- stronger reasoning,
- improved governance,
- semantic consistency, and
- higher-quality knowledge graphs.

16.4.3 The Semantic Shift

Perhaps the most important lesson of Chapters 14–16 is philosophical rather than technical.

This chapter revealed a major distinction between:

> storing data

and:

> encoding meaning.

Traditional graph databases may store relationships such as:

`Pizza → hasTopping → CheeseTopping`

This records:

> a fact.

However, OWL restrictions express something deeper:

```
Pizza and (hasTopping some CheeseTopping)
```

This describes:

> semantic logic.

That distinction is fundamental.

A graph database primarily stores:

> relationships.

An ontology describes:

> logical meaning.

When reasoning is applied, ontology begins enabling:

> machine-interpretable intelligence.

This is why Chapters 14–16 represents an important turning point toward:

> Executable Knowledge Architecture (EKA).

Restrictions begin transforming semantic knowledge into:

> actionable semantic logic!

16.5 Key Concepts

Before moving forward, ensure you are comfortable with the following key concepts introduced in this chapter (14).

Concepts	Description
Property Restriction	Logical constraint applied to property relationships
Existential Restriction (`some`)	Requires at least one qualifying relationship
Universal Restriction (`only`)	Restricts all related values to a specified class

Concepts	Description
Open World Assumption	Missing knowledge is treated as unknown, not false
`EquivalentTo`	Defines necessary and sufficient conditions
`SubClassOf`	Defines necessary conditions only
Semantic Governance	Constraints that preserve semantic integrity
SHACL	Constraint language for graph validation

These concepts will continue to reappear throughout later chapters.

16.6 Protégé Skills Checklist

By completing Chapters 14–16, you should now be able to confidently perform the following tasks in **Protégé**:

- Create existential restrictions using `some`
- Create universal restrictions using `only`
- Understand restriction syntax in OWL
- Distinguish between `EquivalentTo` and `SubClassOf`
- Predict reasoning outcomes under `Open World Assumption`
- Interpret reasoner classifications involving restrictions
- Understand the conceptual distinction between OWL and SHACL
- Explain the governance role of semantic restrictions

If several items still feel unclear, revisiting the practical exercises before moving on is strongly recommended.

16.7 Looking Ahead -- Creating Subclasses

16.7.1 Why Subclasses Matter

In the next Chapter (15), we will return to a concept that may initially appear simpler than restrictions:

> subclasses.

However, subclass modeling is far more important than it may first appear.

Subclasses provide the hierarchical backbone of ontology.

They allow knowledge engineers to organize concepts into:

> semantic taxonomies.

Without hierarchy, ontology quickly becomes:

- flat,
- difficult to reason over, and
- semantically weak.

16.7.2 Restriction and Hierarchy Together

An important insight for you moving into Chapter (15) is this:

> restrictions and hierarchies are not separate modeling techniques.

They work together!

- Class hierarchy provides **structural organization**.
- Restrictions provide **logical refinement**.

Together, they enable ontology to express both:

> semantic structure

and:

> semantic intelligence.

This combination is central to advanced ontology engineering.

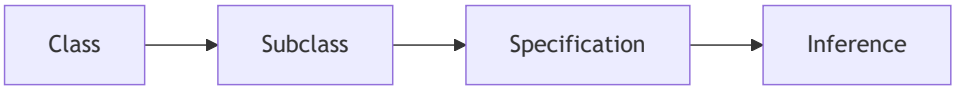
16.7.3 Preview of Exercise 14

In the next chapter (15), based on the continuation of the `Pizza` ontology tutorial, we will explore how subclass creation strengthens:

> semantic specification.

You will see how increasingly specific concepts can inherit meaning from broader parent classes while simultaneously supporting more precise reasoning.

This continues the natural progression of ontology engineering:



As the ontology grows richer, semantic knowledge becomes increasingly structured, governed, and inferable.

This prepares us for the next major step in the journey toward:

> executable semantic intelligence.

16.8 References

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6. Cook, Timothy. *Foreword to Semantic Data Charter*.
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Last updated at: 2026-09-11

Appendix A -- Chapter Mapping with Exercises

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-1	Chapter 00~03	N/A	N/A	N/A	Foundations & EKA Introduction
Volume-1	Chapter 04	Exercise 1, 2, 3	Section 4.0	04	Creating Named Classes
Volume-1	Chapter 05	Exercise 4	Section 4.1	05	Using a Reasoner
Volume-1	Chapter 06	Exercise 5	Section 4.2	06	Disjoint Classes
Volume-1	Chapter 07	Exercise 6	Section 4.3	07	Create Class Hierarchy
Volume-1	Chapter 08	N/A	N/A	N/A	Conceptual Modeling Foundations
Volume-2	Chapter 09	Exercise 7	Section 4.4	09	Object Properties
Volume-2	Chapter 09	Exercise 8	Section 4.5	09	Creating Property Hierarchies
Volume-2	Chapter 10	Exercise 9	Section 4.6	10	Inverse Properties
Volume-2	Chapter 11	Exercise 10	Section 4.7	11	Property Characteristics
Volume-2	Chapter 12	(Added by Author)	N/A	N/A	Property Characteristics Deep Dive

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-2	Chapter 13	Exercise 11, 12	Section 4.9	13	Domain & Range
Volume-3	Chapter 14	Exercise 13	Section 4.10.2	14	Existential Restrictions
Volume-3	Chapter 15	N/A	Section 4.10.2	14	Semantic Description (Theory)
Volume-3	Chapter 16	N/A	Section 4.10.2	14	Description Logic Foundations
Volume-4	Chapter 17	N/A	N/A	N/A	SKDL Introduction
Volume-4	Chapter 18	Exercise 14	Section 4.10.3	15	Stage 1 — Conceptual Modeling
Volume-4	Chapter 19	Exercise 15	Section 4.10.3	16	Stage 2 — Semantic Description
Volume-4	Chapter 20	Exercise 16	Section 4.10.3	17	Stage 3 — Knowledge Reuse (Cloning Pattern)
Volume-4	Chapter 20	Exercise 17	Section 4.10.3	18	Stage 3 — Knowledge Reuse (Practicing Specialization)
Volume-4	Chapter 21	Exercise 18	Section 4.10.3	19	Stage 4 — Semantic Governance (Disjointness)
Volume-4	Chapter 22	Exercise 19	Section 4.10.4	20	Stage 5 — Semantic Validation (Probe Class)
Volume-4	Chapter 23	Exercise 20	Section 4.11	21	Stage 6 — Semantic Reasoning (Primitive Classes)
Volume-4	Chapter 23	Exercise 21	Section 4.11		Stage 6 — Semantic Reasoning (Defined Classes)

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-4	Chapter 24	N/A	N/A		Stage 7 — Executable Knowledge (Introduction)
Volume-5	Chapter 25	Exercise 22	Section 4.12		Universal Restrictions
Volume-5		Exercise 23	Section 4.13		Open World Assumption & Closure Axioms
Volume-5		Exercise 24	Section 4.14		Enumerated Classes
Volume-5		Exercise 25	Section 4.15		hasValue Restrictions
Volume-5		Exercise 26	Section 4.15		Cardinality Restrictions
Volume-5		Exercise 27	Section 5.1		Datatype Properties
Volume-5		Exercise 28, 29, 30, 31, 32	Section 5.2		Data Property Values & Classification
Volume-5		Exercise 33	Section 9.1		DL Queries
Volume-5		Exercise 34, 35, 36	Section 10.0		SWRL & SQwRL Rules
Volume-6					Advanced Topics
Volume-7					EKA & AI Integration

Last Updated at: 2026-09-11

Contributors

Name / GitHub Handle	Contribution Type	Issue/Discussion	Chapter(s)	Date	Memo
Timothy Cook	Issue Report	#66	20 & 21	2026-09-06	Continue technical review
000wahab000	Pull Request Contribution	PR #57	Various	2026-09-04	Contributed updates and improvements via Pull Request #57.
Jah-yee	Pull Request Contribution	PR #54	Various	2026-09-04	Submitted the repository's first community-contributed pull request, helping enhance project collaboration and open-source engagement.
sarcanon	Question / Issue Report	#39-#50	Various	2026-08-28	Many detail issue reports
LaraAcuna	Question / Content Improvement	#26	13	2026-08-12	Raised a thoughtful question about modeling `OR` semantics in Domain and Range (Chapter 13). Her question prompted the addition of a

Name / GitHub Handle	Contribution Type	Issue/Discussion	Chapter(s)	Date	Memo
					complete guide to `owl:unionOf`, separate properties, and local restriction patterns, including new screenshots, RDF examples, and a comparison table. This contribution significantly improved the practical guidance of Chapter 13. Fixed in v0.44.0.
Michael DeBellis	Refinement Suggestion	#19	15	2026-07-01	Proposed revised definition for VegetarianPizza pattern to align with common real-world interpretation (include cheese/dairy). Pending.
Michael DeBellis	Terminology Clarification	#20	Multiple	2026-07-01	Highlighted need to clearly distinguish

Name / GitHub Handle	Contribution Type	Issue/Discussion	Chapter(s)	Date	Memo
					"Annotation Properties" (reasoner-invisible) from logical property axioms. Pending.
Michael DeBellis	Critical Distinction / Error Report	#21	00, 08	2026-07-01	Identified fundamental conflation between Property Graphs (e.g., Neo4j) and Native RDF Graphs (e.g., AllegroGraph, GraphDB, Stardog). Major architectural correction required. Pending.
Michael DeBellis	Question / Technical Review	#9	00, 06, 14	2026-06-13	Clarified reasoner inference materialization distinction (Protégé UI behavior vs. production triplestores). Fixed in v0.39.0.
Michael DeBellis	Error Report	#10, #11, #12, #13, #14, #15	Various	2026-06-16	Multiple error reports including

Name / GitHub Handle	Contribution Type	Issue/Discussion	Chapter(s)	Date	Memo
					OWL modeling errors, contributor attribution, proofreading, and other corrections. See individual issues for details.
mlungsta89	Bug Report	#3	—	2025-09-15	Reported reasoner getting stuck issue. Closed.
nikokaoya	Feedback	#2	—	2024-06-26	Positive feedback and encouragement. Closed.
Your name here	Error report / Pull request / Question	—	X	YYYY-MM-DD	Brief Memo

Last updated: 2026-09-11